KU LEUVEN

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Maximising Edge Density Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work begins with the necessary background to formulate the problem and all its variants. It then discusses their answers or highlights the difficulty behind unsolved cases. Afterwards, computational research on small graphs checks finite instances of these values and supplies dense examples in open cases. This is accomplished by measuring connectivity with a maximum-flow routine and by searching for extremal graphs with metaheuristics. Completed exhaustive computations establish an extremal value in a finite search space. Heuristic searches and unfinished enumerations provide only lower bounds, witnessed by the feasible graphs they find. At the end, the thesis synthesises the results and presents the open problems that remain. This work is theoretical in nature, but its computational methods, constructions and open questions address the relationship between connectivity and size in networks.

I am grateful to my promotor, Dr. Stijn Cambie, for his day-to-day guidance, corrections, and mathematical feedback throughout the project. I could not ask for a more patient and enthusiastic supervisor. I thank my readers, Prof. Jan Goedgebeur and Prof. Karel Dekimpe, for their time and careful reading. In particular, Prof. Goedgebeur's expertise in computational graph theory significantly strengthened the computational part of this thesis. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work.

Contribution Statement

All the work in this thesis is my own, except where I explicitly say otherwise. I used artificial intelligence to implement the code, suggest and verify proofs, format LaTeX and check spelling. Where a result comes from the literature I say so and cite it. The theorems I use but do not claim as mine are those of Mader, Bártfai, Bollobás, Leonard, Sørensen and Thomassen, Menger, Ford and Fulkerson, Whitney, Gomory and Hu, Mantel and Turán, Fekete, and Huang and Lyu. Tutte's triconnected decomposition and its algorithmic form due to Hopcroft and Tarjan are cited once as a signpost for further work and are used in no proof. Related hypergraph-connectivity work includes Bérczi, Chandrasekaran, Király and Kulkarni [1]. Proofs are located in Appendix A and all computational work is available in my personal repository [2].

A result whose proof I have not checked line by line carries the badge [AI] in its heading, both in Chapter 1 where it is stated and in Appendix A where it is proved. Those proofs were written by an AI model, and for several of them I am not the right person to check them. A badged result should be read as a conjecture that comes with a proof attached which is likely correct, rather than as a theorem I can vouch for myself. The proofs that carry no badge are the ones I have worked through myself. Two of them I wrote out in full: the Gomory–Hu double count for the simple undirected edge-disjoint case (Theorem 1.2) and for the undirected multigraph edge-disjoint case (Theorem 1.3), together with the three counting lemmas they run on (Lemmas A.5 to A.7). The other unbadged results of Appendix A are ones whose proofs I have since read and checked. Results taken from the literature carry no badge either, since their proofs are published and are cited where the result is stated. Definitions, conjectures and open questions carry no badge. A construction carries a badge when its stated properties come with a proof that I have not checked line by line. The badge concerns verification of the proof, not whether the object is explicitly defined.

The clique-core counterexample and the research-note argument for eventual exactness were supplied by Stijn Cambie with AI assistance. The counterexample is included in Construction A.27. Only the verified construction counts are retained in Construction 3.1. Eventual optimality is not asserted.

The summary of my original results is:

Variant	Contribution
Hypergraph, vertex separation	Exact value at $m = 2$. The $m = 3$ universal bound and its stated attainment range, via a new incidence-rank lemma (Theorems 1.9 and 1.10).
Directed multigraph	Exact value for all n, m, plus quadratic-branch classification for $n \geq \max(8, 2m)$ (Theorems 1.13 and A.45).
Multigraph, incidence separation	The two multigraph vertex variants posed as problems of their own: exact values at $m \leq 3$, the reduction to a knapsack over 2-connected blocks, thickened bipartite blocks against an upper bound of the same order, and the directed value to its leading term with the same block structure (Theorems A.57, A.63 and A.64, Corollaries A.59 and A.66, and Propositions A.61 and A.68). Every result in this row carries the badge except the $m = 2$ value.
Directed simple graph, arc separation	For fixed $m \geq 3$, unconditional $n^2/4 + \Theta_m(n)$ bound and an $O(\sqrt{m}\, n^{3/2})$ structural distance from a one-way bipartite graph (Proposition A.36 and Theorem 1.11).
Transferred quadratic bounds	For fixed $m \geq 3$, $k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$. For fixed $r \geq 3$ and $m \geq 2$, leading term $(m-1)n^2/(4(r-1))$ for directed hypergraphs under both separations and all orientations (Theorems 1.12, 1.14 and A.74).

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [3], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m-1\}.$$

Mader proved that, for simple graphs and all $n \geq 2$ and $m \geq 2$,

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

and provided a construction achieving this bound when $(m-1)|(n-1)$. This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, hypergraphs and multihypergraphs), the path-separation notion and the presence or absence of directions. The two separation notions are edge-disjointness and internal vertex-disjointness, the second read in the incidence graph, so that separated routes share neither an intermediate vertex nor an edge copy. Parallel copies are therefore separate routes, which makes the two multigraph vertex variants problems of their own rather than restatements of the simple ones. The simple edge-disjoint case is proved independently here using Gomory–Hu trees [4], and a general extremal construction resolves it for all n and m. Other rigorously settled variants include the multigraph value $L_m(n) = (m-1)(n-1)$, the equality of the simple edge and vertex problems for $m \leq 4$, due to Bártfai [5], Bollobás [6] and Leonard [7], and their divergence at $m = 5$, shown by Sørensen and Thomassen [8]. The directed case is resolved for $m = 2$: the value $\ell_2^{\mathrm{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved in a few lines, a feasible digraph at $m = 2$ being its own reachability skeleton and so inheriting the skeleton's arc count. For $m \geq 3$, the thesis gives dense constructions, without asserting an exact eventual value. A twelve-vertex clique-core construction supplied by Stijn Cambie refutes the earlier formula for all orders (Construction A.27).

The centrepiece is then a unified program, which represents non-hypergraphs with a multiplicity matrix and converts the vertex-disjoint problem to an edge-disjoint one by building the capacity network of the vertex-splitting construction, in which a vertex is a unit gate while an adjacency keeps its multiplicity. The hypergraphs are handled as a list of hyperedges, which may use head and tail sets to represent directed hyperedges. Then a maximum-flow algorithm measures connectivity between all pairs of vertices and reports the maximum local connectivity. At each point of the program, optimisations are applied to reduce the search space and traverse it more efficiently. When these optimisations are exhausted and the computation becomes intractable, the program uses metaheuristics to search for extremal graphs, which are then checked for correctness.

The hypergraph vertex problem is settled completely at $m = 2$. At $m = 3$ the formula $\lfloor 2(n-1)/(r-1) \rfloor$ is a universal upper bound for multihypergraphs and is exact whenever $r \geq 3$ and $2 \leq \binom{n-2}{r-2}$, the upper bound following from a new rank bound on incidence graphs proved by induction over blocks and separating pairs. On the directed side the multigraph problem closes completely, $L_m^{\mathrm{dir}}(n) = (m-1)\max(2(n-1), \lfloor n^2/4 \rfloor)$ for every n and every m, by deleting a sparse reachability-preserving subgraph at a time rather than a vertex at a time. For every fixed $m \geq 3$, the extremal value for simple digraphs, arc and vertex alike, is $n^2/4 + \Theta_m(n)$ with no assumption on the shape of the extremiser, and at $m = 2$ it differs from $n^2/4$ by only an $O(1)$ correction. For fixed $r \geq 3$ and m, the directed hypergraph, which had only bounds a factor of four apart, has its leading term settled at $(m-1)n^2/\big(4(r-1)\big)$, so its bipartite construction is asymptotically optimal.

Abbreviations and Symbols

MILP	mixed-integer linear program
[AI]	in a theorem heading: the proof of that result was written by an AI model and is not verified line by line by the author (Contribution Statement)

$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m-1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u-v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u-v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\mathrm{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\mathrm{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\mathrm{dir}}(n)$	simple directed vertex-disjoint extremal function
$K_m(n)$, $K_m^{\mathrm{dir}}(n)$	undirected and directed multigraph vertex-disjoint extremal functions, multiplicity counted (Section 1.3.1)
$\ell_m^{(r)}(n)$, $k_m^{(r)}(n)$	r-uniform simple hypergraph edge-disjoint and incidence-graph vertex-disjoint extremal functions
$L_m^{(r)}(n)$, $K_m^{(r)}(n)$	the same two for r-uniform multihypergraphs, multiplicity counted (Table 1.1)
$M(n)$	$\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$, the directed multigraph value per unit of $m-1$
$M^*(n)$	the m-free directed multigraph optimum, $L_m^{\mathrm{dir}}(n) = (m-1)\, M^*(n)$, equal to $M(n)$ by Corollary A.41
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p-part to a q-part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$h_m(b)$	most edges a feasible 2-connected block on b vertices can carry (Theorem A.13)
$W_m(G_0)$	largest total multiplicity over a fixed underlying simple graph G_0 (Lemma A.62)
$g_m(b)$	most W_m a feasible 2-connected block on b vertices can score (Theorem A.63)
c_m	per-vertex growth constant of the simple undirected vertex problem, $c_m = \lim_{n\to\infty} k_m(n)/(n-1) = \sup_{b\ge 2} h_m(b)/(b-1)$ (Theorem A.13)

Contents

Chapter 1

The Problem and Its Variants

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter gives the foundation for the question and its variants and goes over the results that can be proven using logical arguments.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *(multi)graph* G is a finite set of points, called *vertices*, together with a multiset of *edges*, each joining two distinct vertices. Asking the two ends to be distinct rules out loops, which some texts do admit and which play no role in anything below. We write $V(G)$ for the set of vertices, $E(G)$ for the multiset of edges, and $\mu(u, v)$ for the *multiplicity* of a pair, the number of copies of the edge joining u and v. The *size* $|E(G)| = \sum_{\{u,v\}} \mu(u, v)$, the sum running over unordered pairs of distinct vertices, counts the edges with their multiplicity, and it is the very quantity this thesis tries to make as large as possible under certain constraints. If $\mu(u, v) \leq 1$ for every pair, so that no edge is repeated, G is a *simple graph*. A repeated edge joins two vertices exactly as a single edge does, so what separates the two notions is the multiplicity rather than the number of ends. One pictures such a simple graph by representing the vertices as dots and the edges as lines between them.

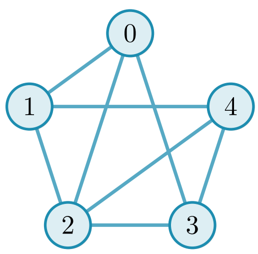


Figure 1.1. The simple graph $K_5 - \{04, 13\}$ has five vertices and eight edges, where K_n is the complete graph on n vertices.

We write $n := |V(G)|$ for the number of vertices and define the *degree* of a vertex as the number of edges meeting it. Unless specified otherwise, graphs are simple. A *path* from a vertex u to a vertex v is an alternating sequence $v_0, e_1, v_1, \ldots, e_q, v_q$ of distinct vertices and specified edge copies, where $v_0 = u$, $v_q = v$ and each e_i joins v_{i-1} to v_i. Thus parallel copies give distinct one-edge paths. In a simple graph the edge copies are determined by their endpoints, so a path may be written as a sequence of vertices alone. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u,v)$. The pair with the most such paths determines the value for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u,v).$$

For a graph with fewer than two vertices, we set this maximum to zero. The same convention applies to the other maximum local connectivities below. A theorem of Menger [9], recalled in Appendix A, gives this quantity a second face: $\lambda_G(u,v)$ also equals the least number of edges one must remove to cut every route from u to v (Figure 1.2). The maximum number of edge-disjoint paths between a pair is exactly the number of edges in the minimum cut between them when every edge has unit capacity.

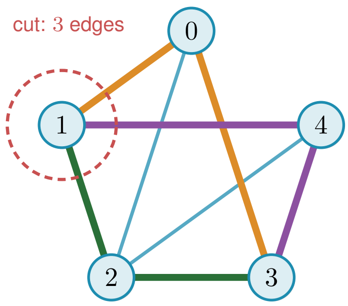


Figure 1.2. Menger's theorem on Figure 1.1: three edge-disjoint 1–3 routes (coloured) meet the minimal three-edge cut around vertex 1, so $\lambda(1,3) = 3$.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a cap on connectivity. Requiring $\lambda_G^{\max} \leq m-1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be separated by removing fewer than m edges. The question is how many edges a graph can hold while staying under that cap.

Question 1.1 (Erdős Problem 915 [10]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{\, |E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m-1 \,\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the cap. Mader proved that no graph under the cap has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound) in the simple-graph case and provided the shared-clique construction, which achieves the (lower) bound but is only defined when $m-1$ divides $n-1$. The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [11]). *For all $n \geq 2$ and $m \geq 2$,*

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Notice how $n \geq m$ is not needed for the inequality itself, and is generally needed to attain equality, since a graph on $n < m$ vertices cannot exceed the edge count $\binom{n}{2}$ of the complete graph on it, which falls short of $\lfloor m(n-1)/2 \rfloor$ once m exceeds n. The floor can still let a small case through by coincidence: K_2 attains $\lfloor 3/2 \rfloor = 1$ at $n = 2$, $m = 3$. The proof given in Appendix A was discovered independently of Mader and uses the Gomory–Hu tree (Theorem A.2). The shared clique graph is then generalised to an extremal construction that exists for all values of $n \geq m$ (Theorem A.10). The very same Gomory–Hu tree returns for the multigraph variant (Theorem 1.3). The hypergraph bound (Proposition 1.8) is proved by an elementary induction on a hyperedge cut instead. Hence its properties are recorded here. Every graph has a weighted spanning tree such that $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v. All $\binom{n}{2}$ local connectivities are therefore encoded by just $n - 1$ weights. The largest of them, $\lambda_G^{\max}$, is simply the heaviest edge of the tree. The proof (Appendix A) uses a double counting argument on that tree.

1.3 Variants of the problem

So far the problem has been stated for simple graphs, but the same question remains unanswered in a more general setting. Each assumption made so far will be toggled to create a lattice of variants and the thesis will explore each node of that lattice.

The first is *arity*: an edge joins exactly two vertices and a *hyperedge* joins $r \geq 2$ vertices. Hence such *hypergraphs* use *Berge paths*, which aligns with the path definition above. The size of a hypergraph counts its hyperedges with their multiplicity, exactly as the size of a graph counts its edges. The second is *multiplicity*: a *simple* graph or hypergraph carries at most one copy of each edge, so $\mu \leq 1$ throughout, while a *multigraph* allows parallel copies. The third is *direction*: a directed edge, called an arc, is an ordered pair of vertices and a directed hyperedge has one tail and $r - 1$ heads. These three toggles create a lattice of eight models, shown in Figure 1.3.

The digraphs do not have a Gomory–Hu tree, but they do have a directed analogue of Menger's theorem. A digraph has three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. Figure 1.4 shows a digraph carrying three arc-disjoint routes between one chosen pair of vertices. The directed form of Menger's theorem says that route count equals the size of the smallest arc-cut separating the pair, and the figure also marks the out-degree and in-degree that bound both quantities from above, which is exactly how Menger reads in a one-way network. A directed r-hyperedge has one tail and $r - 1$ heads.

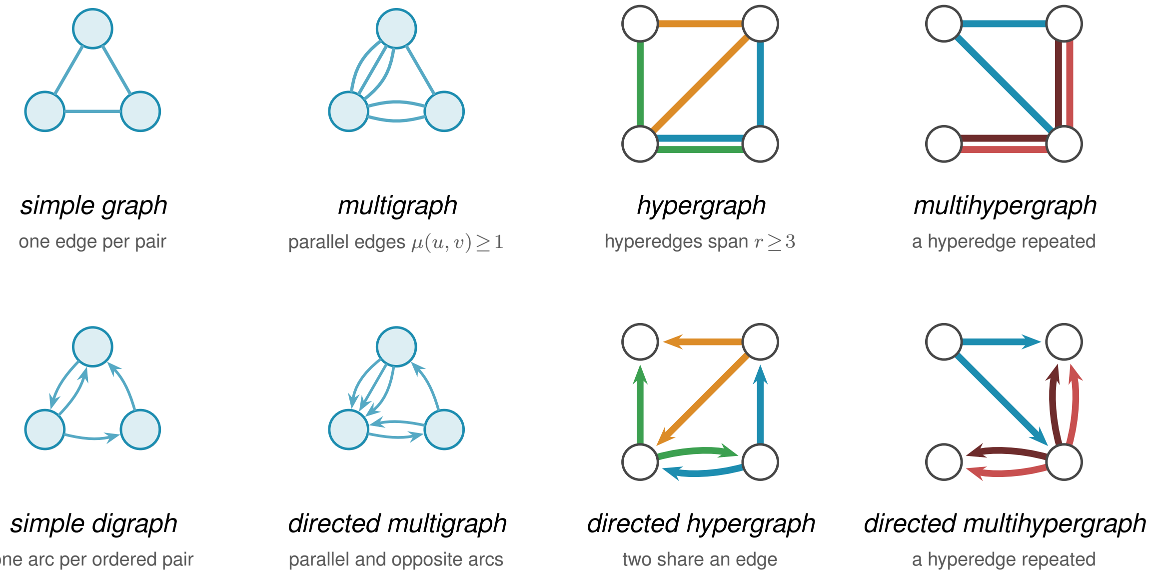


Figure 1.3. The eight models side by side, with the four undirected on top and the four directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges. In the two multihypergraph panels the two shades of red are the two copies of one hyperedge, not two different hyperedges.

In the figures, one colour denotes one hyperedge, drawn either as a star or as a line oriented away from its tail. It is a single edge on several vertices, not a path of ordinary edges. A directed *Berge route* enters that edge at its tail and leaves at one of its heads.

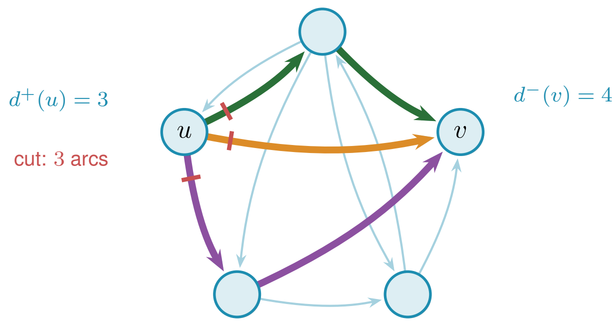


Figure 1.4. Three arc-disjoint $u \to v$ routes (coloured) and the corresponding three-arc directed cut (red) in a *simple* digraph. Thus $\lambda(u,v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

The final assumption that will be considered is *separation*. For graphs and digraphs, routes may be required either to avoid sharing edges, as above, or to be internally vertex-disjoint, sharing no intermediate vertex. This gives the local vertex-connectivity $\kappa_G(u,v)$ and its maximum $\kappa_G^{\max}$. Whitney's inequality $\kappa_G(u,v) \leq \lambda_G(u,v)$ [12] records that avoiding shared vertices is at least as demanding as avoiding shared edges and Figure 1.5 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex. The same inequality holds under the convention of Section 1.3.1 because every family counted by κ is edge-copy-disjoint.

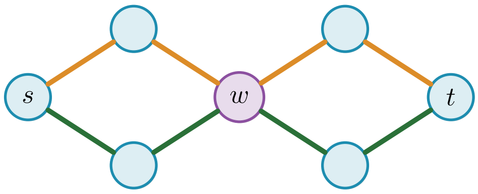


Figure 1.5. The separation axis: two edge-disjoint s–t routes both pass through w, so $\lambda(s,t) = 2$ but $\kappa(s,t) = 1$.

1.3.1 The incidence convention

Every model in this thesis measures vertex separation in the same place, the incidence graph. The incidence graph $I(\mathcal{H})$ has one node for each vertex and one node for each edge copy, and it joins an edge copy to each of the vertices it contains. A route from u to v is a path of $I(\mathcal{H})$, and two routes are separated when they share no interior node. Interior nodes come in both kinds, so separated routes share neither an intermediate vertex nor an edge copy. The second half of that is what the hypergraph needs, since two Berge routes can run through the same hyperedge without ever meeting at a vertex, and a hyperedge that may carry only one route at a time is the whole difficulty of the model. For a simple graph the definition changes nothing, because an edge lies on a route exactly when both of its endpoints do. For a multigraph it says that q parallel copies of uv are q separated routes, exactly as q copies of a hyperedge are, since each copy supplies its own interior edge-node and no interior original vertex. So the two multigraph vertex variants are questions of their own, written $K_m(n)$ and $K_m^{\mathrm{dir}}(n)$, rather than the simple problems under another name. They are finite for the same reason the edge variants are: m parallel copies are already m separated routes, so no multiplicity in a feasible network can exceed $m-1$.

With all this in mind, the thesis considers multihypergraphs and directed multihypergraphs for both edge- and vertex-disjoint routes. Ordinary graphs ($r = 2$) and simple hypergraphs (no repeated hyperedges) are special cases. The final status map appears in Figure 3.3. The notation for all their extremal functions is summarised in Table 1.1.

Table 1.1. Notation for the extremal functions. The third column uses the incidence convention defined above, in every model.

Model	edge-disjoint routes	vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

The results below do not all play the same role. The classical backbone consists of Menger's theorem, the Gomory–Hu tree, and Mader's edge bound. It also includes the vertex-connectivity results of Bártfai, Bollobás, Leonard, Sørensen, and Thomassen. The main results developed in this thesis concern three different frontiers. They are the exact directed multigraph value, the hypergraph vertex problem at $m = 2$ and $m = 3$, and quadratic bounds for directed graphs and hypergraphs. The full sixteen-variant framework shows how those results fit together. Some remaining variants are included to complete that map rather than because they carry equal conceptual weight.

1.4 Proven bounds

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

Appendix A gives the full proof which also uses the Gomory–Hu tree. The lower bound is attained by a spanning tree with every edge thickened to $m-1$ parallel copies.

In simple graphs the edge and vertex problems give the same answer for $m \leq 4$, and three papers together say so. The vertex values came first: k_3 from Bártfai [5] and k_4 from Bollobás [6]. Leonard [7] then proved that the two separations agree throughout $2 \leq m \leq 4$, in the same paper that settled the edge value at $m = 5$, a year before Theorem 1.2 covered every m at once.

Theorem 1.4 (Bártfai [5], Bollobás [6], Leonard [7])**.** *For all $n \geq m$ and $m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

By Whitney's inequality every edge-feasible graph is vertex-feasible, so $k_m(n) \geq \ell_m(n)$ (Figure 1.6). One is tempted to expect the equality forever. This, however, is not the case, and Sørensen and Thomassen proved that the vertex problem diverges from the edge problem at $m = 5$.

Theorem 1.5 (Sørensen–Thomassen [8])**.** *For simple graphs,*

$$k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n) \quad (6 \leq n \leq 13), \qquad k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4 \quad (n \geq 14).$$

The two separations therefore part company at $n = 14$, and $k_5(n) > \ell_5(n)$ for every $n \geq 14$.

Sørensen and Thomassen state the second formula for every $n \geq 6$ apart from $n = 7$ and $n = 12$, and supply those two values separately. On the small range the two formulas agree except at exactly those points, where the second reads one too few at $n = 7$ and one too many at $n = 12$, so splitting the statement by range removes both exceptions. Remark A.12 records the counting convention that turns their numbers into these.

Theorem 1.6 (Directed arc value at $m = 2$)**.** *For simple digraphs,*

$$\ell_2^{\mathrm{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

Two constructions give the lower bound, one on each branch, and Figure 1.7 draws both. The first is a hub joined to every other vertex in both directions, which reaches $2(n-1)$ arcs.

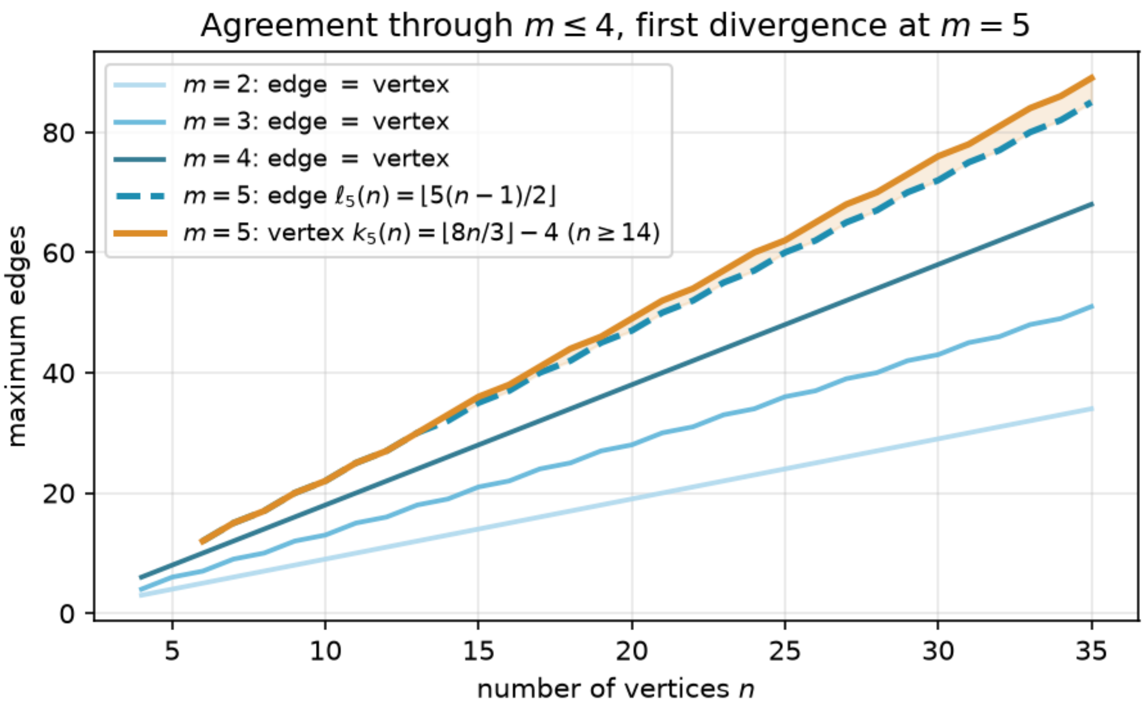


Figure 1.6. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). The vertex function diverges from the edge function at $n = 14$ and grows faster thereafter.

That branch has no unique extremiser, since any tree on n vertices with every edge taken in both directions reaches $2(n-1)$ arcs too, the tree offering each pair only its one path in each direction, and the bidirected star is the case of it drawn here. The second sends every arc from one half of the vertices to the other and none back, which reaches $\lfloor n^2/4 \rfloor$. Section A.3 defines both and checks that neither carries a second route between any pair. The upper bound is proved in Section A.5 and takes a few lines. Every digraph has a thin *reachability skeleton*, a subdigraph carrying exactly the same who-reaches-whom relation, and Section A.4 shows one can always be found with at most $M(n)$ arcs. At $m = 2$ a feasible digraph can spare none of its arcs to that thinning, since an arc whose endpoints are still joined by a longer route inside the skeleton would give that pair two independent routes. So the digraph is its own skeleton and inherits the skeleton's count. The argument never mentions n, so it needs no base cases, and it never mentions the separation, so it is run for κ, the larger feasible family, and the arc value follows from it.

Theorem 1.7 (Directed vertex value at $m = 2$)**.** *For simple digraphs, $k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$.*

The hub leads for $n < 7$ and the bipartite wall leads for $n > 7$. Notice how the directed value is quadratic in n for large n, while the undirected value is linear. This is because digraphs can be lopsided in one direction without creating routes back.

Proposition 1.8 (Hypergraph edge bound)**.** *For $n \geq r \geq 2$ and $m \geq 2$, an r-uniform multihypergraph on n vertices with $\lambda^{\max} \leq m - 1$ has size at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. There are simple*

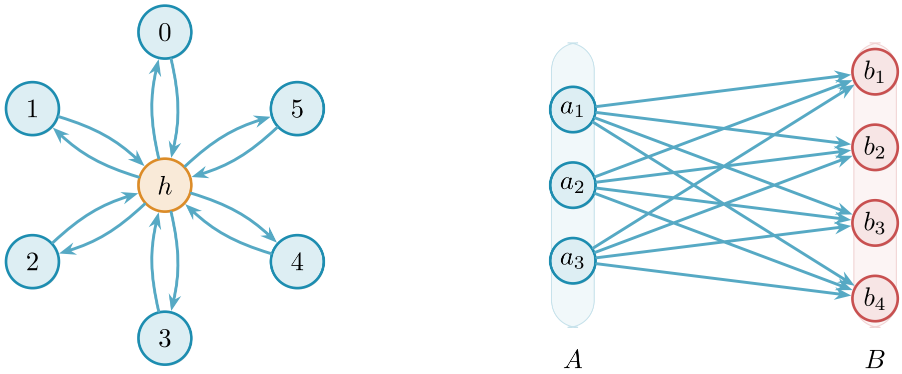


Figure 1.7. The two branches tie at $n = 7$, where each holds 12 arcs at $\lambda^{\max} = 1$. Left: the bidirected star, a hub joined to six spokes both ways, $2(n-1) = 12$. Right: the one-directional complete bipartite digraph $B_{3,4}$, every arc from A to B, $\lfloor n^2/4 \rfloor = 12$.

hypergraphs attaining the bound when $m - 1 \le \binom{n-2}{r-2}$, and multihypergraphs attaining it when $(r-1) \mid n-1$.

The proof is an elementary induction on a hyperedge cut, and Appendix A gives it. It has to be re-run rather than deduced from Theorem 1.3, because the obvious reduction points the wrong way. Replacing every hyperedge by a spanning star on its r vertices gives a multigraph of size $(r-1)|E(\mathcal{H})|$, but two routes may then take two different star edges of one hyperedge, which a family of Berge routes forbids. The star multigraph can therefore carry more connectivity than the hypergraph it came from, and a feasible hypergraph need not produce a feasible multigraph to count. The star hypertree of Figure 1.8 attains the $m = 2$ case: a hub joined to disjoint blocks of $r-1$ vertices, Berge connectivity one between vertices in its nontrivial component, $\lfloor (n-1)/(r-1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. For $r \ge 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge, whenever $m - 1 \le \binom{n-2}{r-2}$ (Theorem A.48). That is a real difference between the two edge sizes. For ordinary graphs simplicity is what separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$. One edge size higher the two are equal, but only while $m - 1 \le \binom{n-2}{r-2}$. That condition says there are enough distinct hyperedges to go around. When it fails the two may part company. At $n = r = m = 3$ there is only one hyperedge to choose from, so a simple hypergraph carries one where the multihypergraph carries two copies of it (Remark A.49).

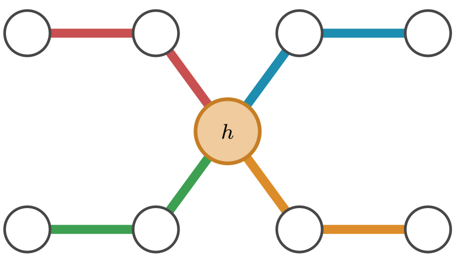


Figure 1.8. The $r = 3$, $n = 9$ star hypertree: four hyperedges meet only at the hub h, attaining $\lfloor (n-1)/(r-1) \rfloor = 4$ with Berge connectivity one.

Theorem 1.9 (Hypergraph vertex value at $m = 2$)**.** *For $r \ge 2$ and $n \ge 1$, the maximum number* *of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 1$ is*

$$K_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree of Figure 1.8, which is simple. In particular $K_2^{(r)}(n) = k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Theorem 1.10 (Hypergraph vertex value at $m = 3$)**.** *For $r \ge 2$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $2 < r < n$, which is exactly the range where $2 \le \binom{n-2}{r-2}$, the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

There are multihypergraphs attaining it when $(r-1) \mid (n-1)$ instead, by repeating one hyperedge rather than needing $\binom{n-2}{r-2}$ distinct ones, so $K_3^{(r)}(n)$ takes the same value on that range (Proposition A.53).

Both proofs read Berge routes as ordinary paths in the incidence graph, and Appendix A gives them. The $m = 3$ case is the harder one and needs a rank bound proved by induction over blocks and separating pairs. At $m \ge 4$ that route stops for want of a 4-connectivity analogue, and the value is open. Proposition 1.8 counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away and it must do this for every line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds and it is why the hypergraph rejoins the family only once the checker is in place.

At $r = 2$ these two theorems speak about multigraphs. Theorem 1.9 gives $K_2(n) = n - 1$, where the cap leaves every multiplicity at one and the extremisers are the spanning trees, and Theorem 1.10 gives $K_3(n) = 2(n-1)$, attained by a spanning tree with every edge doubled (Corollary A.59). The first threshold at which the multigraph vertex value is known to differ from its edge counterpart is $m = 5$, where $K_5(4) = 14$ against $L_5(4) = 12$. Whether they agree at $m = 4$ is open. From $m = 5$ the value also parts company with the thickened tree, which a single grafted triangle already beats (Theorem A.60). Appendix A works the problem out. The measure splits into parallel copies plus longer detours (Lemma A.56), which turns the whole question into a knapsack over the blocks of the underlying simple graph (Theorem A.63). In the block-packing regime, complete bipartite blocks give an asymptotic rate of $(3 - 2\sqrt{2} + o_m(1))m^2$ per added vertex (Theorem A.64), while the underlying simple graph gives the upper bound

$K_m(n) < 2(m-1)^2 n$ (Proposition A.61).

Theorem 1.11 (The quadratic bound with a linear error, [AI])**.** *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3),$$

and, for $m \geq 3$, both the leading constant $\frac{1}{4}$ and the order *of the second-order term of the augmented bipartite count hold unconditionally. At $m = 2$, Theorem 1.6 instead gives $\ell_2^{\mathrm{dir}}(n) = n^2/4 + O(1)$ as $n \to \infty$.*

The proof is in Appendix A. Its engine is a count of two-step routes: in a feasible digraph a vertex cannot have too many out-neighbours that reach one another, since the short detours through them would breach the cap, and that alone forces the arc count down to the wall plus a linear correction.

The routes this count produces are internally vertex-disjoint, not merely arc-disjoint. That is what lets the whole argument be re-run under the stricter separation, and Whitney's inequality alone would not justify the transfer, since it points the other way.

Theorem 1.12 (Directed vertex problem, leading constant and order, [AI])**.** *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\kappa_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

and consequently, for each fixed m,

$$k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3).$$

Theorem 1.13 (Directed multigraph value, every n and m, [AI])**.** *For all $n \geq 2$ and $m \geq 2$,*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\,M(n), \qquad M(n) := \max\!\Big(2(n-1),\; \left\lfloor \frac{n^2}{4} \right\rfloor\Big).$$

The extremisers include the thickened bidirected spanning trees on the linear branch, every edge taken in both directions at multiplicity $m-1$, and the balanced one-way bipartite multigraph at the same multiplicity on the quadratic one. On the linear branch these are constructions exhibited rather than a classification, and Appendix A says explicitly that not every linear-branch extremiser has been classified. The proof deletes a *subgraph* at a time rather than a vertex. Inside any feasible multigraph there is a sparse skeleton that alone preserves which vertices reach which, and removing one copy of each of its arcs costs every reachable pair at least one route. Peeling off $m-1$ such skeletons therefore accounts for every arc, and the parity obstruction that stopped the vertex-deletion induction never arises. Equality also classifies the quadratic extremisers once $n \geq \max(8, 2m)$ (Theorem A.45), and only the smaller range remains open.

Theorem 1.14 (The directed hypergraph constant, [AI])**.** *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)^2(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$, which is the stronger hypothesis by $\kappa \leq \lambda$. Hence for fixed r and m both the edge- and the vertex-disjoint multihypergraph maxima are $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$, and the bipartite construction of Proposition A.69 is asymptotically optimal for both. For simple directed hypergraphs the same conclusion holds when $r \geq 3$. The case $r = 2$ is the simple digraph problem, with leading term $n^2/4$.

The proof, in Appendix A, forms the one-step shadow digraph whose arc $u \to v$ records that some hyperedge has tail u and head v. A matching argument transfers the hypergraph route cap to the shadow's two-step route budget, after which Lemma A.34 controls the shadow and a tail–head incidence count converts its arcs back to hyperedges. Since the bipartite construction already attains that leading term, it is asymptotically optimal.

The same specialisation to $r = 2$ carries the directed arc results across to the directed multigraph vertex problem. At $m = 2$ the cap leaves every multiplicity at one, so $K_2^{\mathrm{dir}}(n) = k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$. For fixed m the arc extremiser of Theorem 1.13 is feasible here as well, since $\kappa \leq \lambda$, and the theorem above at $r = 2$ is the matching upper bound, so $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ (Corollary A.66).

For the simple directed arc problem, a matching upper bound for the constructions of Construction 3.1 remains open here when $m \geq 3$. Nevertheless, Theorem 1.11 proves that the value is $n^2/4 + \Theta_m(n)$. Determining the linear contribution, and then the bounded correction if such an expansion holds, would require sharper bounds. That coefficient is not out of reach either: a feasible digraph has at most $m-1$ two-step routes between any ordered pair, so it contains no m such routes with common endpoints, and the Turán theorem of Huang and Lyu for that family replaces the coefficient $4(m-1)$ above by $m-1$ once n is large in terms of m [13].

Chapter 2

Certifying and Discovering Bounds by Machine

The previous chapter gave insight into the problem and some variants, but left most of the harder questions open. This chapter will serve as a sanity check for the given theorems by testing them on small graphs through brute-force enumeration. It will also be the foundation of conjectures for the open cases, by discovering extremal graphs or at least sufficiently dense graphs. Pattern recognition on the discovered values, or even a family of extremal graphs, then lets conjectures be made, although those statements are naive because they are based on small graphs only. Larger graph spaces will be scoured by a heuristic, namely tabu search. A guided search is needed there because only a small fraction of the search space is extremal: assigning edges at random gives a binomially distributed edge count, so almost every graph lands far from the cap. The archive is the repository at `https://github.com/louisvandenbruwaene/thesis` and every file path quoted in this thesis is relative to it, or to a base directory named where the path is given, with the commands and the audit trail collected in Section A.13. Below is a diagram of the program's architecture, showing how the modelled graphs are fed to the checker and how the results are used to either certify or discover bounds.

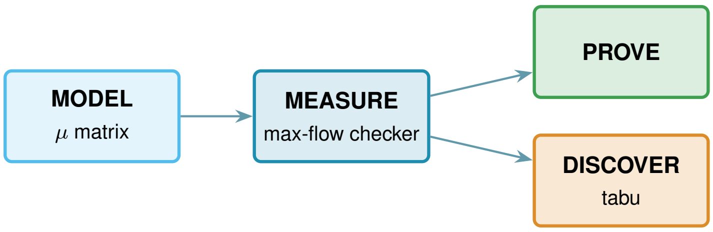


Figure 2.1. A single model feeds the exact max-flow checker. Certification turns that measurement into finite upper bounds, while heuristic search uses it to discover lower bounds. The badge colours on the code cards match the boxes here.

2.1 Modelling

All non-hypergraphs are represented using one multiplicity matrix μ: it is symmetric when undirected and binary when simple. Hypergraphs are stored as (multi)sets of hyperedges, which are sets of vertices with a tail distinguished in the directed model. Figure 2.5 shows this encoding for a directed multigraph and hypergraph. Matrix or tensor representations for hypergraphs are thus not recommended due to their size and sparsity. All possible graphs of a given size will later be checked to see if they are extremal for the given size and connectivity restraint. Matrices are iterated as a depth-first search with entries in $\{0, 1\}$ for simple graphs and in $\{0, \ldots, m-1\}$ for multigraphs, and the diagonal is always zero. Hypergraphs are implemented similarly from the ground up as there are no matrix tools for them. Notice how depth-first search is memory-efficient.

2.2 Measuring

Given n and m, walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ or $\kappa^{\max}$ of each with a max-flow computation and keep the densest graph that stays at or below the cap $m-1$. If the enumeration cannot finish within the time limit, the best graph so far is returned. It is therefore important to distinguish between the densest graph possible and the densest graph found. A completed enumeration gives an exact extremal value. An unfinished enumeration gives only a lower bound. In the vertex-disjoint and hypergraph cases the graph will be transformed before the flow can be measured. This ensures that every variant shares the same integer max-flow engine no matter the implementation. Thus the whole question reduces to a flow computation: A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger's theorem (Theorem A.1) already turned $\lambda(u, v)$ into the number of edges/arcs in a minimum cut. For a simple graph, $\lambda(u, v)$ is therefore also the capacity of a cut over a network with unit capacities. For a multigraph, the parallel copies can be combined into one link whose capacity is their multiplicity $\mu(u, v)$. In the directed case these capacities follow the direction of the arcs, while in the undirected case they are available in either direction. The max-flow/min-cut theorem of Ford and Fulkerson [14] says that this smallest cut has exactly the value of a maximum flow. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity.

The routine that runs this flow and reports the route count is the connectivity *checker*. The program ships an optional compiled C helper that runs this same flow more efficiently.

2.2.1 Transformations

This max-flow computation is built for directed multigraphs (including simple digraphs) with edge-disjoint paths. That is why the other variants will have to be transformed into such a flow network. Undirected multigraphs have a natural identification with a directed multigraph, obtained by replacing every edge with a pair of arcs going in opposite directions. To transform a multihypergraph, replace each distinct hyperedge with a helper gate, an entry point and an exit point joined by a single arc whose capacity is the multiplicity of the hyperedge. Every member of the hyperedge then flows into the entry and out of the exit at unlimited capacity, so the one arc between them is the only bottleneck (Figure 2.3b). A directed hyperedge is wired the same way, except that only its tail reaches the entry and only its heads leave the exit, which is what forces a route to cross it from tail to head. The directed hypergraph's flow network is just a restricted version of the undirected one. Another representation for multihypergraphs is a separate gate with capacity one for each hyperedge.

The hypergraph Menger theorem (Theorem A.46) proves that the resulting maximum flow is exactly the Berge connectivity. Figure 2.2 shows four gate-disjoint routes in a whole hypergraph. For the vertex-disjoint hypergraph checker, only the ordinary vertices are split into unit-capacity gates. Each hyperedge retains its multiplicity-capacity gate.

Lastly, all vertex-disjoint versions can then be transformed into an identical edge-disjoint question by splitting each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3a). Every arc $u \to v$ of the digraph becomes $u^{\text{out}} \to v^{\text{in}}$ in the split network. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u-v paths in the original graph. For a multigraph the arc carrying an adjacency keeps its full capacity $\mu(u, v)$, and only the internal arc of a vertex is a unit. That is the incidence convention of Section 1.3.1 written as a network: parallel copies stay separate routes while the vertex they run through does not, so the split network measures $K_m(n)$ and $K_m^{\text{dir}}(n)$ rather than the value of the underlying simple graph. Notice how the undirected or hypergraph transformation is done beforehand and the gates aren't split by this step.

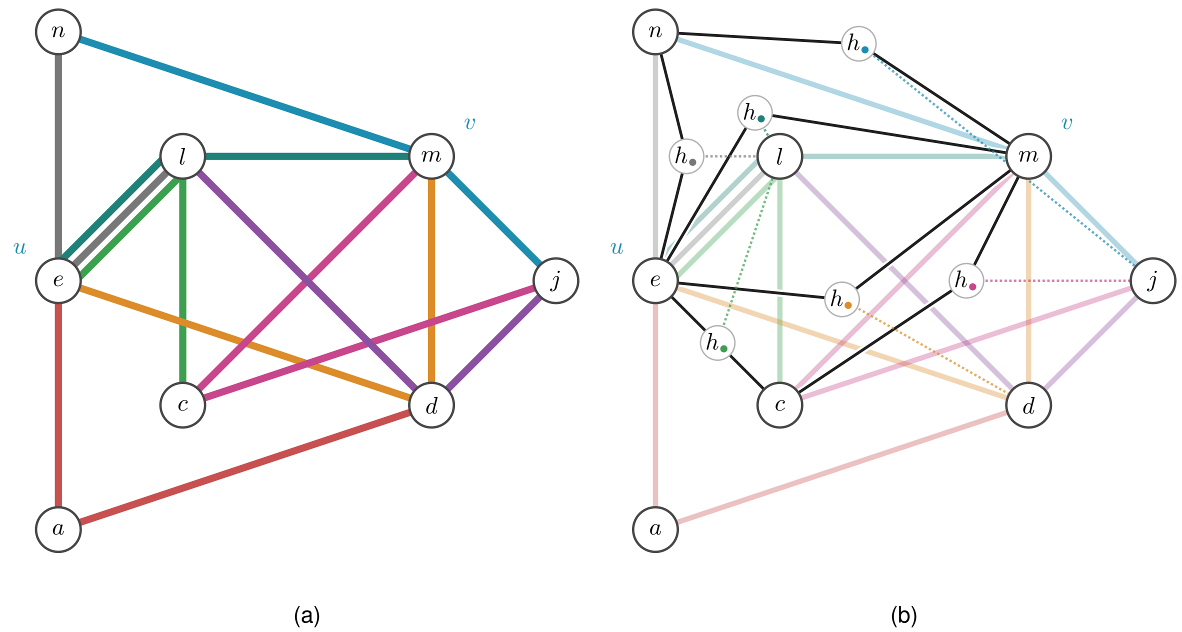
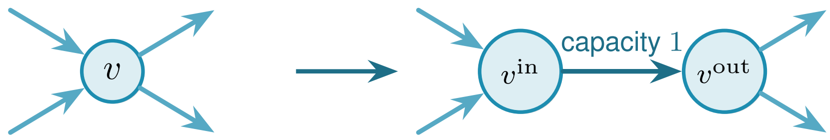


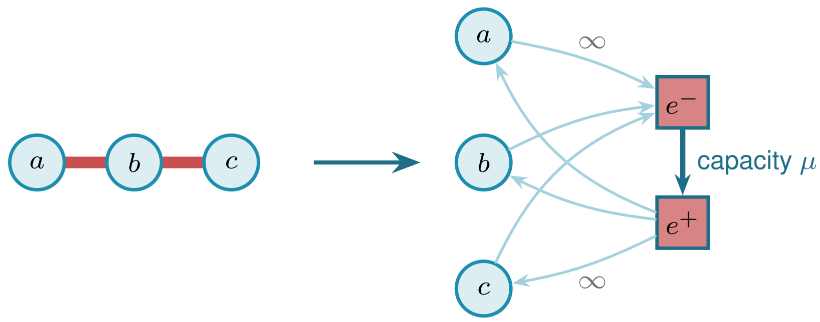
Figure 2.2. An undirected 3-uniform hypergraph and its gate network for u, v. The four black routes use distinct capacity-one gates (they are drawn as one point to reduce visual clutter), so the checker reads $\lambda(u, v) = 4$. A fifth would reuse a gate. The helper points connect to the hyperedge's third member by dotted lines.

2.3 Checking every graph of a fixed size

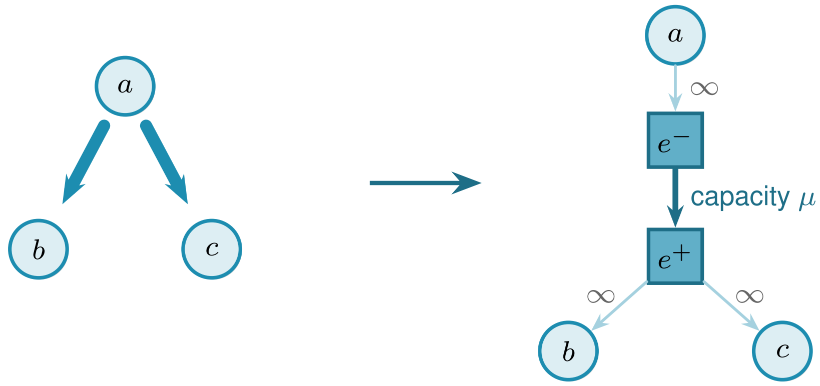
The connectivity checker works on all variants but measures only one graph at a time. To establish an exact value by enumeration, every feasible graph must either be checked or excluded by a sound pruning argument. This is one route to an upper bound. A mathematical theorem can establish it without enumeration. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal entry, a multigraph a choice in $\{0, \ldots, m-1\}$, and a digraph varies every



(a) Split each vertex v into v^{in} and v^{out} joined by a unit-capacity arc to transform vertex-disjointness into edge-disjointness. Only this internal arc is a unit: an adjacency keeps capacity $\mu(u, v)$ and a hyperedge keeps its gate.



(b) Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



(c) The directed hyperedge is transformed into a gate with one tail and two heads. This gate has capacity equal to the multiplicity of the hyperedge and every route must enter at the tail and leave at a head.

Figure 2.3. The building blocks of the connectivity checker: vertex splitting, hyperedge gates and directed hyperedge gates.

ordered pair rather than only the upper triangle. A hypergraph is not a matrix at all, so it is enumerated over its own list: a subset of all $\binom{n}{r}$ possible vertex sets of size r. In the directed model, the tail is specified with a pointer and the multihypergraphs have multiplicities for each hyperedge. Figure 2.5 shows how a directed multigraph and hypergraph are represented. Using these representations for all graph variants means iteration traverses all possible representation states easily. Walking those possibilities, measuring each candidate with the checker and keeping the densest feasible graph settles a bound exactly. If the enumeration hits its deadline before finishing, the returned graph is labelled as a lower bound. Completed enumerations corroborate the small cases reported in the audit. There is one important exception to the driver's terminology: for directed multigraphs under arc separation, `solve(exhaustive=True)` returns the proved closed form and a named witness. That branch is a theorem evaluation, not an independent exhaustive check. It therefore supplies no enumeration markers in the figures. The standalone formula audit calls the enumerator directly when checking that theorem.

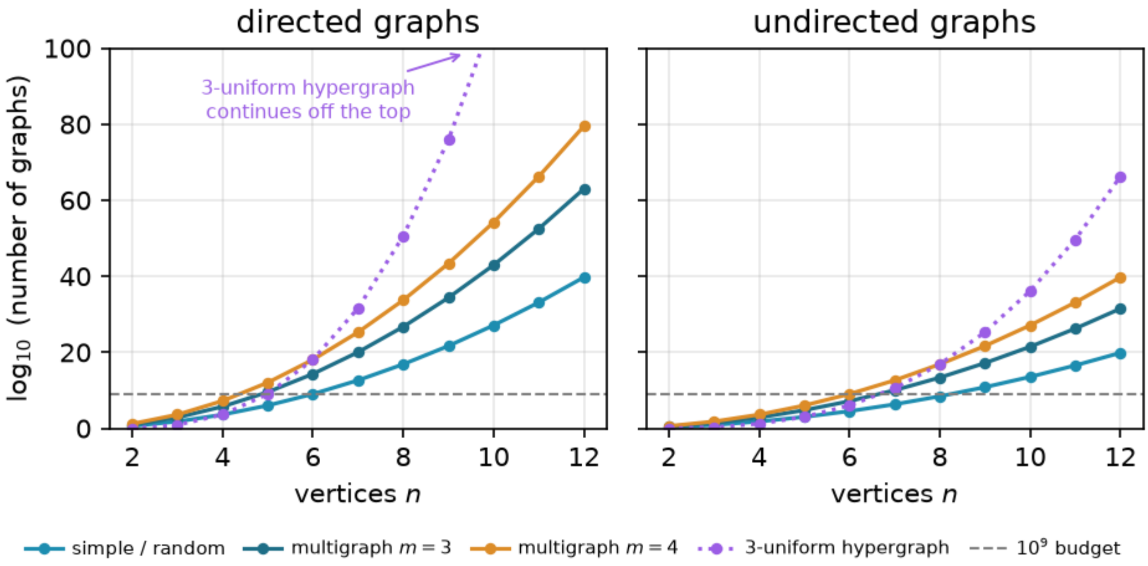


Figure 2.4. Candidate counts for blind enumeration on a logarithmic scale. Every model crosses the generous 10^9 budget in single-digit order, and directed objects grow fastest because they have more independent entries to decide.

The vertex count dominates every other factor. The candidate count has the form (values per entry)$^{\text{(number of entries)}}$. For graphs, the number of entries grows quadratically with n. For fixed-rank r-uniform hypergraphs, it grows as $\binom{n}{r} = \Theta(n^r)$. The $r = 3$ experiments plotted here therefore have cubic growth in the exponent.

Raising the connectivity budget enlarges the number of values available in each entry. A simple graph has two states per entry, while a multigraph has more. Adding direction instead increases the number of entries. It decides both directions of a pair independently, turning two undirected states into four directed states. Which change costs more depends on m and on the model. Directed hypergraphs are especially numerous because each hyperedge also requires a choice of tail and heads. These variants are particularly costly to enumerate, even where a theorem already determines their leading term. The separation, by contrast, does not change the enumeration: edge and vertex disjointness start from the same candidate space, but use different feasibility tests. Their pruning and running times can therefore differ.

The number of graphs the checker must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m - 1$, so that each run ranges over the m values $\{0, \dots, m-1\}$, raises the base from two to m, giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes and it does so only for the smallest sizes. Figure 2.4 measures exactly how fast the search space grows.

2.3.1 Pruned search

The enumeration step will skip whole families of hopeless graphs, via a pruned search. The directed $m = 2$ values are checked independently at every order through $n = 7$, under both

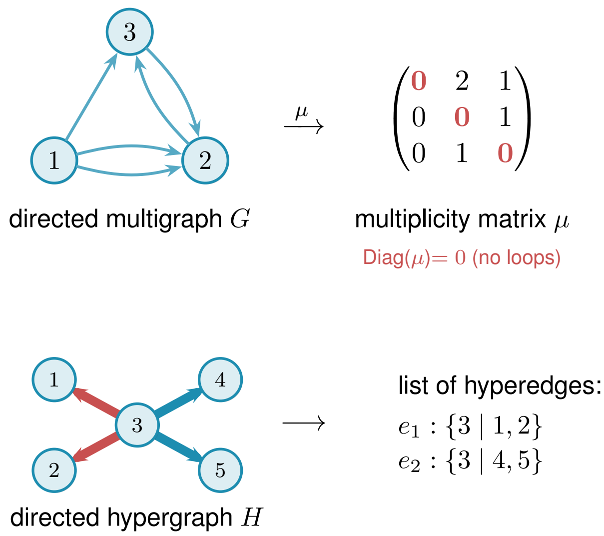


Figure 2.5. The two encodings for directed variants. The multiplicity matrix μ represents a digraph, with μ_{ij} arcs from i to j. A list of hyperedges represents a directed hypergraph and records each tail separately. A multihypergraph repeats entries in that list. A tensor representation would require n^r entries, almost all zero, and is therefore avoided.

separations (Table A.11). Theorems 1.6 and 1.7 are proved at every n at once and use none of those checks, so the sweep corroborates them rather than supporting them. Those orders lie just beyond the range where blind digraph enumeration is practical. They can be reached by enumerating more cleverly. Every variant offers a fixed list of edges its graphs are allowed to use, whether those are pairs, ordered pairs, or oriented sets of r vertices, and a graph is a choice of how many copies each of them gets. Rather than writing out whole choices one after another, fix an order on that list and decide the edges one at a time, leaving the undecided ones absent. Each partly built graph then stands for the whole family of graphs agreeing with it so far, and a pruning rule discards that entire family if it is deemed to bear no fruit.

Adding an edge never lowers connectivity because every existing family of disjoint routes survives (Proposition A.4). Every completion of a partly built graph only adds edges. The partial graph is therefore a lower bound on the connectivity of every completion. If it already breaks the cap, every completion does too. The entire branch can then be abandoned.

A second pruning method starts with the edges already placed. It then adds the largest possible contribution from the undecided edges: their number times the largest allowed multiplicity. The resulting sum is an upper bound on every completion. If this upper bound cannot beat the best feasible graph found so far, the branch is discarded without further exploration (Figure 2.6). Only infeasible graphs and graphs that cannot beat the incumbent are ever discarded, so a run that finishes returns exactly the maximum the basic walk would have returned.

2.3.2 Generating search

Iterating over all possible graphs and measuring each one's connectivity repeats many isomorphic cases. Another method, suggested by Prof. Jan Goedgebeur, instead generates all non-isomorphic graphs. It uses the `geng` program from the `nauty` toolkit of McKay and Piperno [15]

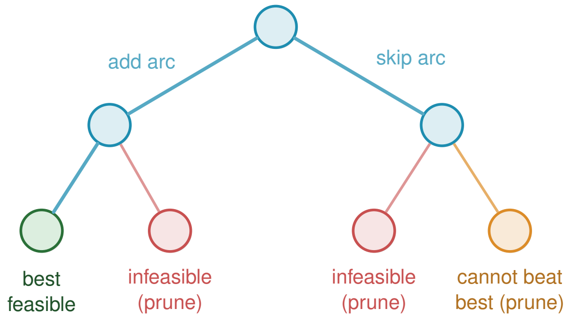


Figure 2.6. Each level of the partially built graph fixes one adjacency. Infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded.

to perform the difficult task of generating one representative of each isomorphism class. The program is called as a subprocess, and its `graph6` output is decoded. For a given n, `geng` produces all non-isomorphic *undirected* graphs. The program then treats each such graph as a support graph and assigns directed arc multiplicities to its edges. Consequently, the search automatically inherits the support graph's isomorphism reduction. For each support edge $\{u, v\}$, considered in a fixed order, it assigns the ordered pair of multiplicities $\mu(u, v)$ and $\mu(v, u)$. Each lies between 0 and $m - 1$, and the two cannot both be zero because every support edge must represent at least one arc. After each assignment, the checker evaluates the partially decorated multigraph. If any pair already exceeds the allowed connectivity, that branch is stopped immediately: adding more arcs cannot decrease connectivity. This is the same monotonicity-based pruning used in Figure 2.6. Decorations that reach the target arc count are retained and converted to canonical form, which ensures that each isomorphism class is reported only once. Because different support graphs produce disjoint search spaces, they can be processed independently on separate processor cores. The related orientation tools `directg` and `watercluster2` are not used. Both allow edges in both directions, but they do not assign the repeated arc multiplicities used here [16]. The custom decoration step also applies the connectivity pruning after each assignment.

Table 2.1 records historical timings for three different single-core tasks. The distinction matters: they are not three equivalent ways of discovering an unknown optimum. The blind sweep maximises the arc count over all matrix assignments. The pruned depth-first search also maximises it, but starts with a supplied feasible construction as an incumbent. The `geng` pipeline is given the already known extremal arc count and lists the feasible isomorphism classes at that target. Finding representatives there does not rule out a larger feasible count, so this last run does not prove the optimum independently.

The blind sweep is omitted after $n = 4$, where $2^{n(n-1)}$ assignments already take 6.3 s. Each additional vertex multiplies the assignment count by 2^{2n}. The estimated runtime is about an hour at $n = 5$ and several weeks at $n = 6$. This is the growth shown in Figure 2.4.

Below $n = 5$, fixed overhead dominates the two remaining methods. The main costs there are Python function calls and subprocess start-up. Those timings do not give a meaningful comparison of search performance. At $n = 7$, the recorded pruned maximisation takes 1156.6 s and the supplied-target classification takes 17.0 s. Their different outputs and prior information prevent interpreting this ratio as a speed-up for proving an unknown optimum. The script gives the pruned search a 1800-second stopping target. The blind and generated routines have no common time limit. These observations illustrate the cost of the respective tasks, not an equal-budget performance comparison.

n	supplied target	blind (s)	pruned (s)	classification (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	–	0.22	0.039
6	10	–	14.5	0.73
7	12	–	1156.6	17.0

Table 2.1. Historical timings for different tasks: blind maximisation, construction-seeded pruned maximisation and geng-based classification at a supplied target. Only the first two independently bound the optimum from above. A dash means not attempted. These are separate runs from Table A.11, whose recorded times differ. The table is not evidence of a like-for-like speed-up.

2.4 Proving every m at once

The search so far treats one pair (n, m) at a time. For a directed multigraph, scaling can instead check every threshold at a fixed n in one pass. It divides all multiplicities by $m-1$. The general theorem of Theorem 1.13 has now replaced this route in `solve`, but the older formulation is still a useful independent finite check.

Divide each multiplicity of a feasible directed multigraph D by $m-1$. The resulting weights satisfy $w(u,v) \in [0,1]$. Every pairwise maximum flow is at most one. Their total weight is the arc count of D divided by $m-1$ (Figure 2.7). Let

$$M^*(n) = \max\Big\{\textstyle\sum_{u\neq v} w(u,v) : w(u,v) \in [0,1],\ \operatorname{maxflow}(s,t;w) \leq 1 \text{ for all } s \neq t\Big\}$$

denote the largest total weight of such a matrix. Scaling then gives

$$L_m^{\mathrm{dir}}(n) \;\leq\; (m-1)\, M^*(n) \qquad \text{for every } m \geq 2.$$

The reverse inequality needs a $\{0,1\}$ matrix of weight $M^*(n)$. Multiplying such a matrix by $m-1$ preserves integer arc multiplicities.

On the linear branch, Construction A.38 supplies the double star. It has one hub joined bidirectionally to every other vertex and has $2(n-1)$ arcs. For $n \leq 6$, this is the larger term of $M(n)$ and realises $M^*(n)$. At the crossover $n = 7$, the quadratic branch catches the linear one (Theorem 1.6). It then overtakes it. The relevant construction becomes the one-directional bipartite digraph from Construction A.38. Thus no single shape witnesses $M^*(n)$ at every order.

What does generalise is integrality. Corollary A.41 proves that some $\{0,1\}$ matrix always attains $M^*(n)$. An explicit construction gives attainment. Clearing denominators from a rational optimum rules out a better fractional matrix. Scaling the integral maximiser by $m-1$ gives the lower bound $(m-1)M^*(n)$. The preceding scaling argument gives the matching upper bound. The finite optimisation is retained only as an independent check of the cases $n \leq 6$, and Section A.13 records what it ran.

$$L_m^{\mathrm{dir}}(n) = (m-1)\, M^*(n) \qquad \text{for every } m \geq 2.$$

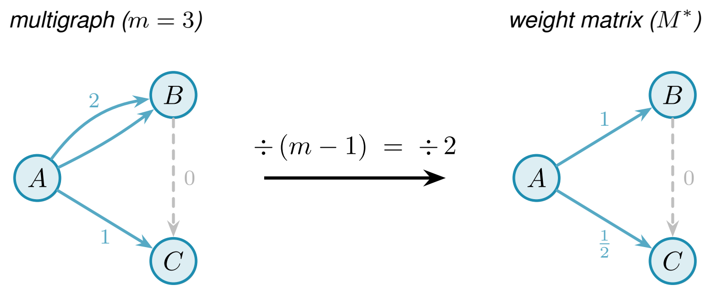


Figure 2.7. Scaling at $m = 3$: dividing multiplicities by $m - 1 = 2$ turns the integer multigraph into a weight matrix with entries in $[0, 1]$ and pairwise maximum flow at most one. Hence one m-free optimisation bounds every threshold.

2.5 The heuristic search

So far, iterating all graphs of a fixed size that meet the connectivity constraint and counting their edges was a blind walk. Apart from pruning, there is nothing that steers the walk towards better candidates because they all need to be checked anyway. From a certain point onwards, that will become impossible due to the massive number of possible graphs. The goal then becomes finding the densest possible graphs in such an efficient and guided way that they are likely to be (almost) extremal. The guide that pulls the walk towards a (local) maximum is called a heuristic. To avoid getting stuck in a local maximum, a metaheuristic allows worsening moves to discover new optima.

To implement the metaheuristic, give each graph one number: its energy

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max\bigl(0,\; c^{\max}(G) - (m-1)\bigr),$$

which rewards edges (arcs, hyperedges) and penalises any excess connectivity, where $c^{\max}(G)$ denotes $\lambda^{\max}(G)$ or $\kappa^{\max}(G)$, matching whichever separation the run measures. The search starts from the empty graph and moves by toggling one adjacency: an edge, an arc, or a hyperedge, depending on the model. The method used is *tabu search* [17], which escapes a local optimum by forbidding the pairs it has just touched. Among the moves not forbidden, the walk always takes the one that lowers Φ most, even when every candidate raises it. This is unlike a greedy walk that only ever accepts an improving move. After each move the pair it changed is put on a tabu list for a fixed number of steps, so the walk cannot immediately undo what it just did and cycle back into the optimum it came from. A move that would beat the best graph seen so far overrides the tabu flag regardless of its status, an *aspiration criterion* that keeps the tabu list from ever blocking genuine progress, and the walk perturbs itself when it stalls.

Simulated annealing is an alternative on the same energy and neighbourhood [18]. It accepts worsening moves with the probability given by the Metropolis rule [19]. That probability decreases as the run cools. At an equal wall-clock budget, annealing is the weaker engine here. For $n = 7$ and $m = 3$, tabu reaches the 24-arc directed multigraph extremiser, while annealing reaches 20 arcs. For the simple digraph, tabu reaches 18 arcs and annealing reaches 16. The reported discovery bounds use tabu search, supplemented by explicit constructions. Annealing is retained for the comparison above and as an independent cross-check in the program's self-test, where agreement between two unrelated walks is worth more than the speed of either.

Allowing the walk above the cap is a search strategy, not a requirement for reaching every feasible graph. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the cap, and reading that path backward builds the target up the same way. The feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is. We let the walk become infeasible because reachability and efficiency are different questions. A feasible graph can be locally maximal even when it is far from globally optimal. Escaping it may require exchanging one edge for another. Removing first and then adding stays feasible, but the removal temporarily worsens the objective. Reversing the order keeps the edge count high, but the addition may temporarily violate the connectivity cap. Allowing both orders gives the walk another short route across a local barrier. Infeasibility is a search device, not a target. A single walk depends on its starting point and random seed. The program therefore runs several independent restarts until the wall-clock budget expires. It keeps the densest witness found by any restart. The restarts share no state and can run on separate processor cores.

2.5.1 Measuring only on the possible bounds

Removing an edge, arc, or hyperedge can never raise $\lambda^{\max}$, while adding one raises it by at most one. The same holds for $\kappa^{\max}$. This relies on the monotonicity and unit-step bounds of Proposition A.4. Therefore the max-flow algorithm only needs to be run every once in a while, depending on how close to the constraint the graph sits. Most steps need no measurement at all. A removal from a feasible graph leaves it feasible. An addition made while the graph sits strictly below the cap lifts $\lambda^{\max}$ by at most one, so it lands at worst on the cap. Every other step calls the checker. There are two of those: an addition whose running bound sits at or above the cap, and a removal from a graph already above it. The search keeps that running bound and raises it by one for each addition it accepts. At a fixed step interval it measures the exact value again. Section A.12 gives the bookkeeping, and the regression test that holds the fast walk to the naive one step for step.

2.6 An equal search allocation across all variants

A separate experiment gives each of the sixteen variants one hour of search in total. This is one hour per variant, not per parameter case. Each variant is tested at $n = 6, 8, 10, 12$ and $m = 3, 6$. Each of these eight cases receives three runs with initial seeds 0, 1000000 and 2000000, each with a 150-second stopping target. Thus every variant receives $8 \cdot 3 \cdot 150 = 3600$ seconds of allocated search.

All runs start empty. No construction, known optimum or earlier incumbent is supplied to the search. Graph variants use tabu search and hypergraph variants use random greedy growth. Hypergraphs have rank 3 and directed hypergraphs use the forward convention. Runs are serial, with the variant order rotated between cases. Validation and reporting take place outside the search budget.

The original solver used a calendar clock for its stopping rule. Nineteen runs had a discrepancy between that clock and the independently measured monotonic duration. They are retained in the data but excluded from equal-budget statistics. Each flagged trial was repeated with the same frozen search code and seed, changing only its deadline clock to a monotonic one. The replacement takes its original slot regardless of which run found more edges. Selecting the better of the two would give those slots an unfair additional chance. The discarded attempts are counted separately as extra computation.

The timing rule is cooperative, so a connectivity calculation already in progress can overrun its target. Monotonic time also has platform-specific behaviour during system suspension. The records therefore include calendar duration, monotonic duration and CPU time. The computer was not isolated from other work, so this is a comparison under equal allocated search durations, not a controlled comparison of processor efficiency.

Table 2.2 compares the unaided outcomes with a separately supplied construction count C. Reaching C need not mean reaching the optimum. Failing to reach it measures a limitation of these particular runs, not a failure of the construction. The detailed tables in Section A.13.1 retain the minimum, median and maximum across the three seeds instead of replacing a search value by C.

Across the 128 parameter cases, the best run reaches or exceeds C in 74 cases and falls short in 54. All three seeds reach C in 68 cases. In 8 cases the search improves on the supplied construction, which confirms why C is a comparison value rather than an asserted optimum. The limitations are substantial in some directed models. For example, at $n = 12$ and $m = 6$ the directed multigraph arc search returns 110 in every run, while the supplied construction has 180 arcs. Thus the successful small validation cases below do not imply reliable recovery of the dense families at larger orders.

The selected runs used 57657.7 seconds of measured monotonic search time against the 57600-second total allocation. Including the discarded originals raises the saved search time to 58690.8 seconds. These totals exclude validation and reporting and are not calendar durations of the experiment.

Variant	Best reaches C	All seeds reach C
simple graph, undirected, edge	8/8	7/8
simple graph, undirected, vertex	8/8	8/8
simple graph, directed, arc	3/8	3/8
simple graph, directed, vertex	3/8	3/8
multigraph, undirected, edge	8/8	8/8
multigraph, undirected, vertex	4/8	4/8
multigraph, directed, arc	2/8	2/8
multigraph, directed, vertex	2/8	1/8
simple hypergraph, undirected, edge	5/8	4/8
simple hypergraph, undirected, vertex	5/8	4/8
simple hypergraph, directed, arc	4/8	4/8
simple hypergraph, directed, vertex	4/8	4/8
multihypergraph, undirected, edge	5/8	4/8
multihypergraph, undirected, vertex	5/8	4/8
multihypergraph, directed, arc	4/8	4/8
multihypergraph, directed, vertex	4/8	4/8

Table 2.2. Eight parameter cases per variant under a one-hour total search allocation. The first count records cases where the best of three seeds reaches or exceeds the supplied construction count C. The second requires all three seeds to do so. Neither column counts proved optima.

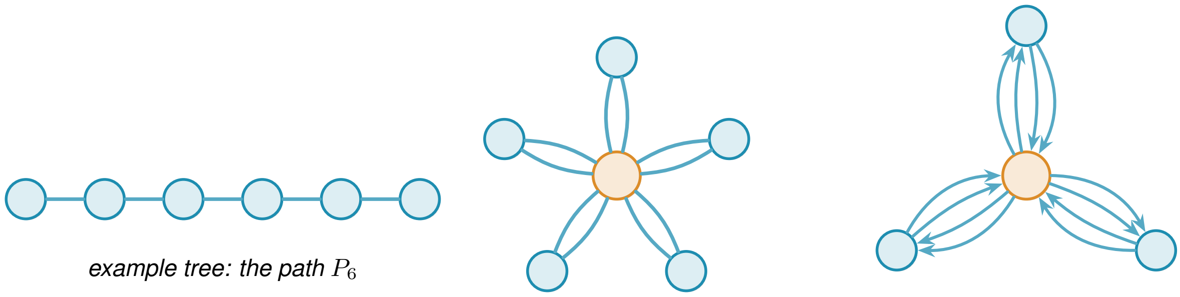
2.7 Rediscovering the known extremisers

Before addressing open cases, the search is tested on settled ones. Each run starts from the empty graph. The search is given the edge-count objective and the connectivity cap, and retains only feasible witnesses. Its intermediate states may be infeasible. It receives no description of the known extremal construction. In all seven validation cases in Table 2.3, it reaches the established value. This is a statement about those runs, not a claim that it succeeds at every size or throughout the sixteen-variant grids. Table 2.3 compares the densest witness with the theoretical value.

Variant	n	m	Limit (s)	Elapsed (s)	Search	Optimum
simple undirected	6	2	2	2.006	5	5
multi undirected	6	3	2	2.004	10	10
simple directed	4	2	2	2.000	6	6
simple directed	6	2	2	2.020	10	10
simple directed	7	2	2	2.022	12	12
multi directed	4	3	2	2.000	12	12
multi directed	5	3	2	2.011	16	16

Table 2.3. Seven short validation runs. Each uses unaided tabu search, initial seed zero and a two-second stopping target per case. Elapsed times include the search and may slightly exceed the target because deadline checks are cooperative. The search count and known optimum are separate columns. These are not one-hour benchmarks.

The recorded witnesses can be inspected, rather than inferring their shapes from their edge counts. They are saved with methods and elapsed times in `program/data/rediscovery.json`. `program/scripts/rediscovery.py` runs the experiment and `make_figures.py` renders the table without rerunning it. Each later restart uses the next integer seed. The supplied optimum is not passed to the search. All seven saved witnesses have star-shaped support, with the direction and multiplicity specified below. Figure 2.8 draws three examples. The undirected simple run at $m = 2$ settles on a tree. The example drawn is the path P_6. Every tree on n vertices has $n - 1$



example tree: the path P_6

$n = 6$, $\ell_2(n) = 5$

multigraph star

$n = 6$, $m = 3$, $L_3(n) = 10$

double star

$n = 4$, $m = 3$, $L_3^{\mathrm{dir}}(n) = 12$

Figure 2.8. Examples from the extremal families discussed alongside Table 2.3, drawn at the reported sizes. The path illustrates the tree family but is not the saved search witness. Figure 1.7 shows a bidirected star, which ties the one-directional wall at $n = 7$.

edges and local edge-connectivity at most one. The search therefore recovers an extremal family rather than a unique shape.

The undirected multigraph run settles on a star whose spokes have multiplicity $m - 1$. This is an instance of the multi-tree extremiser of Theorem 1.3. The directed multigraph run finds the same star with both directions present at that multiplicity. It is the thickened tree of Construction A.38.

The simple directed run finds a bidirected star at $n = 4$. At $n = 7$ it finds twelve arcs, where the hub and the one-directional wall tie (Figure 1.7). This is the crossover predicted by Theorem 1.6. The search carries none of these names. It finds the constructions solely by optimising under the connectivity cap.

The seven successful cases do not establish reliable performance past $n = 7$. At $n = 8$ and $m = 2$, a bidirected star has fourteen arcs, whereas a one-directional balanced wall has sixteen. Moving from a completed hub to the one-directional wall requires many coordinated arc changes (Figure 1.7). Single-arc moves can therefore make the star a difficult state to escape. This explains a possible search obstruction, not a proof that every run must stall there. The explicit construction and its connectivity check supply the sixteen-arc lower bound independently of whether an unaided run finds it.

The double star of Figure 2.8 provides another witness beyond the search range. At $n = 7$ and $m = 4$, it has 36 arcs and $\lambda^{\max} = 3$. Its count equals $2(n-1)(m-1)$. Neither witness alone proves optimality. Each is an exactly measured feasible graph and therefore a lower bound.

Chapter 3

Synthesis, Results, and Open Problems

Chapter 1 established the theoretical results and stated the remaining gaps. Chapter 2 tested small cases and searched larger ones for dense examples. This chapter brings those strands together. It identifies the main conclusions and separates the central open problems from questions of classification and completeness.

3.1 The directed frontier

The simple directed arc problem has several verified dense constructions, but their feasibility must not be confused with optimality. A construction proves a lower bound: it shows that at least that many arcs are possible. Proving the extremal value would additionally require an upper bound for every feasible digraph.

Construction 3.1 (Competing dense directed families)**.** For $n \geq m \geq 2$, choose the densest of the directed hub, the augmented bipartite digraph and the clique-core digraph (Constructions A.15, A.17 and A.27). Their respective arc counts are

$$H_m(n) = m(n-1), \qquad Q_m(n) = \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor, \qquad B_m(n) = \left\lfloor \frac{(n+m-3)^2}{4} \right\rfloor + 2(m-1).$$

These three letters name arc *counts* of the three directed constructions rather than the digraphs themselves. Two similar symbols elsewhere name graphs: $H_{m,n}$ is the undirected hub-and-spokes graph of Construction A.8, which carries $\lfloor m(n-1)/2 \rfloor$ edges and is distinct from the directed hub count $H_m(n)$ above, and $B_{p,q}$ is the one-directional complete bipartite digraph of Construction A.16. Each of the three directed constructions defining $H_m(n)$, $Q_m(n)$ and $B_m(n)$ has maximum local arc-connectivity $m-1$. Consequently,

$$\ell_m^{\mathrm{dir}}(n) \geq \max\{H_m(n), Q_m(n), B_m(n)\}.$$

This is a lower bound, not a proposed formula for the optimum.

The earlier conjecture asserted $\ell_m^{\mathrm{dir}}(n) = \max\{H_m(n), Q_m(n)\}$ for every $n \geq m$. It is false. The clique-core construction, supplied by Stijn Cambie, has 57 arcs at $m = 5$, $n = 12$, with maximum local arc-connectivity 4. The proposed formula gives only 56. More generally, this construction beats both earlier families whenever $m \geq 5$ and $m+7 \leq n \leq 3m-3$ (Construction A.27).

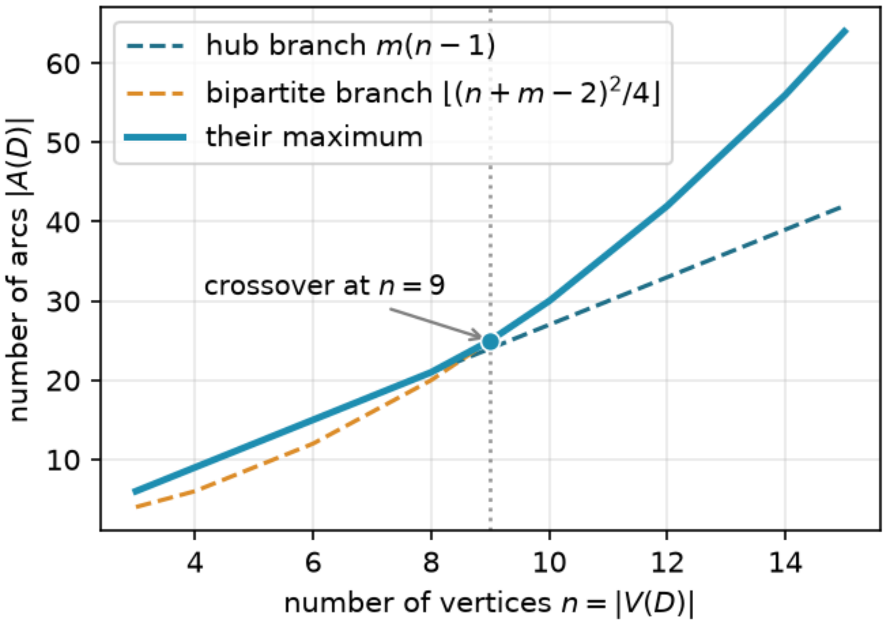


Figure 3.1. The hub and augmented bipartite lower bounds at $m = 3$, and their maximum. The plot asserts no upper bound.

It does not prove that 57 is optimal at $(12, 5)$, nor that the maximum of all three displayed counts is always optimal.

For fixed m, the augmented bipartite family eventually has more arcs than the other two displayed families. That comparison only ranks these constructions. Another feasible family could still be denser. Cambie also supplied an AI-assisted argument for eventual optimality. It is retained separately as a research note, not used as a theorem or adopted as a conjecture in this thesis. At $m = 2$, the exact value is already established by Theorem 1.6: the bipartite count is optimal for $n \geq 7$. Figure 3.1 compares the hub and augmented bipartite counts at $m = 3$.

At $m = 3$, exhaustive search proves the values $6, 9, 12, 15$ for $n = 3, 4, 5, 6$. The two separation conditions agree at every one of these orders (Remark A.25). The search does not settle $n = 7$.

The earlier decomposition conjecture has been removed as well. It would force the quadratic upper bound as soon as that branch exceeds the hub bound, which already happens at $m = 5$, $n = 12$. The counterexample therefore refutes that structural assertion too.

For fixed $m \geq 3$, the augmented bipartite count has the same leading term and the same order of error as the optimum. Theorem 1.11 proves this without assuming the shape of an extremal graph. The linear coefficient remains unknown among the results proved in this thesis. Even identifying it would leave the exact bounded correction to establish. More precisely, the construction gives a linear contribution $(m-2)n/2 + O_m(1)$ above $n^2/4$, while the proved upper bound allows $4(m-1)n$. These inequalities do not show that a unique limiting linear coefficient exists. Once n is large in terms of m, the Turán theorem of Huang and Lyu [13] lowers the upper end to $m - 1$ (Chapter 1). The next section considers the separate question of extremal shape.

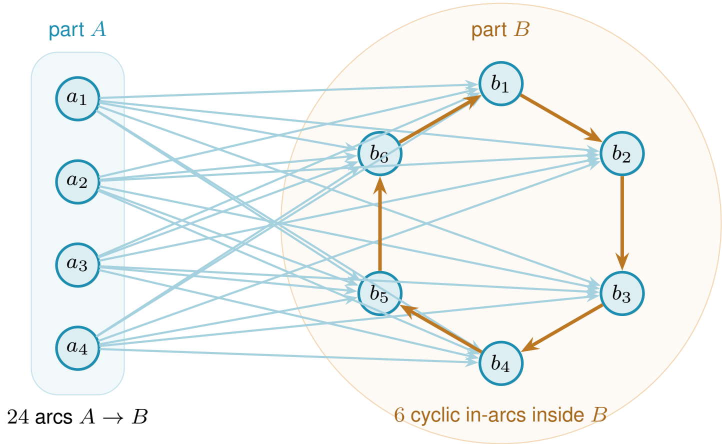


Figure 3.2. The augmented bipartite digraph of Construction A.17 at $m = 3$, $n = 10$. Every cross-part arc runs from A to B, and inside B each vertex takes one arc from its cyclic predecessor, which is $m-2$ arcs per vertex. A pair a_i, b_j then has the direct arc and one detour, so $\lambda^{\max} = 2$, and the count is $24 + 6 = 30$ against the hub branch's 27.

For fixed $m \geq 3$, the directed vertex problem has the same leading term and error order by Theorem 1.12. Appendix A treats it separately from the arc problem. It remains open whether the two values are exactly equal for $m \geq 3$.

Allowing parallel arcs closes the arc-disjoint problem completely. Theorem 1.13 gives the exact value for every $n \geq 2$ and $m \geq 2$.

3.2 Why direction makes the value quadratic

In an undirected graph, every edge contributes to the degree of both endpoints. Mader's theorem turns this symmetry into the linear cap $\lfloor m(n-1)/2 \rfloor$. A digraph has no corresponding symmetry.

In Construction A.16, all arcs go from one part to the other and none return. A directed route can therefore cross between the parts only once. The construction has $\lfloor n^2/4 \rfloor$ arcs at $\lambda^{\max} = 1$. No undirected graph of connectivity one can have quadratic size. The linear cap does not survive the introduction of direction. At $m = 2$ and $n = 8$, the one-directional construction has 16 arcs while the hub has 14. The directed value must therefore follow the larger branch.

Construction A.17 extends this idea to every m. Each vertex of the larger part receives $m-2$ extra in-arcs from within that part. Figure 3.2 draws the construction. The added arcs provide $m-2$ short detours for each relevant pair. They raise $\lambda^{\max}$ to $m-1$ and the arc count to $\lfloor (n+m-2)^2/4 \rfloor$. At $m = 3$ and $n = 10$, the construction has thirty arcs while the hub has twenty-seven.

The quadratic term is what the directed upper bound has to account for. For eventual exactness, the argument must explain why no configuration has more arcs than the augmented bipartite construction once n is sufficiently large in terms of m. The clique-core examples show why a restriction on n is needed. Proposition A.36 and Theorem 1.11 support this structure in two ways. In *size*, the extremal value lies within $O_m(n)$ of the wall's count. The remaining uncertainty includes the linear coefficient and the bounded correction. In *shape*, every feasible digraph

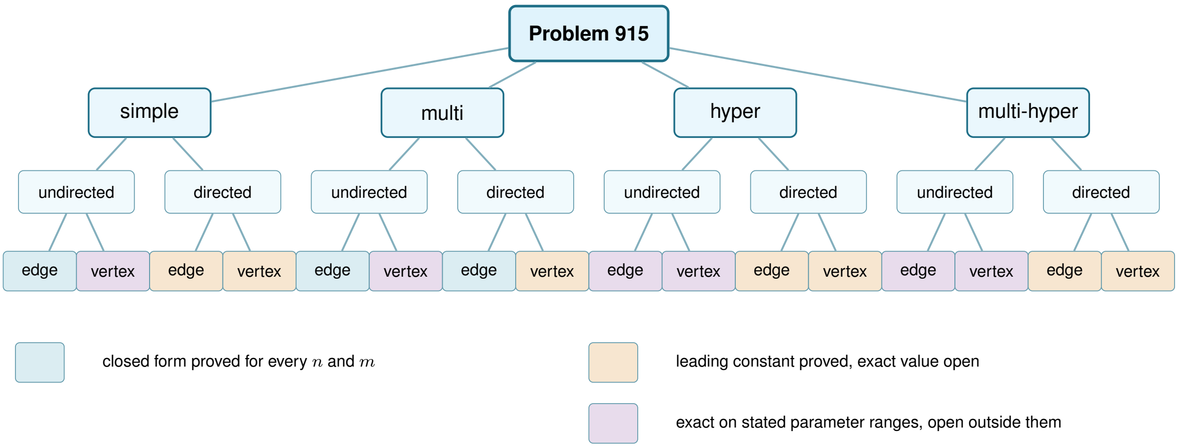


Figure 3.3. Status of the sixteen variants. Blue marks a closed form proved for every n and m. Purple marks an exact value proved on stated parameter ranges and open outside them. Amber marks a proved leading term with the general exact value unknown. For simple directed arcs, Construction 3.1 records lower bounds without asserting optimality. The model and direction rows carry no status. Table 3.1 is authoritative for exceptions and conventions.

becomes a subgraph of a one-directional bipartite graph after deleting $O(\sqrt{m}\, n^{3/2})$ arcs. This weaker error comes from an argument that does not induct on a low-degree vertex. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore the reason the directed and undirected problems differ in difficulty.

3.3 Summary of the sixteen variants

Figure 3.3 sorts all sixteen structural variants by how much is known, and Table 3.1 states the strongest result for each. The entries distinguish exact values, asymptotic bounds and constructions for cases whose exact value remains open. Each exact value is supported by a proof or by a completed exhaustive search. Construction entries provide lower bounds, not claims of optimality. Heuristic witnesses are never presented as upper bounds. The markings follow the standard set in Chapter 2. For results with an AI badge, the author's verification caveat in the Contribution Statement still applies. The summary does not upgrade their proof status.

Here $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. The hypergraph rows use $r \geq 3$. At $r = 2$, use the corresponding graph rows, in particular the different leading constant for simple digraphs when $m \geq 3$. The multigraph incidence estimate uses $m \to \infty$ with $n/m \to \infty$. Simple-graph formulas assume $n \geq m$. Below that range the complete object gives the trivial maximum. The $m = 5$ vertex row is quoted from $n \geq 6$ because that is the range [8] states, not because it fails below it: extending the small-range branch to $n = 5$ reads $\lfloor 5 \cdot 4/2 \rfloor = 10$, which is what K_5 gives at $\kappa^{\max} = 4$. The edge-hypergraph bound is attained when $(r-1) \mid (n-1)$ for multihypergraphs, and for simple hypergraphs when $r \geq 3$ and $m - 1 \leq \binom{n-2}{r-2}$. The $m = 3$ vertex-hypergraph bound is attained by a simple hypergraph when $2 < r < n$, and by a multihypergraph whenever $(r-1) \mid (n-1)$ (Proposition A.53). Full statements and exceptions are in Theorems 1.2, 1.4, 1.5, 1.9 to 1.14 and A.48 and Proposition 1.8.

Model	Edge or arc-disjoint	Vertex-disjoint
Undirected		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \leq 4$, and at $m = 5$ the edge value for $6 \leq n \leq 13$ then $\lfloor 8n/3 \rfloor - 4$ from $n = 14$, while it is open for $m \geq 6$
multigraph	$(m-1)(n-1)$, proved	$n-1$ at $m = 2$ and $2(n-1)$ at $m = 3$ (proved), while it is open for $m \geq 4$, where it is a knapsack over 2-connected blocks and has order $\Theta(m^2 n)$ in the joint regime stated below
r-hypergraph	$\leq \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\leq \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$, while it is open for $m \geq 4$
r-multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, with repeated hyperedges settling cases missed by the simple model at $m = 3$
Directed		
simple digraph	$M(n)$ at $m = 2$ (proved), and $n^2/4 + \Theta_m(n)$ for $m \geq 3$, with dense constructions but no general exact formula	$M(n)$ at $m = 2$ (proved), and $n^2/4 + \Theta_m(n)$ for $m \geq 3$ with equality to the arc value open
multigraph	$(m-1)M(n)$, proved	$M(n)$ at $m = 2$ (proved), and $(m-1)n^2/4 + O_m(n)$ for $m \geq 3$ with the exact value open
r-hypergraph	$(1+o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m, leading term proved	the same leading term for fixed r, m
r-multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Table 3.1. The sixteen variants and the strongest result known for each. The two multigraph vertex rows count multiplicity, since q parallel copies are q separated routes under the incidence convention of Section 1.3.1. Ranges of validity, including the one at $m = 5$, are stated below the table.

3.3.1 Detailed finite values

Table 3.1 and Figure 3.3 give the conclusion for every variant. The computational audit records the underlying finite values at $m = 3$ and $m = 6$. It also contains the four detailed grids (Section A.13.2). Keeping those repeated data in the appendix lets this chapter focus on the mathematical distinctions between the variants.

3.3.2 Orientations of a directed hyperedge

Theorem 1.14 pins the leading term in the directed hypergraph rows of Table 3.1. The table uses the *forward* convention: one tail and $r - 1$ heads. Reversing every hyperedge gives the equivalent one-head *backward* convention (Lemma A.71). Theorem A.74 gives the same leading term for the general model, which allows all nonempty tail–head splits. It covers both separation conditions. Orientation therefore creates no new asymptotic row in Table 3.1. The conventions may still differ at small orders. Whether they agree exactly once $n > r$ remains open.

3.4 Contributions and limitations of the program

The program replaces separate implementations with one shared interface. It represents graph models by a multiplicity matrix. It represents undirected hypergraph models by collections of vertex sets and directed ones by tail and head sets. An exact checker measures connectivity. A certifier proves upper bounds for small cases. Tabu search for graphs and random greedy search for hypergraphs find examples and hence lower bounds. The seven validation runs in Section 2.7 recover their known answers.

The three tools provide different levels of evidence. The checker measures one supplied graph exactly. A completed exhaustive search proves the best value at one finite order. The certifier is also a finite check. Without a replayable bound, it is not a reusable proof. Heuristic search supplies lower bounds only. A value is settled only when its lower and upper bounds agree. Directed vertex values require separate checks because Whitney's inequality transfers witnesses but not upper bounds. The repository contains the code and commands for rerunning the computational workflows. This does not guarantee identical outcomes for time-limited searches. Some historical results also lack saved witnesses or measured runtimes, as recorded in the accompanying data. The principal checks run from `program/erdos915_unified.py`. Several individual claims are backed by standalone scripts in `program/scripts/` instead. Each is named where its result appears. Some implement separate checks, while experiment drivers reuse the shared solver. Section A.13 gives the environment, commands and audit evidence. No theorem here depends only on an unchecked solver result.

The program also has clear limits. The heuristic search cannot prove upper bounds. The certifier proves an upper bound only at one fixed order. Without a replayable certificate, that solver check does not become a reusable argument (Section 2.3). For directed multigraphs, the recorded successful solver checks reach only $n = 6$. At larger orders, the hand proof in Appendix A establishes the value. The certifier did not finish at $n = 7$.

Computation also leaves the structural gaps untouched. Small-order agreement did not prevent the old directed arc formula from failing. Even when one displayed construction eventually beats the others, its optimality among all feasible digraphs remains open here. The directed vertex problem for $m \geq 3$ does not follow from the arc case. The undirected vertex problem for $m \geq 6$ has been open for half a century. The program can explore these problems, but it cannot settle them alone.

3.5 Open problems

Table 3.2 separates the proved results from the remaining frontiers. Each row gives the strongest result established here. It then states the next mathematical target.

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem 1.11)	Determine when, if ever, the constructions of Construction 3.1 attain the optimum.
Directed simple vertex, $m \geq 3$	$k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$ (Theorem 1.12)	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Directed multigraph incidence, $m \geq 3$	$K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ (Corollary A.66)	Sharpen the next-order term and decide whether the arc extremiser is optimal here too.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$ (Theorem A.13)	Determine c_m by finding sharper bounds on feasible blocks.
Undirected multigraph incidence, $m \geq 4$	Reduction to a knapsack over 2-connected blocks. The value is $\Theta(m^2 n)$ as $m \to \infty$ with $n/m \to \infty$ (Theorems A.63 and A.64 and Proposition A.61)	Determine the best 2-connected block of each size and sharpen the asymptotic bounds.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$ (Theorems 1.9 and 1.10)	Determine which edge-separation bounds remain valid for vertex separation when $r \geq 3$.
Directed hypergraphs	For fixed $r \geq 3$ and $m \geq 2$, leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

Table 3.2. The remaining problems, separated from the results already proved in this thesis.

Four directions carry most of the remaining mathematical weight. The first is the simple directed arc problem. Construction 3.1 supplies three dense families, but no matching general upper bound. The clique-core counterexample leaves the smaller orders as a separate question.

The second direction is the classical undirected vertex problem for $m \geq 6$. It remains open despite the results of Sørensen and Thomassen. The third is the hypergraph vertex problem beyond $m = 3$. There the incidence-rank argument has no known higher-connectivity replacement. The fourth is the multigraph incidence problem beyond $m = 3$, which the convention of Section 1.3.1 poses as a question of its own. Its value is a knapsack over 2-connected blocks, and what is missing is the best block of each size. As $m \to \infty$ with $n/m \to \infty$, the ratio $K_m(n)/(m^2 n)$ is at least $3 - 2\sqrt{2} - o(1)$ and less than 2. These bounds do not establish that the ratio converges.

The other rows of Table 3.2 remain natural questions. They ask for finer classifications or finite exact values. They complete the map, but they are secondary to the four frontiers above.

Appendix A

Full Proofs and Computational Audit

This appendix carries every proof in the thesis. The body states each result and briefly explains its main idea. The full arguments are collected here. So are the supporting lemmas and propositions that are not stated in the body. Figures appear beside the arguments that need them.

How to read this appendix. The shared graph, route, and connectivity terminology is introduced in Chapter 1. Terms specific to the appendix are defined when they first become necessary. The appendix has four parts, and they do not carry equal weight.

PART I contains classical machinery: Theorems A.1 and A.2 and Mader's theorem. A reader who trusts the cited results may skip it. PARTS II and III contain the original arguments. Readers who want a selective route through them can focus on three proof ideas. The first is the reachability skeleton of Lemma A.19, which settles Theorems 1.6 and 1.7 in a few lines and Theorem 1.13 for every n and m. The second is the incidence-rank argument of Lemma A.54 and Theorem 1.10. It settles the hypergraph vertex problem at $m = 3$ by induction over blocks and separating pairs. The third is the two-step budget of Lemma A.34. It drives Theorems 1.11, 1.12, 1.14 and A.74 and pins six variants asymptotically.

PART IV contains the search's connectivity bookkeeping and the computational audit. Every construction named in the body is defined in this appendix. The directed constructions appear in Section A.3. A heading with the badge [AI] contains a proof written by an AI model that the author has not verified line by line. The Contribution Statement explains this status. An unbadged heading is either a cited classical result or an argument the author has checked. The latter group includes the Gomory–Hu double count of Section A.2. That count proves Theorems 1.2 and 1.3 from Lemmas A.5 to A.7.

Part I. Classical machinery

The two theorems every later argument runs on, stated with their citations and not reproved, and Mader's theorem proved here in full through the Gomory–Hu tree.

A.1 Menger, Gomory–Hu, and the degree bound

We use two classical results in their capacitated form. Unit capacities represent simple graphs, integer capacities represent multiplicities, and cut capacity is the total capacity crossing the cut.

Theorem A.1 (Menger and max-flow/min-cut [9, 14])**.** *In any capacitated digraph, and for distinct vertices u, v, the maximum flow from u to v equals the minimum capacity of a cut separating u from v. This is the max-flow/min-cut theorem. With integer capacities, integral flow decomposition recovers Menger's path–separator theorem. At unit capacities it gives the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u–v paths equals the minimum number of edges whose removal separates them,*

$$\lambda_G(u, v) = \min\{\,|C| : C \subseteq E(G),\ G - C \text{ has no } u\text{–}v \text{ path}\,\}.$$

Undirected graphs become symmetric networks, multigraph multiplicities become capacities, hypergraphs use the gate network of Figure 2.3b, and vertex-disjoint routes use the vertex splitting of Figure 2.3a. Thus one maximum-flow theorem supports every checker variant.

Theorem A.2 (Gomory–Hu [4])**.** *Every* undirected *capacitated graph G has a weighted tree T on the same vertex set, its* Gomory–Hu tree*, with two properties. First, for all distinct u, v, the value $\lambda_G(u, v)$ is the minimum edge weight on the u–v path in T. Second, deleting a tree edge $e = xy$ partitions the vertices into two sets whose cut in G is a minimum x–y cut and has capacity equal to the weight of e. Thus the tree records both pairwise connectivity values and the specific cuts used below.*

The theorem applies to undirected simple graphs and multigraphs, but not to asymmetric directed cuts or vertex connectivity. Its standard construction is proved in [4]. The second property is essential to the double count: reproducing pairwise values alone would not justify treating every tree-edge partition as a cut of that weight.

We also use the immediate endpoint bound.

Lemma A.3 (Degree bound)**.** *Every pair u, v of distinct vertices of a graph G satisfies $\lambda_G(u, v) \leq \min(\deg u, \deg v)$, and every ordered pair of a digraph D satisfies $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u–v path uses a distinct edge at u, so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. □

Chapter 2 leans on two monotonicity facts to keep the search's connectivity checks cheap. Both follow directly from max-flow / min-cut.

The *binding connectivity* is the maximum local connectivity $\lambda^{\max}$, namely the value of the pair that binds the feasibility cap.

Proposition A.4 (Monotonicity of the binding connectivity)**.** *Let G be any graph, digraph, hypergraph, or directed hypergraph model used in this thesis, simple or with multiplicities, and let G' be obtained from G by changing one edge, arc, or hyperedge by a single copy. Then*

(1) (monotonicity) *removing the unit cannot raise $\lambda^{\max}$, and adding it cannot lower $\lambda^{\max}$, and*

(2) (unit step) *adding the unit raises $\lambda^{\max}$ by at most one.*

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Write G^+ for G with one edge, arc, or hyperedge copy added, and fix a pair (s,t), ordered in a directed model. Removing a copy only removes routes, so neither local connectivity can increase under removal. This proves the monotonicity part after taking the maximum over all pairs.

For the unit step in edge connectivity, take a largest edge-, arc-, or hyperedge-disjoint family of s–t routes in G^+. At most one member of the family uses the new copy. Discard that member if it exists. Every remaining route lies in G, so

$$\lambda_G(s,t) \le \lambda_{G^+}(s,t) \le \lambda_G(s,t) + 1.$$

The vertex measure has the same unit step, and under the incidence convention of Section 1.3.1 it has the same one-line proof in every model. A route is a path of the incidence graph, and every copy the route uses is an interior node of that path, so two internally disjoint routes never share a copy. At most one member of a largest family therefore uses the new copy. Deleting that route leaves a family in G and gives

$$\kappa_G(s,t) \le \kappa_{G^+}(s,t) \le \kappa_G(s,t) + 1.$$

Taking the maximum over all pairs proves the two asserted unit-step bounds. A copy parallel to an existing adjacency is one of the cases, not an exception to it: a new copy of uv is one further route from u to v, so it raises $\kappa(u,v)$ by exactly one, and by at most one at every other pair. □

A.2 Mader's theorem in full

Let T be a Gomory–Hu tree of G and let $\mathrm{dist}_T(e)$ count the tree edges on the path between the endpoints of $e \in E(G)$, as Figure A.1 illustrates. The upper bound is the following double count. Simplicity enters only because at most $n-1$ graph edges can join adjacent vertices of T.

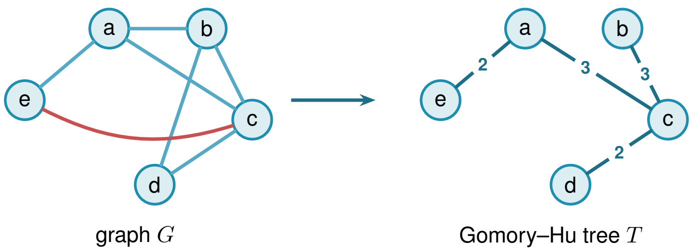


Figure A.1. A graph G and a Gomory–Hu tree T on the same vertices. Each graph edge crosses $\mathrm{dist}_T(e)$ tree cuts, and at most $n-1$ edges of a simple graph have tree distance one.

Lemma A.5 (Distance-one edges)**.** *At most $n-1$ edges of a simple graph G satisfy $\mathrm{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\mathrm{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T. The tree T has exactly $n-1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G. □

Lemma A.6 (Sum of tree distances)**.** $\sum_{e\in E(G)} \operatorname{dist}_T(e) \;\geq\; 2|E(G)| - (n-1).$

Proof. Partition $E(G)$ into $E_1 = \{e : \operatorname{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \operatorname{dist}_T(e) \geq 2\}$. Then

$$\sum_{e\in E(G)} \operatorname{dist}_T(e) \;\geq\; |E_1| + 2\,|E_{\geq 2}| \;=\; 2|E(G)| - |E_1| \;\geq\; 2|E(G)| - (n-1),$$

using Lemma A.5 in the last step. □

Lemma A.7 (Double-counting identity)**.** *Let G carry* unit capacities*, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain* number *of edges of G crossing the cut that t induces. Then*

$$\sum_{t\in E(T)} c(t) \;=\; \sum_{e\in E(G)} \operatorname{dist}_T(e).$$

This is the form used for Mader's theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By Theorem A.2 each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e\in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t\in E(T)} c(t) = \sum_{e\in E(G)} \sum_{t\in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e\in E(G)} \operatorname{dist}_T(e),$$

where the last equality holds because an edge $\{u,v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u–v path in T. □

Proof of Theorem 1.2, upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m-1$ and let T be a Gomory–Hu tree of G. Each tree edge $t = \{x,y\}$ has weight $c(t) = \lambda_G(x,y) \leq m-1$, so summing over the $n-1$ tree edges, $\sum_t c(t) \leq (m-1)(n-1)$. Combining with Lemmas A.6 and A.7,

$$2|E(G)| - (n-1) \;\leq\; \sum_{e\in E(G)} \operatorname{dist}_T(e) \;=\; \sum_{t\in E(T)} c(t) \;\leq\; (m-1)(n-1).$$

Rearranging gives $2|E(G)| \leq m(n-1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. □

The factor two comes from charging all but at most $n-1$ edges to at least two tree cuts. Parallel edges destroy that refinement, leaving the multigraph bound $(m-1)(n-1)$ of Theorem 1.3.

Proof of Theorem 1.3. For the lower bound, take a spanning tree of K_n, that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n-1$ edges, and give every tree edge multiplicity $m-1$, meaning $m-1$ parallel copies. The result has $(m-1)(n-1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m-1$ copies, so $\lambda(u,v)=m-1$ for every pair and $\lambda^{\max}=m-1$. The graph sits right at the cap.

For the upper bound, suppose $\lambda_G^{\max}\le m-1$ and build a Gomory–Hu tree T of G (Theorem A.2), the same $(n-1)$-edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G, of capacity $c(e)=\lambda_G(u,v)\le m-1$ for the pair (u,v) it represents. Now charge each edge of G to a tree edge: an edge $\{x,y\}$ crosses the cut of the lightest tree edge on the tree path from x to y, so it is counted inside that tree edge's capacity. Summing the $n-1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)|\le\sum_{e\in T}c(e)\le(m-1)(n-1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity-$(m-1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m-1)(n-1)$ and not a floor. □

Construction A.8 (Hub-and-spokes graph)**.** For $n\ge m\ge 2$, let $H_{m,n}$ have vertex set $\{h,v_1,\dots,v_{n-1}\}$ with (i) all $n-1$ *hub edges* $\{h,v_i\}$, and (ii) a *spoke graph* on $\{v_1,\dots,v_{n-1}\}$ with $\left\lfloor\frac{(m-2)(n-1)}{2}\right\rfloor$ edges in which every v_i has spoke degree at most $m-2$. Such a spoke graph exists by Lemma A.9. This is the general extremal construction. If $(m-1)\mid(n-1)$, take the spoke graph to be disjoint copies of K_{m-1}, and adjoining h turns them into copies of K_m with one shared vertex, the shared-clique construction.

Lemma A.9 (Near-regular graphs exist)**.** *For integers $N\ge 1$ and $0\le r\le N-1$ there is a graph on N vertices with $\left\lfloor\frac{rN}{2}\right\rfloor$ edges and maximum degree at most r.*

Proof. Recall that r-regular means that every vertex has degree r. Here a *circulant graph* is one whose edges are prescribed by fixed cyclic distances modulo N. A *matching* is a set of edges with no shared endpoint, and it is *near-perfect* when it covers all but one vertex. If rN is even we build an r-regular circulant graph on vertex set $\{0,\dots,N-1\}$: for even r connect each i to $i\pm j \pmod N$ for $j=1,\dots,r/2$, and for odd r (then N is even) use $j=1,\dots,(r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both are r-regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r-1)$-regular circulant with distances $1,\dots,(r-1)/2$ and add the near-perfect matching that pairs vertex i with $i+(N+1)/2 \pmod N$ for $i=0,\dots,(N-3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2}+\frac{N-1}{2}=\frac{rN-1}{2}=\left\lfloor\frac{rN}{2}\right\rfloor$ edges, with one vertex of degree $r-1$ and all others of degree r. □

Theorem A.10 (Extremal graphs exist)**.** *For all $n\ge m\ge 2$ the graph $H_{m,n}$ has $\left\lfloor\frac{m(n-1)}{2}\right\rfloor$ edges and $\lambda^{\max}\le m-1$.*

Proof. The edge count is $(n-1)+\left\lfloor\frac{(m-2)(n-1)}{2}\right\rfloor=\left\lfloor\frac{m(n-1)}{2}\right\rfloor$, since $(m-2)(n-1)$ and $m(n-1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i, whose total degree is at most $1+(m-2)=m-1$. Lemma A.3 then gives $\lambda(u,v)\le m-1$. □

The upper bound above proves Theorem 1.2, and it uses no relation between n and m: the Gomory–Hu tree has $n-1$ edges whatever m is. Theorem A.10 adds that the bound is attained once $n \geq m$, which is where that hypothesis is needed and where it belongs. For $n < m$, the complete graph is itself feasible and the bound is slack except at $(n, m) = (2, 3)$, where the floor creates the isolated equality noted in the main text. Equality then holds systematically from $n = m$ onward. The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality)**.** *Let G be simple with $\lambda_G^{\max} \leq m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G. If $m(n-1)$ is even, then every tree edge has weight exactly $m-1$, every tree edge is also an edge of G, and every edge of G joins vertices at tree distance at most 2. If $m(n-1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m-2$, or one tree edge is not an edge of G, or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three non-negative integer slacks, one for each inequality it uses:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \qquad S_2 = \sum_{e:\, \mathrm{dist}_T(e) \geq 2} \big(\mathrm{dist}_T(e) - 2\big), \qquad S_3 = (n-1) - |E_1|.$$

In words, S_1 counts tree weights below $m-1$, S_2 counts edges beyond tree distance two, and S_3 counts tree edges that support no edge of G. Retracing the proof gives two expressions for the same quantity,

$$\sum_e \mathrm{dist}_T(e) \;=\; 2|E| - |E_1| + S_2 \qquad \text{and} \qquad \sum_e \mathrm{dist}_T(e) \;=\; \sum_t c(t) \;=\; (m-1)(n-1) - S_1.$$

Substituting $|E_1| = (n-1) - S_3$ into the first and equating it with the second,

$$\begin{aligned} 2|E| - (n-1) + S_2 + S_3 \;&=\; (m-1)(n-1) - S_1, \\ \text{so} \qquad 2|E| \;&=\; m(n-1) - (S_1 + S_2 + S_3). \end{aligned}$$

The whole characterisation reads off that last identity. If $m(n-1)$ is even, equality $|E| = m(n-1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. □

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G, so G contains its own Gomory–Hu tree and has only $n-1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod 3$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of Construction A.8 that the search of Chapter 2 rediscovers.

Remark A.12 (The counting convention, and the audit it forces)*.* This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. Sørensen and Thomassen, and the Erdős Problems database [10] after them, state the same results the other way round, as the *smallest* number of edges that *forces* such a pair. Those two quantities are adjacent integers by construction: if k is the largest feasible edge count then no graph on n vertices with $k+1$ edges is feasible, so $k+1$ is exactly the forcing threshold.

What follows is the bookkeeping that convention forces on every constant quoted from the literature.

The shift is visible in every entry the database and this thesis both state. Write $f_m(n) = k_m(n)+1$ for the forcing count, which is the form the database and [8] both use, so that the symbols k_m and ℓ_m keep their meaning here throughout. The database gives $f_2(n) = n$ where a tree has $n-1$ edges, $f_3(2n) = 3n-1$ where this thesis has $k_3(2n) = \lfloor 3(2n-1)/2 \rfloor = 3n-2$, and $f_4(n) = 2n-1$ where this thesis has $2n-2$. The edge entries shift the same way: the database quotes $5n-2$ where $\ell_5(2n) = 5n-3$ here, and $3n-2$ [20] where $\ell_6(n) = 3n-3$ here. It also writes the Bollobás–Erdős conjecture as $1+\binom{m}{2}n$ *on* $1+(m-1)n$ *vertices* against $\binom{m}{2}n$ here. Every one is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that size.

The conjecture is the one entry that must keep its vertex count attached, and an earlier draft of this thesis lost it. On $N = 1+(m-1)n$ vertices the conjectured value is $\binom{m}{2}\frac{N-1}{m-1} = \frac{m(N-1)}{2}$, which is the rate $c_m = m/2$ of Theorem A.13. Read instead as a count on n vertices, $\binom{m}{2}n$ overstates it by a factor of $m-1$, giving $3n$ at $m=3$ where the true value is about $3n/2$.

An inconsistent conversion between forcing and avoiding counts can also produce a false comparison. The comparison $k_5 > \ell_5$ is invariant under the shift, since *both* functions move by one, so no internal check can detect a half-converted statement. What does detect it is evaluating the two sides in different conventions, which manufactures a spurious divergence one size early. Reading $\lfloor 8 \cdot 13/3 \rfloor - 3 = 31$ against this thesis's $\ell_5(13) = 30$ appears to separate the two problems at $n = 13$, but the 31 is a forcing count and the 30 is an avoiding count. Done consistently, $k_5(13) = 30 = \ell_5(13)$ and the separation begins at $n = 14$, which is also what [8] proves directly by showing $f_5(n) = \lfloor 5(n-1)/2 \rfloor + 1$ throughout $6 \le n \le 13$.

The range likewise comes from the paper rather than from the database, which records only the clean tail $n \ge 13$. Theorem 4 there states $f_5(n) = \lfloor 8n/3 \rfloor - 3$ for every $n \ge 6$ apart from $n = 7$ and $n = 12$, and supplies $f_5(7) = 16$ and $f_5(12) = 28$ separately. Its closing line then covers the whole small range at once, $f_5(n) = \lfloor 5(n-1)/2 \rfloor + 1$ for $6 \le n \le 13$. That is why Theorem 1.5 is stated in two ranges and needs no exceptions: the two sizes the paper sets aside are the only points of $6 \le n \le 13$ at which the large-n formula misses the small-n one, low by one at $n = 7$ and high by one at $n = 12$.

A.2.1 The vertex problem is a block problem

The general *vertex* problem $k_m(n)$, from $m \ge 6$ onwards, has remained open since the 1970s, and this short section records its block structure. It changes the object being searched: from graphs on n vertices, with n growing, to 2-connected graphs alone.

More generally, a graph on more than k vertices is *k-connected* if deleting any set of at most $k-1$ vertices leaves it connected. Thus a graph is 2-connected when it has at least three vertices and deleting any one vertex leaves it connected. A *cut vertex* is a vertex whose deletion disconnects the graph. In this subsection a *block* means a maximal connected subgraph that contains an edge and has no cut vertex of its own, so in a simple graph it is either a single bridge edge or a maximal 2-connected subgraph. Isolated vertices contribute no edge and are not counted as blocks. A *bridge* is an edge whose deletion disconnects its component.

The word *knapsack* below means the displayed integer optimisation: each chosen block has cost $b-1$ vertices beyond the shared one and value $h_m(b)$ edges, and the total cost may not exceed $n-1$.

Theorem A.13 (The vertex problem is a knapsack over blocks, [AI]). *For $b \geq 2$ let*

$$h_m(b) \;:=\; \max\big\{\, |E(B)| \;:\; B \text{ 2-connected or a single edge},\ |V(B)| = b,\ \kappa_B^{\max} \leq m-1 \,\big\},$$

with $h_m(b) := -\infty$ when no such B exists, as happens for every $b \geq 3$ at $m = 2$, where a cycle already carries two internally disjoint routes. Then for all $m \geq 2$ and $n \geq 2$,

$$k_m(n) \;=\; \max\Big\{ \sum_i h_m(b_i) \;:\; b_i \geq 2,\ \sum_i (b_i - 1) \leq n-1 \Big\}.$$

Consequently $k_m(n) \geq k_m(n_1) + k_m(n_2)$ whenever $n = n_1 + n_2 - 1$, and

$$c_m \;:=\; \lim_{n\to\infty} \frac{k_m(n)}{n-1} \;=\; \sup_{b \geq 2} \frac{h_m(b)}{b-1}$$

exists, the limit being approached from below. The supremum is finite: Remark A.14 gives $k_m(n) < 2(m-1)n$, so $c_m \leq 2(m-1)$.

Proof. Let G be feasible. Adjacent vertices u, v lie in a common block B, and no u-v path leaves B: leaving means crossing a cut vertex, and returning means crossing that same cut vertex a second time, which no path does. So $\kappa_G(u,v) = \kappa_B(u,v)$, every block inherits feasibility, and since every edge lies in exactly one block, $|E(G)| = \sum_B |E(B)| \leq \sum_B h_m(|V(B)|)$. Within one component the block sizes satisfy $\sum_B(|V(B)|-1) =$ (order of the component) $-\,1$, and summing over components loses one further vertex apiece, so $\sum_B(|V(B)|-1) \leq n-1$. That is $\leq$.

For $\geq$, take blocks $B_1, \ldots, B_t$ with $\sum(b_i - 1) \leq n-1$, pick one vertex, and attach all the B_i to it, pairwise disjoint otherwise. The result has $1 + \sum(b_i - 1) \leq n$ vertices. Add isolated vertices if necessary to bring its order to exactly n. Its nontrivial blocks are exactly the B_i, so pairs inside a block keep their κ and pairs split between two blocks are separated by the shared vertex and have $\kappa \leq 1 \leq m-1$. It is feasible with $\sum_i |E(B_i)|$ edges.

Superadditivity is the special case of gluing two extremal graphs at a single vertex. Write $a_n := k_m(n)$ and $b_t := a_{t+1}$ for $t \geq 1$. The gluing inequality gives $b_{s+t} \geq b_s + b_t$ for $s, t \geq 1$. Fekete's lemma [21] therefore gives $b_t/t \to \sup_{t\geq 1} b_t/t$, or equivalently $a_n/(n-1) \to$

$\sup_{n\geq 2} a_n/(n-1)$. Also $a_n/(n-1) \geq h_m(b)/(b-1)$ whenever $(b-1) \mid (n-1)$, while $a_n \leq \sup_b h_m(b)/(b-1)\cdot(n-1)$ from the display. Hence the limit is $\sup_b h_m(b)/(b-1)$. □

Three things follow at once.

At $m = 3$ it *proves* Theorem 1.4's value. A block with $\kappa^{\max} \leq 2$ is an edge or a cycle, since a block that is neither contains a *theta subgraph*, the union of three internally disjoint paths with the same two endpoints. Those three paths violate the cap. So $h_3(2) = 1$ and $h_3(b) = b$ for $b \geq 3$, the rate $b/(b-1)$ is largest at $b = 3$, and the knapsack packs triangles: $k_3(n) = 3\lfloor(n-1)/2\rfloor + ((n-1) \bmod 2) = \left\lfloor \frac{3(n-1)}{2} \right\rfloor$.

At $m = 4$ the picture is $h_4(b) = 2(b-1)$ for every $b \geq 4$, so the rate $h_4(b)/(b-1)$ reaches 2 at $b = 4$ and stays there, against 1 at $b = 2$ and $3/2$ at $b = 3$. The knapsack therefore spends its whole budget on blocks of size at least four as soon as one fits, giving $k_4(n) = 2(n-1)$ for $n \geq 4$. Below that only the small blocks are available and the formula overshoots, $k_4(2) = 1$ against its 2 and $k_4(3) = 3$ against its 4, which explains why Theorem 1.4 requires $n \geq m$. This one is a repackaging rather than a second proof. The upper bound $h_4(b) \leq k_4(b) = 2(b-1)$ is Theorem 1.4 itself read on a block, so it cannot turn round and reprove it, and nothing as elementary as the theta argument above is on offer for the blocks with $\kappa^{\max} \leq 3$. What the reformulation adds is the extremal block, one the appendix has already built: the hub-and-spokes graph $H_{4,b}$ of Construction A.8, a hub joined to every other vertex with a cycle on the rest. Unlike the shared-clique graph it is a single block, and it meets $2(b-1)$ at every size. An exhaustive sweep of the 2-connected graphs confirms that it is optimal, and the unique optimum, at every b from 4 to 8.

At every m it identifies the Bollobás–Erdős construction, as many copies of K_m sharing one vertex as fit, as the case $b = m$: $h_m(m) = \binom{m}{2}$ and the rate is exactly $m/2$. Their conjecture is precisely the assertion $c_m = m/2$, that no block beats K_m. Exhaustive computation over all 2-connected graphs on at most nine vertices, of which there are 194066 at the top size alone, confirms that none does, for every $m \leq 8$: the rate $m/2$ is attained and never exceeded. That sweep runs from its own standalone script, `program/scripts/simple_vertex_blocks.py`, rather than from the main program, so it corroborates the checker instead of sharing code with it. A *bouquet* is a collection of blocks sharing one common vertex and otherwise disjoint. Since its rate is a weighted average of the rates of its blocks, a counterexample must contain a single 2-connected block that beats $m/2$ on its own, and by the computation such a block needs at least ten vertices.

The reason the computation does not reach further is structural rather than a matter of processor time. The conjecture is false. Leonard disproved it first, at $m = 5$, with an explicit graph on 57 vertices [22], and Mader showed that the difference $k_m(n) - mn/2$ is unbounded for $m \geq 6$ [11]. What replaces it, and what the rest of this section leans on, is a lower bound of Sørensen and Thomassen [8]: for each fixed m,

$$c_m \;\geq\; \frac{m(m-1)-2}{2m-3} \;=\; \frac{m}{2} + \frac{1}{4} + O(1/m),$$

so this construction improves the conjectured rate by an amount tending to $1/4$ as m grows. It does not bound the improvement from above. That display alone refutes the conjecture for every $m \geq 5$, without a second reading of either earlier paper: subtracting gives

$$2\big(m(m-1)-2\big) - m(2m-3) \;=\; m-4,$$

which is positive exactly when $m > 4$, so the Sørensen–Thomassen rate exceeds $m/2$ from $m = 5$ onwards.

A 2-*cut*, also called a *separating pair*, is a pair of vertices whose removal disconnects a graph. A *triconnected decomposition* splits a 2-connected graph along its 2-cuts into pieces with no such separator. A graph is 3-*connected* when it remains connected after any two vertices are deleted.

Their witness is built by a recursion, described here in the language of this section, since it says why the blocks above cannot find it. Start from $G^0 = K_m$ minus an edge, and form G^j from two fresh copies of G^0 together with G^{j-1}, identifying one vertex of each with one vertex of the next, the three identifications arranged in a *cycle*. Each step adds $2m-3$ vertices and $m(m-1)-2$ edges, which is the rate above. The arrangement is cyclic, and three graphs glued in a cycle at three vertices have no cut vertex at all, so G^j is 2-connected and is a *single block*. Theorem A.13 therefore sees it as one indivisible object of size $m+(2m-3)j$ and gets no purchase on it. What the graph does have is 2-cuts, one pair of identification vertices separating each copy from the rest, so the decomposition that would see it is the classical triconnected one along separating pairs [23, 24]. Sørensen and Thomassen also treat the 3-connected case at $m = 5$ [8]. This does not supply a general theorem for every m. A decomposition along separating pairs would have to control both its pieces and the routes created when those pieces are reassembled.

It also explains the sizes. The recursion beats the rate $m/2$ only once $j > 2/(m-4)$, so the smallest member of the family that outruns a bouquet of K_m's has 26 vertices at $m = 5$, 24 at $m = 6$ and 18 at $m = 7$, against the nine that exhaustive enumeration reaches. The construction was rebuilt and checked here at $m = 5, 6, 7$ and $j \leq 3$, by the standalone script `program/scripts/st_construction.py`: the vertex and edge counts match the formulas exactly, every member is 2-connected, and every member has $\kappa^{\max} = m-1$, so each is feasible and none contains the forbidden m internally disjoint paths.

Remark A.14 (A degeneracy question, and what it would give). A general density bound is $k_m(n) < 2(m-1)n$: Mader's density theorem [25] would otherwise produce an m-connected subgraph. Feasibility says much more than that. It gives, for instance, that any two adjacent vertices have at most $m-2$ common neighbours, since each common neighbour is a route of length two beside the edge.

The natural strengthening is that a feasible graph always has a vertex of degree at most $m-1$. Since feasibility passes to induced subgraphs, that would make every feasible graph $(m-1)$-degenerate, meaning that every non-empty induced subgraph has a vertex of degree at most $m-1$, and give

$$k_m(n) \;\le\; (m-1)n - \binom{m}{2} \qquad (n \ge m-1),$$

halving the constant in the general upper bound used here. The range is where that count applies. Ordering the vertices by repeated deletion of one of degree at most $m-1$ gives $\sum_i \min(i-1,\, m-1) = (m-1)n - \binom{m}{2}$ once $n \ge m$, and at $n = m-1$ the same expression collapses to $\binom{n}{2}$, which is the complete graph's count and so still correct. Below $n = m-1$ the display drops under $\binom{n}{2}$, and for small enough n under zero, so $\binom{n}{2}$ is the bound there instead.

The degeneracy statement itself holds for $m = 3$, where it is classical: a graph of minimum degree at least three contains a *subdivision* of K_4, obtained by replacing its edges with internally disjoint paths. The four vertices corresponding to those of K_4 are its *branch vertices*, and two of them are joined by three internally disjoint paths. Exhaustive search over all graphs with minimum degree at least m finds no feasible one for $m \le 6$ and $n \le 10$, run from the standalone script `program/scripts/vertex_min_degree.py`. It is stated here as a question rather than a conjecture, since the sizes reached are well below those at which the Bollobás–Erdős conjecture itself first fails.

Part II. The directed results

The three constructions the body names but does not define, the reachability skeleton that both exact directed results run on, the value at $m = 2$ under both separations, the structural propositions that bound the frontier at $m \ge 3$, and the directed multigraph problem closed for every n and every m.

A.3 The directed constructions

Three basic shapes supply the directed lower bounds used below. The additional clique-core construction appears in Construction A.27. Chapter 1 shows the first two side by side in Figure 1.7 and records what they reach. Here each is defined precisely and its connectivity checked.

Construction A.15 (Directed hub)**.** For $n \ge m \ge 2$, take a hub h and label the other $N = n-1$ vertices $0, \dots, N-1$. Add the $2N$ hub arcs $h \to i$ and $i \to h$, and on the non-hub vertices the $(m-2)N$ ring arcs $i \to i + j \bmod N$ for $j = 1, \dots, m-2$. The total is $m(n-1)$ arcs and every non-hub vertex has $d^+ = d^- = m-1$.

Every ordered pair has a non-hub member, whose relevant degree is exactly $m-1$ and, since $n \ge m$, at most the hub's own $n-1$, with equality only at $n = m$. That member is therefore the binding one and the degree bound gives $\lambda^{\max} \le m-1$. At $m = 2$ the ring is empty and what remains is the *bidirected star*, in which two spokes can only reach each other through the hub.

Construction A.16 (One-directional bipartite digraph)**.** Split the vertices into parts A and B of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, and include every arc from A to B and none the other way. The total is $\lfloor n^2/4 \rfloor$ arcs.

Nothing leaves B and nothing enters A, so no route has a second step and every pair is joined by at most the one direct arc, giving $\lambda^{\max} = 1$.

Construction A.17 (Augmented bipartite digraph)**.** For $n \geq m \geq 2$, set $|B| = \lceil (n+m-2)/2 \rceil$ and $|A| = n - |B|$, include every arc from A to B, and inside B give each b_j the $m-2$ in-arcs from its cyclic predecessors $b_{j-1}, \ldots, b_{j-(m-2)}$. The total is $\lfloor (n+m-2)^2/4 \rfloor$ arcs.

The only in-arcs of a $b \in B$ that an $a \in A$ can reach are the direct arc $a \to b$ and the $m-2$ two-step detours through b's in-neighbours inside B, so such a pair carries exactly $m-1$ routes and no more, while a pair inside B is held down by the $m-2$ in-arcs of its head that lie inside B, those being the only in-arcs another vertex of B can reach, and pairs from B to A or inside A have no routes at all.

At $m = 3$, $n = 10$ this is the digraph that refutes the natural first guess $\max(m(n-1), \lfloor n^2/4 \rfloor)$. Here $|B| = \lceil 11/2 \rceil = 6$ and $|A| = 4$, so its 24 arcs from A to B carry a six-cycle inside B, one arc per b_j since $m-2=1$, giving every b_j one extra in-arc, so each pair a_i, b_j has the direct arc and one detour and $\lambda^{\max} = 2$. That is $24+6 = 30$ arcs against the guess of 27, and the correction is the shift of the partition away from a balanced split of n, which the guess assumes, towards a balanced split of $n+m-2$. At this size that quantity is 11 and odd, so $|B| = 6$ and $|B| = 5$ tie, the latter giving $25+5 = 30$ as well.

A.4 The reachability skeleton

One tool serves both directed results that follow, so it is proved here before either of them. The object is a thin subdigraph that carries the whole reachability relation of D and nothing else. Section A.5 uses it once, to settle $m = 2$ in a few lines under both separations, and Section A.7 uses it $m-1$ times over, peeling one copy of each of its arcs per unit of the cap.

Definition A.18 (Reachability skeleton)**.** Let D be a directed multigraph. A spanning subdigraph $R \subseteq D$, using at most one copy of each arc it uses (no parallel copies, even if D has them), is a *reachability skeleton* of D if, for every pair of vertices $u \neq v$, u reaches v by some directed path in R exactly when u reaches v by some directed path in D.

A reachability skeleton keeps every yes-or-no fact about who can reach whom and discards everything else, in particular every parallel copy of every arc. It is a caricature of D: thin, and faithful on reachability, but not on the number of disjoint paths. The lemma below says such a skeleton always exists and is never too big. That one fact is what both later arguments run on. At $m = 2$ the skeleton turns out to be the whole digraph, which is the content of Theorem 1.7, and at a general m deleting it repeatedly is what Lemma A.39 makes pay.

Lemma A.19 (A sparse reachability skeleton always exists, [Al])**.** *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Proof. Building R takes three steps: handle the traffic *inside* each strongly connected piece of D, handle the traffic *between* pieces, then count. A *strongly connected component* (an SCC) is a maximal set of vertices any two of which reach each other, and every directed multigraph partitions uniquely into its SCCs, since mutual reachability is an equivalence relation. Figure A.2 carries one small running example through all three steps, and every object named below appears in it concretely.

Step 1: inside each component. Fix one SCC of order $n_i \geq 1$ and pick any vertex r inside it as a root. Since r reaches every vertex of the component, a standard fact about digraphs gives a spanning *out-arborescence*: a tree of $n_i - 1$ arcs of D, every one pointing away from r, along which r reaches every other vertex of the component (grow it one hop at a time, always available since the component is strongly connected). Symmetrically, because every vertex of the component reaches r, there is a spanning *in-arborescence* of $n_i - 1$ arcs, every one pointing toward r, along which every vertex reaches r. Taking both trees together, any two vertices u, v of the component are joined by the route $u \to r \to v$, first the in-arborescence, then the out-arborescence. This concatenation can revisit vertices, but deleting closed portions leaves a directed u–v path. Thus the at most $2(n_i - 1)$ arcs of the two trees reconstruct *every* reachability relation inside this one component.

Step 2: between components. Contract each SCC to a single point and, between two contracted points, keep one arc whenever D has at least one arc from the first component to the second (dropping any further parallel arcs between the same two components at this stage). Call the result Q_0, a digraph on q points, one per component. Two different SCCs can never reach each other in both directions, since that would make them one strongly connected component after all, so Q_0 has no directed cycle: it is a DAG (a directed acyclic graph). Reachability between two different components of D is, by this very construction, exactly reachability between the two corresponding points of Q_0. Thin Q_0 down to an *inclusion-minimal* spanning subdigraph $Q \subseteq Q_0$ that still has the same reachability relation, deleting one arc at a time for as long as some arc is redundant, which must stop since Q_0 has only finitely many arcs.

The contracted DAG Q_0 is called the *condensation* of D. The reason to thin Q_0 down to Q is that Q's *underlying undirected graph*, meaning Q with the arrows erased, has no triangle. Suppose three points x, y, z of Q were pairwise joined by some arc of Q. Since $Q \subseteq Q_0$ and Q_0 has no cycle, Q has none either, so between any two of x, y, z the arc runs in only one direction, never both. Three points pairwise joined with a consistent direction on each pair admit only one pattern, up to relabelling: a linear order, say $x \to y$, $y \to z$, and $x \to z$. But then $x \to z$ is redundant, since x already reaches z via y, so deleting it would leave the reachability relation of Q unchanged, contradicting that Q is inclusion-minimal. So no three points of Q can be pairwise joined at all: that is triangle-freeness of the underlying graph.

A triangle-free graph on q vertices has at most $\lfloor q^2/4 \rfloor$ edges, by Mantel's theorem [26], the $r = 2$ case of the more general theorem of Turán [27]. Applied to Q,

$$|A(Q)| \;\leq\; \left\lfloor \frac{q^2}{4} \right\rfloor .$$

Step 3: assembling R and counting. Build R from the two arborescences of Step 1 inside every SCC, together with one witness arc of D for every arc of Q (some arc of D realises it, by the very definition of Q_0). By Step 1, every pair inside one component reaches each other in R exactly as in D. By Step 2, and because within-component reachability is already secured, every pair in two different components reaches each other in R exactly as in D too: to travel from u to v across the components Q's path visits, reach each witness arc's tail inside its own component (Step 1), cross the witness arc, and repeat. Since $R \subseteq D$ throughout, R can never certify a reachability D does not already have, so R is a genuine reachability skeleton. Writing $n_1, \ldots, n_q$ for the SCC orders, so $\sum_i n_i = n$, its arc count is at most the $2(n_i - 1)$ arborescence arcs of each component plus one witness per arc of Q. The first count is an upper bound rather than an equality, since the in- and out-arborescence of one component may share an arc, which only helps. So

$$|A(R)| \;\leq\; \sum_{i=1}^{q} 2(n_i - 1) \;+\; |A(Q)| \;\leq\; 2(n-q) \;+\; \left\lfloor \frac{q^2}{4} \right\rfloor \;=:\; f(q).$$

It remains to bound $f(q)$ over the range $1 \leq q \leq n$ that the number of components can actually take. Its increments are

$$f(q+1) - f(q) \;=\; -2 \;+\; \left\lfloor \frac{q+1}{2} \right\rfloor,$$

using the identity $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor = \lfloor (q+1)/2 \rfloor$, checked on the two parities of q (at $q = 2j$ both sides are j, at $q = 2j+1$ both are $j+1$). Since $\lfloor (q+1)/2 \rfloor$ never decreases as q grows, neither do the increments of f, which is to say f is convex. A convex function on an interval attains its largest value at one of the two ends, because a value at an interior point is a weighted average of the two ends or less. Hence

$$f(q) \;\leq\; \max\bigl(f(1),\, f(n)\bigr) \qquad \text{for } 1 \leq q \leq n.$$

At $q = 1$ the whole digraph is one component, so $f(1) = 2(n-1) + 0$. At $q = n$ every component is a single vertex, so $f(n) = 0 + \lfloor n^2/4 \rfloor$. Hence $f(q) \leq \max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr) = M(n)$ for every q in range, which is the lemma. $\square$

The inclusion-minimal reachability subdigraph Q is also called the *transitive reduction* of the DAG Q_0. A *transitively closed triangle* is the three-vertex pattern $x \to y$, $y \to z$, $x \to z$.

A.5 The directed arc value at $m = 2$

This section proves the two $m = 2$ values, and it proves them in the order the logic runs rather than the order they are stated. The *vertex* case comes first, in a few lines from Lemma A.19. The *arc* case then follows from it through Remark A.24, because an upper bound on the larger feasible family is an upper bound on the smaller one, and the same argument also proves it

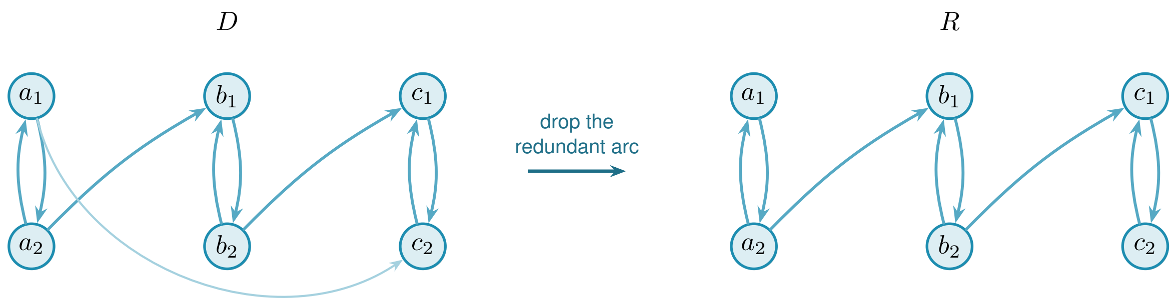


Figure A.2. A small instance of Lemma A.19. In D (left) three bidirected pairs $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$ are the strongly connected components, joined by $a_2 \to b_1$, $b_2 \to c_1$, and $a_1 \to c_2$. Contracting each pair to a point gives a condensation Q_0 on three points, carrying all three arcs $A \to B$, $B \to C$ and $A \to C$. That is exactly the transitively-closed triangle the proof rules out of the thinned Q. The arc $A \to C$ is redundant, since A already reaches C through B, so the transitive reduction Q keeps only $A \to B$ and $B \to C$. The underlying graph of Q is then a path on three points, triangle-free as claimed. The reachability skeleton R (right) reflects exactly this: every SCC keeps both of its internal arcs (an in- and an out-arborescence on two vertices is just the pair itself), while of the three component-crossing arcs of D only the two witnesses of Q's two arcs survive. The direct arc $a_1 \to c_2$ is dropped, yet a_1 still reaches c_2 in R: first $a_1 \to a_2$, then $a_2 \to b_1$, then $b_1 \to b_2$, then $b_2 \to c_1$, then $c_1 \to c_2$.

directly. Neither proof uses induction on n, so neither needs a base case, and neither theorem rests on a machine run. The lemmas below are used elsewhere rather than here: the first two show that a hub forces a linear bound for every m, which explains the hub lower bound discussed in Chapter 3, the third rules out transitive triangles, and the fourth is the deletion monotonicity that Section A.6.1 runs on. These auxiliary statements and the two $m = 2$ theorems carry no badge. Table A.11 records an exhaustive sweep over $2 \le n \le 7$ under both separations. It agrees with both theorems at every size and is corroboration rather than an ingredient.

Lemma A.20 (Two-hop lemma). *Let D be a simple digraph with $\lambda_D^{\max} \le m - 1$ containing a vertex r with $d^+(r) = n - 1$. Then every $v \ne r$ has at most $m - 2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r, that is, distinct vertices each sending an arc to v, the paths $r \to v$ and $r \to u_i \to v$ for $i = 1, \dots, m-1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n - 1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \le m - 1$. □

Theorem A.21 (Hub upper bound). *If a simple digraph D with $\lambda_D^{\max} \le m - 1$ has a vertex r with $d^+(r) = n - 1$, then $|A(D)| \le m(n - 1)$.*

Proof. Split the arcs into those leaving r (exactly $n - 1$), those entering r (at most $n - 1$), and the internal arcs avoiding r. By Lemma A.20 each of the $n - 1$ other vertices receives at most $m - 2$ internal arcs, so there are at most $(m - 2)(n - 1)$ internal arcs, and the total is at most $m(n - 1)$. □

Lemma A.22 (No transitive triangle)**.** *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \to z$ and $x \to y \to z$ would be two arc-disjoint routes.*

Lemma A.23 (Deletion monotonicity)**.** *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0,*

$$\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D) \qquad \textit{and} \qquad \kappa^{\max}(D - v_0) \leq \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D, and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D. The two statements are proved by the same one-line observation because deleting a vertex only removes routes, never creates any, and it cannot make two surviving routes share anything they did not share before.

In words: deleting a vertex can never manufacture independent routes. Every route surviving in $D - v_0$ was already a route in D, so no pair comes out of the deletion better connected than it went in. It is stated here because it is elementary and directed, and it is used in Section A.6.1, where the sparse case of Lemma A.34 deletes a vertex of low degree and needs to know the budget survives. It holds for both separations alike.

Remark A.24 (Which way Whitney runs)*.* Whitney's inequality is easy to misread: the tempting direction is the wrong one. From $\kappa_D(u, v) \leq \lambda_D(u, v)$ for every pair it follows that $\kappa_D^{\max} \leq \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \leq m - 1\} \subseteq \{D : \kappa_D^{\max} \leq m - 1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal numbers it hands over for free is

$$k_m^{\mathrm{dir}}(n) \geq \ell_m^{\mathrm{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ, so there really are digraphs in the difference of the two families. Whitney therefore gives one thing, that an arc construction is a vertex *lower* bound, and the vertex upper bound has to be proved separately. The proof below does that, and Chapter 3 treats the two separations the same way for $m \geq 3$, where the vertex problem stays open without assuming that the two optima coincide.

Proof of Theorem 1.7. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$.

Lower bound. The directed hub construction (Construction A.15 at $m = 2$, the bidirected star) and the one-directional bipartite construction (Construction A.16) both have $\lambda^{\max} = 1$, so by Remark A.24 both have $\kappa^{\max} \leq 1$ and are feasible for the vertex problem as well. Between them they witness $\ell_2^{\mathrm{dir}}(n) \geq M(n)$ and $k_2^{\mathrm{dir}}(n) \geq M(n)$ at once.

Upper bound. Let D be a simple digraph on $n \geq 2$ vertices with $\kappa_D^{\max} \leq 1$, and let $R \subseteq D$ be a reachability skeleton with $|A(R)| \leq M(n)$, which Lemma A.19 supplies for every loopless directed multigraph and so in particular for this one.

We claim $A(D) = A(R)$. One inclusion is the definition of R. For the other, take an arc $(u, v) \in A(D)$ and suppose $(u, v) \notin A(R)$. The arc itself shows that u reaches v in D, so u reaches v in R as well, along some directed path P using only arcs of R. Since $(u, v) \notin A(R)$, the path P is not that single arc, so it has length at least two. Now P and the one-arc route $u \to v$ are two distinct u-v routes, and they are internally vertex-disjoint, because the one-arc route has no interior at all and the empty set meets nothing. Hence $\kappa_D(u, v) \geq 2$, against feasibility. So $(u, v) \in A(R)$.

Therefore $|A(D)| = |A(R)| \leq M(n)$, which with the lower bound gives $k_2^{\mathrm{dir}}(n) = M(n)$. □

In words: at $m = 2$ a feasible digraph can afford no arc that some other route already makes redundant, so it is its own reachability skeleton, and the count Lemma A.19 proves for skeletons in general is already a count for D itself. Nothing here inspects n, and no base case is needed.

The arc value costs one further line.

Proof of Theorem 1.6. The two constructions above give $\ell_2^{\mathrm{dir}}(n) \geq M(n)$. For the upper bound, Remark A.24 places every arc-feasible digraph inside the vertex-feasible family, so Theorem 1.7 bounds it. Chaining the three,

$$M(n) \;\leq\; \ell_2^{\mathrm{dir}}(n) \;\leq\; k_2^{\mathrm{dir}}(n) \;\leq\; M(n),$$

so all three are equal. The same argument also proves it directly: in the claim above, P shares no *arc* with (u, v) either, since P runs inside R and (u, v) does not, so $\lambda_D(u, v) \geq 2$ follows by the identical sentence. □

Hence $\ell_2^{\mathrm{dir}}(n) = k_2^{\mathrm{dir}}(n)$, and the two separations agreeing at $m = 2$ is a consequence of that chain rather than a coincidence of two searches. What is a computed coincidence is narrower and is recorded in Table A.11: the two sweeps also visit the same number of search nodes at every size.

Three independent confirmations sit beside these proofs and none is needed by them. The exhaustive search of Chapter 2 was run under both feasibility tests, reporting the same maxima $2, 4, 6, 8, 10, 12$ at $n \leq 7$, and the two smallest of those are immediate by hand: at $n = 2$ there are at most 2 arcs, and at $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the remaining arcs form a transitive triangle, which Lemma A.22 forbids. Separately, Theorem 1.13 proves $L_2^{\mathrm{dir}}(n) = M(n)$ for directed multigraphs, and Corollary A.42 records that at $m = 2$ the multigraph and the simple digraph problems are the same problem. Those runs and that theorem corroborate the values, and the proofs above are what establish them.

Remark A.25 (The directed vertex problem at $m \geq 3$). The skeleton argument settles $m = 2$. For $m \geq 3$, the vertex problem is separate: the arc constructions belong to the smaller arc-feasible family and give no upper bound on $k_m^{\mathrm{dir}}(n)$. What can be said is computational. Running the exhaustive search of Chapter 2 under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\mathrm{dir}}(n)$	2	6	9	12	15
$k_3^{\mathrm{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the construction lower bound $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$. The table stops at $n = 6$ because that is where the exhaustion stops: at $n = 7$ neither sweep closes inside an hour and a half, and each returns the construction's 18 as a bare lower bound. That construction formula applies only for $n \geq m$. At $n = 2$, the complete digraph instead gives the maximum of 2 arcs. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

Lemma A.26 (The augmented bipartite digraph is feasible)**.** *For $m \geq 2$ and $n \geq m$, the digraph of Construction A.17 satisfies $\lambda^{\max} = m - 1$ and has $\lfloor (n+m-2)^2/4 \rfloor$ arcs. Hence $\ell_m^{\mathrm{dir}}(n) \geq \lfloor (n+m-2)^2/4 \rfloor$.*

Proof. The arc count is $|B|\,(|A| + m - 2)$, one arc from each of the $|A|$ vertices of A plus $m-2$ internal in-arcs, for each of the $|B|$ vertices of B. Now $|A| + m - 2 = \lfloor (n+m-2)/2 \rfloor$ and $|B| = \lceil (n+m-2)/2 \rceil$, and the product of the two halves of an integer N is $\lfloor N^2/4 \rfloor$, which gives the stated total at $N = n + m - 2$.

For the connectivity, fix $a \in A$ and $b \in B$. Consider the following routes from a to b: the direct arc, and the two-step detours $a \to b' \to b$ that pass through an in-neighbour b' of b inside B. There are $m-2$ such detours, giving a family of $m-1$ arc-disjoint routes. Longer routes may exist, but they cannot enlarge this family beyond the cut bound that follows. No larger arc-disjoint family is possible, because every route from a to b must finish with one of those $m-1$ in-arcs. Deleting them therefore cuts every route from a to b. For a pair b_i, b_j inside B the same cut argument gives an upper bound of $m-2$: the total in-degree of b_j is $|A| + m - 2$, but nothing leaves B towards A, so the only in-arcs of b_j that a route starting at b_i can reach are the $m-2$ inside B, and deleting those cuts the pair. Pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$. □

A.6 Structural propositions on the directed frontier

The augmented bipartite construction gives a quadratic lower bound, but it does not describe every extremal digraph. The following construction, supplied by Stijn Cambie, refutes the earlier formula and the decomposition proposed alongside it.

Construction A.27 (Clique-core counterexample)**.** Let $n \geq m \geq 2$, put $p = n - m$, $a = \lceil p/2 \rceil$

and $b = \lfloor p/2 \rfloor$. Take disjoint sets S, C, T of sizes a, m, b, with $C = \{c_1, \ldots, c_m\}$. Include exactly the arcs

$$c_i \to c_j \ (i \neq j), \qquad S \to \{c_1, \ldots, c_{m-1}\}, \qquad \{c_1, \ldots, c_{m-2}\} \to T, \qquad S \to T.$$

An arrow between sets means that all arcs in that direction are present.

The following cuts verify feasibility. In each row, X contains the starting vertex and excludes the terminal vertex, and the capacity counts arcs leaving X.

ordered pair	X	capacity
$c_i, c_j \ (i \neq j)$	$\{c_i\} \cup T$	$m-1$
s, c_j	$\{s\} \cup T$	$m-1$
c_i, t	$C \cup (T \setminus \{t\})$	$m-2$
s, t	$\{s\} \cup C \cup (T \setminus \{t\})$	$m-1$

All other ordered pairs are unreachable. Each pair of distinct core vertices also has the direct path and $m-2$ two-step paths inside C, so $\lambda_D^{\max} = m-1$. The arc count is

$$|A(D)| = m(m-1) + (m-1)a + (m-2)b + ab = \left\lfloor \frac{(n+m-3)^2}{4} \right\rfloor + 2(m-1) =: B_m(n).$$

Indeed, $a(b+1) = \lfloor (p+1)^2/4 \rfloor$. Subtracting the old branches gives

$$B_m(n) - m(n-1) = \left\lfloor \frac{(p+1)^2}{4} \right\rfloor - 2p, \qquad B_m(n) - \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = m - 1 - \lfloor p/2 \rfloor.$$

Both differences are positive when $m \geq 5$ and $7 \leq p \leq 2m-3$. At $m = 5$, $n = 12$, the four arc classes contribute $20 + 16 + 9 + 12 = 57$, whereas the earlier formula gives $\max(55, 56) = 56$. This establishes a lower bound, not the exact value of $\ell_5^{\mathrm{dir}}(12)$. The independent script `program/scripts/clique_core_check.py` checks the displayed cuts and identities and enumerates all 4094 nontrivial cuts of the twelve-vertex example.

The same example refutes the earlier decomposition conjecture. A partition with all arcs from a nonempty A to B, no arcs inside A, and no arcs from B to A would give

$$|A(D)| \leq |B|(n - |B|) + (m-2)|B| \leq \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor.$$

For the first inequality, fix $a \in A$ and $b \in B$. The direct path and the two-step paths through the internal in-neighbours of b force $d_B^-(b) \leq m-2$. At $m = 5$, $n = 12$, this would bound every extremiser by 56, contradicting the feasible graph with 57 arcs. At $n = 3m-3$, the excess over the quadratic value is still one, so any threshold beyond which the augmented bipartite count is always optimal would have to be at least $3m-2$ for $m \geq 5$. This necessary condition does not establish that such a threshold exists.

A.6.1 The two-step budget

Definition A.28 (Two-step budget)**.** Let D be a simple digraph and let $C \geq 1$. Write $p_2(u, v)$ for the number of vertices $x \notin \{u, v\}$ with both (u, x) and (x, v) present, so that each such x is the midpoint of a two-step route $u \to x \to v$. Call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v) \;\leq\; C.$$

Lemma A.29 (Feasible digraphs are budgeted)**.** *Let D be a simple digraph and $m \geq 2$. If $\lambda_D^{\max} \leq m-1$, or if $\kappa_D^{\max} \leq m-1$, then D is $(m-1)$-budgeted.*

Proof. Fix an ordered pair (u, v) with $u \neq v$ and take the direct arc (u, v), when it is present, together with the two-step route through each of the $p_2(u, v)$ midpoints. A midpoint is automatically distinct from u and from v, since D has no loops.

These routes are pairwise arc-disjoint. Two of them with different midpoints $x \neq x'$ share no arc, since the first uses (u, x) and (x, v) and the second uses (u, x') and (x', v), and neither uses the arc (u, v), which has no midpoint at all. They are also pairwise internally vertex-disjoint. A two-step route has interior exactly $\{x\}$, one vertex, distinct midpoints give disjoint interiors, and the direct arc's interior is disjoint from all of them for want of any element.

The family is therefore capped by $\lambda_D(u, v)$ and by $\kappa_D(u, v)$ alike, so either hypothesis gives $\mathbf{1}_{(u,v)\in A(D)} + p_2(u, v) \leq m-1$. □

The next lemma is where the budget stops being about pairs and becomes a statement about degrees alone. In words, a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n.

Lemma A.30 (The budget, read on degrees)**.** *Every C-budgeted simple digraph D on n vertices satisfies*

$$\sum_{x\in V} d^+(x)\, d^-(x) \;\leq\; C\, n(n-1).$$

Proof. Summing the budget over all $n(n-1)$ ordered pairs of distinct vertices gives $|A(D)| + \sum_{u\neq v} p_2(u, v) \leq C\, n(n-1)$, since the indicators sum to the arc count. Write Δ for the number of digons of D, meaning the pairs of vertices carrying an arc in each direction. Three small facts finish it.

(a) Walks of length two, counted by their midpoint.

$$\sum_{u,\, v\in V} p_2(u, v) \;=\; \sum_{x\in V} d^-(x)\, d^+(x),$$

where the left-hand sum runs over *all* ordered pairs, the diagonal $u = v$ included. Fixing a vertex x, a walk $u \to x \to v$ is nothing more than an independent choice of an in-neighbour u of x and an out-neighbour v of x, so x is the midpoint of exactly $d^-(x)\, d^+(x)$ of them.

(b) What the diagonal counts. A diagonal term $p_2(u,u)$ counts the closed walks $u \to x \to u$, and each digon contributes two such walks, one based at each of its endpoints. Hence $\sum_{u\in V} p_2(u,u) = 2\Delta$.

(c) Digons are few. Distinct digons use disjoint pairs of arcs, so $2\Delta \le |A(D)|$.

Splitting the all-pairs sum of (a) along its diagonal and feeding in the budget,

$$\begin{aligned}
\sum_{x\in V} d^+(x)\, d^-(x) \;&=\; \sum_{u\neq v} p_2(u,v) \;+\; 2\Delta && \text{by (a) and (b),}\\
&\le\; \big[C\, n(n-1) - |A(D)|\big] \;+\; 2\Delta \\
&\le\; \big[C\, n(n-1) - |A(D)|\big] \;+\; |A(D)| && \text{by (c),}\\
&=\; C\, n(n-1).
\end{aligned}$$

The last two lines absorb the digon term exactly against the slack the budget left behind, so nothing is given away in passing from pairs to degrees. □

Lemma A.31 (Small side of a high-degree vertex)**.** *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min\big(d^+(x), d^-(x)\big) \;\le\; \frac{2\, d^+(x)\, d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x) d^-(x)$. Since $\mu \le M$ we have $M \ge d(x)/2$, hence $d^+(x)d^-(x) = \mu M \ge \mu\, d(x)/2$, which rearranges to the claim. □

Lemma A.32 (Arithmetic identity)**.** *For all $n \ge 2$,* $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.

Proof. For $n = 2k$ the left side is $k(k-1) + k = k^2$, and for $n = 2k+1$ it is $k^2 + k = k(k+1)$. Both match the right side. □

Lemma A.33 (Deleting the smaller half of every vertex, [Al])**.** *Every simple digraph D on n vertices can be made* one-directional bipartite*, meaning that every remaining arc runs from one part to the other and none returns, by deleting at most $\sum_x \min(d^+(x), d^-(x))$ arcs. Consequently*

$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sum_{x\in V} \min\big(d^+(x), d^-(x)\big).$$

Proof. Call x *source-like* if $d^+(x) \ge d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x marks $d^-(x) = \min(d^+(x), d^-(x))$ arcs for deletion and a sink-like x marks $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number deleted is at most the displayed sum, an arc counted twice by that estimate only making it safer. An arc (a, b) survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T and has at most $|S|\,|T| \le \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the display. □

Lemma A.34 (The counting core, [AI]). *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4C(n-1).$$

Proof. Induct on n. The claim is immediate at $n = 2$, where a simple digraph has at most 2 arcs and the right-hand side is $1 + 4C \geq 5$. Let $n \geq 3$ and split on the minimum total degree.

If some vertex v has $d(v) < n/2$, then $d(v) \leq \lfloor n/2 \rfloor$ because $d(v)$ is an integer. Deleting a vertex only removes routes, so $D - v$ is C-budgeted too, and the induction hypothesis together with Lemma A.32 gives

$$\begin{aligned} |A(D)| &\leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4C(n-2) + \left\lfloor \frac{n}{2} \right\rfloor \\ &= \left\lfloor \frac{n^2}{4} \right\rfloor + 4C(n-2) \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4C(n-1). \end{aligned}$$

Otherwise $d(x) \geq n/2$ for every vertex. By Lemma A.31 and then Lemma A.30,

$$\sum_x \min\bigl(d^+(x), d^-(x)\bigr) \;\leq\; \frac{2}{\min_x d(x)} \sum_x d^+(x) d^-(x) \;\leq\; \frac{4}{n}\, C\, n(n-1) \;=\; 4C(n-1),$$

and Lemma A.33 converts that into the claim. □

Theorem 1.11, stated in Chapter 1, says in words that a feasible simple digraph is never more than a linear amount denser than the balanced bipartite wall, which holds $\lfloor n^2/4 \rfloor$ arcs. The wall's quadratic term is therefore the leading term, and what remains in question is how large the linear correction is. Neither proof below assumes anything about the shape of D.

Proof of Theorem 1.11 [AI]. By Lemma A.29 a simple digraph with $\lambda_D^{\max} \leq m-1$ is $(m-1)$-budgeted, so Lemma A.34 at $C = m-1$ gives the stated bound.

For the two-sided asymptotic statement, Construction A.17 gives

$$\ell_m^{\mathrm{dir}}(n) \;\geq\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \;=\; \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

The second application is the directed *vertex* problem. Theorem 1.7 settles it at $m = 2$, and for $m \geq 3$ this thesis has been careful to record that it does not follow from the arc case: Whitney's $\kappa \leq \lambda$ makes the vertex-feasible family the larger of the two, so an arc upper bound restricts nothing there. That remains true of the *bound*. What carries across is the route-counting, because the family Lemma A.29 exhibits happens to be disjoint in the stronger sense as well, which is why that lemma is stated for both separations at once and why the argument never needs the arc value.

Proof of Theorem 1.12 [AI]. By Lemma A.29 again, now under its second hypothesis, a simple digraph with $\kappa_D^{\max} \le m-1$ is $(m-1)$-budgeted, so Lemma A.34 at $C = m-1$ gives the stated bound.

For the lower bound, the augmented bipartite digraph of Construction A.17 has $\lambda^{\max} \le m-1$, hence $\kappa^{\max} \le m-1$ by Whitney, so for $n \ge m$ it witnesses $k_m^{\mathrm{dir}}(n) \ge \lfloor (n+m-2)^2/4 \rfloor$. For fixed $m \ge 3$ this is $n^2/4 + \Theta_m(n)$, and at $m = 2$ Theorem 1.7 gives the sharper $n^2/4 + O(1)$ asymptotic. □

These bounds leave a gap in the linear coefficient. Exact eventual equality would also determine the bounded correction. The augmented bipartite construction supplies a lower linear contribution of $(m-2)n/2 + O_m(1)$, whereas Theorem 1.11 allows $4(m-1)n$ above the quadratic term. For $m \ge 3$, the ratio of these linear coefficients is $8(m-1)/(m-2)$: sixteen at $m = 3$, and tending to eight as m grows. That is the factor at every n. Once n is large in terms of m, the Turán theorem of Huang and Lyu [13] cited in Chapter 1 lowers the upper end to $m-1$, and the factor with it to $2(m-1)/(m-2)$: four at $m = 3$, and tending to two. Remark A.35 explains why the existing degree estimates alone do not close that gap.

Remark A.35 (The dense case's coefficient needs a new inequality). One might hope to cut the coefficient of the dense case, the one where every $d(x) \ge n/2$, using the observation that in the augmented bipartite construction one whole side has $\min(d^+, d^-) = 0$, so summing Lemma A.31's worst case at every vertex looks wasteful. That observation alone does not cut the coefficient. The value four is the best universal coefficient obtainable from the two facts that case uses, the budget $\sum_x d^+(x)\, d^-(x) \le Cn(n-1)$ of Lemma A.30 and the floor $d(x) \ge n/2$. Read $C = m-1$ throughout what follows.

Think of the budget as money and of $t_x = \min(d^+(x), d^-(x))$ as what each vertex buys with it. At a vertex with $d(x) = s \ge n/2$, one also has $s \ge 2t$. Hence the cheapest possible cost for a given t is the two-branch function

$$c(t) = \min_{s \ge \max(n/2, 2t)} t(s-t) = \begin{cases} t(n/2 - t), & 0 \le t \le n/4, \\ t^2, & t \ge n/4. \end{cases}$$

On the first branch the value per unit cost is $t/c(t) = 1/(n/2 - t)$, which increases to $4/n$, whereas on the second it is $1/t$, which decreases from $4/n$. Thus $t \le (4/n)c(t)$ for every t, with equality in the continuous relaxation at $t = n/4$. Summing and using the shared budget $Q = (m-1)n(n-1)$ gives

$$\sum_{x \in V} t_x \le \frac{4}{n} \sum_{x \in V} d^+(x) d^-(x) \le \frac{4Q}{n} = 4(m-1)(n-1).$$

Conversely, fix m and take n large. Assign $t = n/4$ to as many abstract vertices as the budget permits, assign one further $t \in [0, n/4]$ so that its continuous cost absorbs the remainder, and assign zero to the rest. This uses at most $16(m-1) + 1 \le n$ vertices. The remainder is then within $O_m(n)$ of the full cost $n^2/16$, and writing that last t as $n/4 - \delta$ puts it on the first branch, where $c(t) = (n/4 - \delta)(n/4 + \delta) = n^2/16 - \delta^2$, so δ^2 is the shortfall and $\delta = O_m(\sqrt{n})$. The resulting total is $4(m-1)n - O_m(\sqrt{n})$, matching the coefficient four asymptotically. Therefore no smaller universal coefficient follows from these two inequalities alone.

The crude per-vertex bound therefore loses nothing within this relaxation, and the reason is that its two sources of slack close at that very point. Lemma A.31 is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \to n/2$ is an equality only when $d(x) = n/2$. Both hold at once exactly when $d^+(x) = d^-(x) = n/4$, which is the profile the relaxed optimum concentrates on. Cutting the constant therefore needs a second and independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

One thing Theorem 1.11 does not say is how far a feasible digraph is from the bipartite shape, since its dense case deletes arcs but its sparse case argues by induction instead. Lemma A.33 gives that structural statement directly, at the cost of a weaker bound, and the two are not comparable: at $m = n = 3$ the display below reads 11 against the theorem's 18, and the theorem only becomes the smaller of the two past the crossover of its $4(m-1)(n-1)$ with the $\sqrt{m}\, n^{3/2}$ below. Dropping the lower-order terms puts that crossover near $16(m-1)^2/m$, a rough estimate reading 21.3 at $m = 3$ and 36 at $m = 4$. Solved exactly, it falls between 19 and 20 at $m = 3$ and between 33 and 34 at $m = 4$. The second conclusion is a *bipartisation* bound: a lower-order number of deletions leaves the digraph a subgraph of a one-directional bipartite digraph. It does not say that the partition is balanced or that the surviving arcs fill it, so it stops short of a stability statement in the usual sense, which would need a near-extremality hypothesis to start from.

Proposition A.36 (Unconditional quadratic bound and bipartisation, [AI])**.** *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m - 1$. Then deleting at most $\sqrt{m}\, n^{3/2}$ arcs leaves D a subgraph of a one-directional bipartite digraph, and*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sqrt{m}\, n^{3/2}.$$

Proof. D is $(m-1)$-budgeted by Lemma A.29. Since $\min(a,b) \leq \sqrt{ab}$ for non-negative a, b, and by the Cauchy–Schwarz inequality in the form $\sum_x \sqrt{a_x} \leq \sqrt{n \sum_x a_x}$, Lemma A.30 gives

$$\begin{aligned}\sum_{x \in V} \min\bigl(d^+(x), d^-(x)\bigr) \;&\leq\; \sum_{x \in V} \sqrt{d^+(x)\, d^-(x)} \;\leq\; \sqrt{n \sum_{x \in V} d^+(x)\, d^-(x)} \\ &\leq\; \sqrt{n \cdot (m-1)\, n(n-1)} \;=\; n\sqrt{(m-1)(n-1)} \;\leq\; \sqrt{m}\, n^{3/2}.\end{aligned}$$

Lemma A.33 turns that count into both conclusions at once. □

Equality in both $\min(a,b) \leq \sqrt{ab}$ and Cauchy–Schwarz would require balanced in- and out-degrees and the same degree product at every vertex. Saturating the total budget would then give degrees of order $\sqrt{mn}$, incompatible with quadratic size for fixed m. The budget is a sum over vertices, not a bound of $(m-1)n$ on each individual product. Lemma A.34 obtains its linear error by combining that sum with a lower bound on the minimum total degree, and deleting a vertex when that lower bound fails.

The vertex problem at $m \geq 3$ is therefore open in the same narrow sense as the arc problem rather than in the wide one. Its leading constant is $\frac{1}{4}$ unconditionally, the term after it is linear in n, and since $\ell_m^{\mathrm{dir}}(n) \leq k_m^{\mathrm{dir}}(n)$ with both now inside $n^2/4 + \Theta_m(n)$, the two values agree to first order. The linear coefficient and the bounded correction remain undetermined here, as does the question whether the two values coincide exactly.

A.7 The directed multigraph problem, solved in full

At $m = 2$ the value can also be reached by induction on n, deleting one vertex of least total degree and bounding what it carried by the average. That route very nearly settles the multigraph case at every m as well. Dividing every multiplicity by $m-1$, as in the scaling step of Chapter 2, turns a feasible multigraph into a weighting with values in $[0, 1]$ whose pairwise maximum flows are all at most one. Deleting a vertex of least weighted degree from that weighting drives the averaging argument through cleanly on the linear branch, and on the quadratic branch as well whenever n is even. What breaks it is a single fractional remainder at odd n: a leftover of size $\frac{k}{2k-1}$ that an integer count could round away but a genuinely fractional weight cannot. Finding a specific vertex that is provably light enough to absorb that remainder turns out to resist every direct attempt.

This section abandons that attempt and proves the full value the other way, along the road Section A.4 has already laid. Instead of deleting one vertex at a time, it deletes one whole *sparse subgraph* at a time, chosen so that the deletion is guaranteed to lower every positive pairwise connectivity by at least one in a single stroke. That change removes the fractional step altogether, along with the restriction to a particular m or a particular parity of n.

A.7.1 The lower bound: thickening

The lower bound rests on one observation: multiplying every arc's multiplicity by a fixed integer multiplies every cut's capacity, and hence every pairwise connectivity, by that same integer.

Lemma A.37 (Thickening)**.** *Let D_0 be a directed multigraph and $k \geq 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u, v)$ by k. Then $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$ for every ordered pair $u \neq v$, and $|A(k \cdot D_0)| = k\,|A(D_0)|$.*

Proof. By Theorem A.1, $\lambda_{D_0}(u, v)$ equals the minimum capacity, over all u–v cuts, of the total multiplicity crossing that cut. Multiplying every multiplicity by k multiplies the capacity of *every* cut by k at once, so it multiplies the minimum over all cuts by k as well, which is precisely $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$. The arc count scales by k because every one of its summands, each multiplicity, does. □

Construction A.38 (Thickened tree and thickened bipartite digraph)**.** For $m \geq 2$, take any spanning tree T of K_n and give every edge multiplicity $m-1$ in each direction. Between two vertices,

T offers only its one tree path, so this is $m-1$ copies of the $m=2$ bidirected tree of Theorem 1.6, feasible at $\lambda^{\max}=1$ there, and Lemma A.37 with $k=m-1$ gives $\lambda^{\max}=m-1$ here and an arc count of $2(n-1)(m-1)$. Alternatively, take the one-directional complete bipartite digraph of Construction A.16 (balanced parts A, B with $|A|=\lfloor n/2\rfloor$, $|B|=\lceil n/2\rceil$, feasible at $\lambda^{\max}=1$) and give every arc multiplicity $m-1$: the same lemma gives $\lambda^{\max}=m-1$ and $(m-1)\lfloor n^2/4\rfloor$ arcs.

Taking whichever of the two constructions is larger gives $L_m^{\mathrm{dir}}(n) \geq (m-1)M(n)$ for every n and m, which is the lower bound half of Theorem 1.13. The rest of this section proves the matching upper bound.

A.7.2 The upper bound: a reachability skeleton

An upper bound needs a deletion that is guaranteed to lower every positive connectivity, paired with a bound on how much it removes. Deleting a vertex preserves feasibility, but need not lower every positive pairwise connectivity. Once multiplicities are free to sit anywhere between 0 and $m-1$, no single vertex is guaranteed to carry a controllable share of the arcs, which is exactly the fractional remainder above. The fix is to stop insisting on a vertex and delete a whole *subgraph* instead, and the reachability skeleton of Section A.4 is one: general enough to always exist, sparse enough to bound by Lemma A.19, and, as the next lemma shows, specific enough that removing it lowers every positive pairwise connectivity by at least one.

Lemma A.39 (Deleting a skeleton lowers every positive connectivity)**.** *Let D be a directed multigraph with $\lambda^{\max}(D) \leq k$ for some $k \geq 1$, and let R be a reachability skeleton of D. Write $D-R$ for the multigraph obtained by deleting, from each multiplicity $\mu(u,v)$, the one copy (if any) that R uses on that pair. Then, for every ordered pair $u \neq v$,*

$$\lambda_{D-R}(u,v) \ \leq \ \max\bigl(\lambda_D(u,v)-1, 0\bigr),$$

and consequently $\lambda^{\max}(D-R) \leq k-1$.

In words: subtracting one copy of each skeleton arc costs every reachable pair at least one route (a single pair can lose more, if some of its own routes happened to run through the deleted arcs too). The skeleton preserves reachability, so however many arc-disjoint routes survive the deletion, one further route can always be run entirely inside the arcs that were deleted, and that extra route is what the original cap has to pay for.

Proof. Fix an ordered pair $u \neq v$ and put $t=\lambda_{D-R}(u,v)$. If $t=0$, then $t \leq \max\bigl(\lambda_D(u,v)-1,0\bigr)$ immediately. Otherwise $D-R$ carries $t \geq 1$ pairwise arc-disjoint u–v paths. These use arcs of $D-R$ and hence of D, since $D-R \subseteq D$, so u reaches v in D. Because R is a reachability skeleton, u then also reaches v in R: some directed path P from u to v uses only arcs of R. Every arc of R has been subtracted out of $D-R$, so P shares no arc with any of the t paths already found inside $D-R$. Together, P and those t paths are $t+1$ pairwise arc-disjoint u–v paths in D, so $t+1 \leq \lambda_D(u,v)$, which gives the displayed pointwise bound. Since $\lambda_D(u,v) \leq k$

for every pair and $k \geq 1$, maximising gives $\lambda^{\max}(D - R) \leq k - 1$. $\square$

Proof of Theorem 1.13 [AI]. The lower bound is Construction A.38. For the upper bound, fix n and prove by induction on $k = m - 1 \geq 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq k$ has $|A(D)| \leq k\, M(n)$. At $k = 0$, every pair has $\lambda_D(u, v) \leq 0$, so no arc (u, v) can even be present, since a single arc is already one route, and $|A(D)| = 0 = 0 \cdot M(n)$.

For $k \geq 1$, take D with $\lambda^{\max}(D) \leq k$ and let R be a reachability skeleton, which exists with $|A(R)| \leq M(n)$ by Lemma A.19. By Lemma A.39, $D - R$ has $\lambda^{\max}(D - R) \leq k - 1$, and it is again a directed multigraph on the same n vertices, so the induction hypothesis gives $|A(D - R)| \leq (k - 1)M(n)$. Since R uses at most one copy of each arc it touches, the arcs of D split exactly into those of R and those left in $D - R$, so

$$|A(D)| \;=\; |A(R)| \;+\; |A(D - R)| \;\leq\; M(n) \;+\; (k - 1)M(n) \;=\; k\, M(n).$$

Taking $k = m - 1$ gives $L^{\mathrm{dir}}_m(n) \leq (m - 1)M(n)$, matching the lower bound. $\square$

Nothing above depended on m or on the parity of n: the same three-step construction of R and the same one-line deletion bound run identically at every level. In particular, the induction closes exactly the finite computation the rest of this appendix once needed help to settle.

Corollary A.40 (The directed multigraph value at $n = 7$, $m = 3$, [AI])**.** $L^{\mathrm{dir}}_3(7) = 24$.

Proof. Since $M(7) = \max(12, 12) = 12$, Theorem 1.13 at $m = 3$ gives $L^{\mathrm{dir}}_3(7) = 2M(7) = 24$. $\square$

This hand proof replaces an unfinished computation once discussed in Chapters 2 and 3. The mixed-integer call `prove_integral_arc_bound(7, 3, 25)` reached its time limit on the bundled CBC solver, and a fourteen-hour uncapped enumeration was stopped without a verdict. It was never run on a stronger solver. The computation and the proof do not independently confirm one another: only the proof settles the claim.

Corollary A.41 (The fractional relaxation is integral)**.** *For every $n \geq 2$, the m-free optimum $M^*(n)$ of Chapter 2 equals $M(n)$, and it is attained by a $\{0, 1\}$ weight matrix. Consequently the scaling identity*

$$L^{\mathrm{dir}}_m(n) \;=\; (m - 1)\, M^*(n)$$

holds for every n and every $m \geq 2$, with no hypothesis on the optimum.

In words: allowing fractional weights does not raise the optimum. The relaxed optimum is already attained by a multigraph with weights 0 and 1, so the scaling identity that carries one computed number to every m at once needs no integrality hypothesis attached to it.

Proof. For $M^*(n) \geq M(n)$, use either construction from Construction A.38 at multiplicity one. Both give $\{0, 1\}$ matrices in which every pairwise maximum flow is at most one. Their total weights are $2(n - 1)$ and $\lfloor n^2/4 \rfloor$, respectively.

For the reverse, note first that the optimum is attained at a *rational* matrix. A *rational polytope* is a bounded region described by finitely many linear inequalities with rational coefficients, and a *vertex* of that polytope is one of its corner points. A weight matrix is feasible exactly when, for each ordered pair, some cut separating that pair has capacity at most one, so the feasible region is the union, over the finitely many ways of choosing one cut per pair, of the rational polytopes cut out by those choices together with $0 \le w \le 1$. A linear objective over such a polytope attains its maximum at a vertex, which is rational, and $M^*(n)$ is the largest of finitely many such maxima.

So let w be a feasible rational optimum and let q be a common denominator of its entries. Then $q\,w$ has integer entries, and multiplying every weight by q multiplies every cut capacity by q and hence every maximum flow by q, exactly as in Lemma A.37. So $q\,w$ is a directed multigraph with $\lambda^{\max} \le q$, that is, feasible at $m-1=q$, and Theorem 1.13 bounds its arc count by $q\,M(n)$. Dividing by q gives $M^*(n) = \sum_{u \ne v} w(u,v) \le M(n)$.

The identity follows because Theorem 1.13 gives $L_m^{\mathrm{dir}}(n) = (m-1)M(n)$. □

This settles the integrality question Chapter 2 raises when it introduces $M^*(n)$, namely whether an optimal weight matrix can always be chosen with entries in $\{0,1\}$. It can, and the argument runs backwards from the usual direction. Nothing here inspects the fractional problem at all. An integral theorem, proved for every m at once, is strong enough to be applied to the scaled-up matrix $q\,w$, and the fractional statement falls out of the integral one rather than the other way round. That is only possible because Theorem 1.13 holds at every m, however large, so no denominator is out of reach.

Corollary A.42 (The $m=2$ case collapses to the simple digraph)**.** *At $m=2$ (so $k=1$), Theorem 1.13 reads $L_2^{\mathrm{dir}}(n) = M(n)$, matching Theorem 1.6 exactly, and this is not a coincidence. A directed multigraph with $\lambda^{\max} \le 1$ can carry no parallel arcs at all, since two parallel copies of one arc are already two arc-disjoint routes between their endpoints, so at $m=2$ the multigraph problem and the simple digraph problem are literally the same problem. Theorem 1.13 recovers Theorem 1.6 as its $k=1$ case, rather than needing it as a separate input.*

A.7.3 Extremal classification on the quadratic branch

Theorem 1.13 settles the value at every n and m, but a value is not a full description of the extremal digraphs. The question addressed here is which multigraphs attain it on the quadratic branch at a general n. The reachability-skeleton proof looks unpromising for this, since it is designed never to need to know which multigraph it is taking apart. But that indifference is only in the direction of the *bound*. Run backwards, from the assumption that the bound is met exactly, equality in the same construction imposes three restrictions: it pins down the number of strongly connected components, then the shape of the skeleton, and finally the multigraph.

Two consequences of equality do the work, and both come from reading the count of Lemma A.19 as an equation rather than an inequality.

Lemma A.43 (What equality forces, [AI])**.** *Let $m \ge 2$ and $n \ge 8$, so that $\lfloor n^2/4 \rfloor > 2(n-1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \le m-1$ and*

$|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. *Then*

(1) D *is acyclic, and*

(2) D *has a reachability skeleton* R*, inclusion-minimal for reachability in the sense that no arc of it is redundant, whose underlying undirected graph is exactly the balanced complete bipartite graph* $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$*, so* R *is an acyclic orientation of it.*

Proof. The proof of Theorem 1.13 peels reachability skeletons $R_1, \ldots, R_{m-1}$, each deletion dropping every positive pairwise connectivity by at least one, so that after $m-1$ peels no arc survives and $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \le M(n)$, so equality in the hypothesis forces $|A(R_i)| = M(n)$ for every i, and in particular for $R := R_1$.

Now read the count of Lemma A.19 backwards. It gives $|A(R)| \le f(q) := 2(n-q) + \lfloor q^2/4 \rfloor$, where q is the number of strongly connected components of D, so $f(q) \ge \lfloor n^2/4 \rfloor = f(n)$. The increments $f(q+1) - f(q) = -2 + \lfloor (q+1)/2 \rfloor$ are non-decreasing, so f is convex on $q \in \{1, \ldots, n\}$. Write $d_i := f(i+1) - f(i)$. Suppose toward a contradiction that $q < n$. If $q = 1$ then $f(q) = f(1) = 2(n-1)$, and the hypothesis $n \ge 8$ gives $\lfloor n^2/4 \rfloor > 2(n-1)$, so $f(q) < f(n)$, contradicting $f(q) \ge f(n)$. So $1 < q < n$. Since $f(1) < f(n) \le f(q)$, the sum $d_1 + \cdots + d_{q-1} = f(q) - f(1)$ is positive, and since the d_i are non-decreasing this forces the last term d_{q-1} to be positive too (otherwise every term up to it would be at most $d_{q-1} \le 0$ and the sum could not be positive). Non-decreasing again gives $d_q, \ldots, d_{n-1} \ge d_{q-1} > 0$, so $f(n) - f(q) = d_q + \cdots + d_{n-1} > 0$, that is $f(n) > f(q)$, contradicting $f(q) \ge f(n)$. Both cases are impossible, so $q = n$: every strongly connected component is a single vertex, which is (1).

With $q = n$ the two arborescences of Step 1 of that proof are empty, so R consists of the witness arcs of Q alone and $|A(Q)| = \lfloor n^2/4 \rfloor$. The underlying undirected graph of Q is triangle-free, proved there from the inclusion-minimality of Q. A triangle-free graph on n vertices with $\lfloor n^2/4 \rfloor$ edges meets Mantel's bound [26] with equality, and the extremal graph for Mantel's theorem is unique: it is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$. That is (2), and R is acyclic because D is. Its inclusion-minimality is the minimality of Q, the two objects coinciding once $q = n$ leaves the arborescences empty, and it is what Lemma A.44 uses next. □

The next lemma is where the skeleton's minimality, which had only been used to make it small, is used to make it *shallow*. This restricts the paths that can occur in an extremal skeleton.

Lemma A.44 (A minimal bipartite skeleton has no long path, [AI])**.** *Let* R *be as in Lemma A.43, with parts* A *and* B*. Then* R *contains no directed path with three arcs.*

Proof. Suppose $u \to v \to w \to x$ were such a path. A path in a bipartite graph alternates sides, so u and x lie on opposite sides, and since R's underlying graph is the *complete* bipartite graph, R contains an arc between them. It cannot be $x \to u$, which would close a directed cycle against the path, and D is acyclic. So $u \to x$ lies in R. But then deleting $u \to x$ from R changes no reachability, since u still reaches x along the path, contradicting the inclusion-minimality of Q. □

Theorem A.45 (Extremal classification on the quadratic branch, [Al]). *Let $m \geq 2$ and $n \geq \max(8,\, 2m)$, so that $M(n) = \lfloor n^2/4 \rfloor$ and n is on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = L_m^{\mathrm{dir}}(n) = (m-1)\lfloor n^2/4 \rfloor$, then*

$$D \;\in\; \left\{(m-1)\cdot B_{\lceil n/2\rceil,\,\lfloor n/2\rfloor},\; (m-1)\cdot B_{\lfloor n/2\rfloor,\,\lceil n/2\rceil}\right\},$$

where $B_{a,b}$ has all arcs from its part of size a to its part of size b, and every displayed arc has multiplicity exactly $m-1$. Thus there is one isomorphism class when n is even and two when n is odd, interchanged by reversing every arc.

In words: past $n = \max(8, 2m)$ the underlying bipartite shape is forced. Split the vertices as evenly as possible, choose either part as the source, run every arc from it to the other part, and repeat each arc $m-1$ times. When n is odd the two choices give two non-isomorphic digraphs, because the source has different size, and they are reversals of one another.

Proof. Take R, A and B from Lemma A.43, and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose $a \to b \to a'$ with $a, a' \in A$ and $b \in B$. If a had an incoming arc $b_0 \to a$ in R, then $b_0 \to a \to b \to a'$ would be a path with three arcs, which Lemma A.44 forbids. So a has no incoming arc in R, and since R's underlying graph is complete bipartite, $a \to b''$ for *every* $b'' \in B$. Symmetrically a' has no outgoing arc, so $b'' \to a'$ for every $b'' \in B$. The β routes $a \to b'' \to a'$, one through each $b'' \in B$, use pairwise distinct arcs, and each of their arcs is present in D because $R \subseteq D$. Hence $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, against feasibility. The case of a path $b \to a \to b'$ with middle vertex in A is identical with the sides exchanged and gives $\lambda_D(b, b') \geq \alpha \geq \beta \geq m$, equally impossible. This is the one step that uses $n \geq 2m$.

Step 2: R is one-directional. A *source* has no incoming arc and a *sink* has no outgoing arc. By Step 1 no vertex of R has both an incoming and an outgoing arc, so every vertex is a source or a sink. Fix $a \in A$. It is adjacent in R to every vertex of B, so it is not isolated. If a is a source then every $b \in B$ has an incoming arc, hence is a sink, hence has no outgoing arc, so every arc at every b runs into it: all arcs of R run from A to B. If instead a is a sink the same argument gives all arcs from B to A. Thus R is either $B_{\alpha,\beta}$ or $B_{\beta,\alpha}$, according to which part is the source. The remaining steps apply identically to either orientation, so from now on use A for the source part and B for the sink part without imposing an order on their sizes.

Step 3: D has no arc outside R. R has the same reachability as D, and in a one-directional bipartite digraph the only ordered pairs joined by a directed path are the pairs $(a, b) \in A \times B$, each by the single arc $a \to b$. So if D had an arc (u, x), then u reaches x in R, forcing $(u, x) \in A \times B$, and R already contains that arc. Hence D and R use exactly the same $\alpha\beta = \lfloor n^2/4 \rfloor$ ordered pairs.

Step 4: every multiplicity is $m-1$. A pair joined by $\mu(a, b)$ parallel arcs has $\lambda_D(a, b) \geq \mu(a, b)$, so feasibility gives $\mu(a, b) \leq m-1$ on each of the $\lfloor n^2/4 \rfloor$ pairs. Their total is $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$, so every one of them attains the cap. Hence D is the multiplicity-$(m-1)$ one-directional complete bipartite digraph on these two parts, and their product $\lfloor n^2/4 \rfloor$ forces their sizes to be $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. Either size may be the source size, giving exactly the two displayed possibilities (which coincide up to isomorphism when n is even). $\square$

Three remarks. First, on where the hypothesis $n \geq 2m$ is used and what it costs. It enters once, in Step 1, where a two-arc path in the skeleton is converted into $\lfloor n/2 \rfloor$ arc-disjoint routes between one pair, and feasibility kills that only when $\lfloor n/2 \rfloor$ reaches m. For $n < 2m$ the theorem is not proved, and it is not known to fail either. Nothing in the argument suggests a boundary there: the step is a lower bound on one pair's connectivity, and losing it removes a contradiction rather than supplying a counterexample. Second, a degree fact falls out of the same equality, and it is what makes a direct search for extremisers finite in practice. Deleting a vertex leaves at most $(m-1)M(n-1)$ arcs, so every vertex of an extremiser carries degree at least $(m-1)\lfloor n/2 \rfloor$. For even n the degree sum $2(m-1)\lfloor n^2/4 \rfloor = n \cdot (m-1)(n/2)$ leaves no slack at all, so every vertex has degree exactly $(m-1)n/2$: an extremiser at even n is regular. Third, on the shape of the argument. Theorem 1.13 is indifferent to which multigraph it dismantles, and Chapter 3 records that indifference as the reason the value says nothing about uniqueness. That is true of the proof read forwards. Read backwards from equality, the same skeleton is a rigid object, and the two facts that had been mere bookkeeping, that f is maximised at one end and that a minimal skeleton is triangle-free, become a determination of q and an application of the equality case of Mantel's theorem.

Part III. The hypergraph results

The edge bound through an elementary cut induction, the vertex problem at $m = 2$ and $m = 3$ by way of an incidence rank bound, and the directed hypergraph leading term.

A.8 The hypergraph bounds

Connectivity in a hypergraph admits several inequivalent definitions, surveyed with their cut algorithms by Chekuri and Xu [28] and by Dewar, Pike and Proos [29]. This thesis fixes on the Berge notion throughout, which is the one the checker of Chapter 2 measures and the one the bounds below are stated for.

The body's hypergraph claims first need a Menger theorem for Berge routes, which also certifies the helper-point checker of Chapter 2. Its cut formulation then supplies the separator used by the elementary induction for the edge bound.

Theorem A.46 (Menger for multihypergraphs, [AI]). *For a multihypergraph $\mathcal{H}$ and distinct vertices u, v, regard parallel hyperedge copies as distinct. The maximum number of hyperedge-copy-disjoint Berge u–v paths equals the minimum number of individual copies whose deletion separates u from v. Equivalently, if a cut C is specified as a set of distinct hyperedge types,* *its cost is $\sum_{e\in C}\mu(e)$, because every copy of each selected type must be deleted. A simple hypergraph is the case where every multiplicity is one.*

Proof. Build the helper-point network of Figure 2.3b: for each distinct hyperedge e a helper arc $e^- \to e^+$ whose capacity is the multiplicity $\mu(e)$, which is one for a simple hypergraph, and arcs $x \to e^-$ and $e^+ \to x$ of unbounded capacity for each member $x \in e$. The proof matches the number of disjoint Berge paths to the size of a minimum cut by translating between Berge-path language and flow language in each direction: a family of hyperedge-copy-disjoint Berge paths becomes a feasible integral flow, and conversely such a flow becomes a family of Berge paths, as the next two paragraphs spell out in turn.

Berge paths become flow. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the directed walk $u \to e_1^- \to e_1^+ \to v_1 \to \cdots \to v$, crossing one helper arc per hyperedge it uses, and carrying one unit of flow. The e_i along a single Berge path are distinct, and a hyperedge-disjoint family rides no copy twice, so at most $\mu(e)$ of its paths cross the helper arc of e. Each helper arc therefore carries at most its own capacity and the family is a feasible flow whose value is its number of paths. At multiplicity greater than one these paths need not be pairwise arc-disjoint, and the argument never needs them to be.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k decomposes into k unit paths. Reading a helper crossing $\cdots x \to e^- \to e^+ \to y \cdots$ back as the single hyperedge step x, e, y, that is, *suppressing* the two helper nodes, turns each unit path into a Berge path. At most $\mu(e)$ of them cross the helper arc of e, since that is its capacity, so each may be given a copy of e of its own and no copy serves two routes. The k Berge paths are therefore hyperedge-disjoint.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedge types separate them in $\mathcal{H}$. Its capacity is $\sum \mu(e)$ over those types, which is the number of copies removed. Conversely, deleting only some copies of a type while leaving another copy changes no reachability, so a minimum copy cut may be taken to delete either all copies of a type or none. Thus network-cut capacity and the minimum number of deleted copies are the same count. The integral max-flow/min-cut theorem equates the largest feasible integral flow value with this smallest cut capacity, which is precisely the claim. □

Lemma A.47 (Cut induction for multihypergraphs, [AI]). *Let $\mathcal{H}$ be a multihypergraph on $n \geq 1$ vertices, not assumed uniform, with $\lambda_{\mathcal{H}}^{\max} \leq k$. Then, the sum running over hyperedge copies,*

$$\sum_e \bigl(|e| - 1\bigr) \;\leq\; k\,(n-1).$$

Proof. Induction on n. At $n = 1$ every hyperedge is a single vertex and both sides are 0.

Let $n \geq 2$ and fix any two vertices $u \neq v$. By Theorem A.46 some set F of at most $\lambda_{\mathcal{H}}(u,v) \leq k$ hyperedge copies separates them. Let S be the set of vertices reachable from u in $\mathcal{H} - F$ and write $\bar{S} = V \setminus S$, so $u \in S$, $v \in \bar{S}$, and every copy meeting both sides lies in F. Both sides are non-empty and at most k copies cross.

Form the two *trace* multihypergraphs $\mathcal{H}[S]$ and $\mathcal{H}[\bar{S}]$ by replacing every copy e by $e \cap S$, respectively $e \cap \bar{S}$, and discarding the intersections of size 0 or 1. A Berge route of a trace lifts, copy by copy, to a Berge route of $\mathcal{H}$ on the corresponding original copies, and distinct trace copies come from distinct original copies, so a hyperedge-disjoint family in a trace lifts to one of the same size in $\mathcal{H}$. Hence $\lambda^{\max} \leq k$ on both traces and the induction hypothesis applies to each.

Now split the copies of $\mathcal{H}$. A copy lying wholly inside one side contributes its whole $|e| - 1$ to that side's trace sum. A crossing copy splits as

$$|e| - 1 \;=\; \big(|e \cap S| - 1\big) \;+\; \big(|e \cap \bar{S}| - 1\big) \;+\; 1,$$

each bracket being the contribution its trace makes, and zero exactly when that trace was discarded. Summing over all copies,

$$\sum_e \big(|e| - 1\big) \;\leq\; k\big(|S| - 1\big) + k\big(|\bar{S}| - 1\big) + \big|\{\text{crossing copies}\}\big| \;\leq\; k(n-2) + k \;=\; k(n-1). \qquad \square$$

For a hypergraph, its *incidence graph* is the bipartite graph joining each vertex to the hyperedges that contain it. A *hypertree* is a hypergraph whose incidence graph is a tree.

Proof of Proposition 1.8. Upper bound. Every hyperedge copy has $|e| = r$, so Lemma A.47 at $k = m-1$ reads $(r-1)|E(\mathcal{H})| \leq (m-1)(n-1)$. Dividing and rounding down, because $|E(\mathcal{H})|$ is an integer, gives the stated floor.

Lower bound. When $(r-1) \mid (n-1)$, take a hub h and split the other $n-1$ vertices into groups $G_1, G_2, \ldots$ of size $r-1$ each. Form the hypergraph whose hyperedges are $\{h\} \cup G_i$, one per group. Its incidence graph is a tree, so this shape is called a *star hypertree*. Give each hyperedge multiplicity $m-1$. Every non-hub vertex lies in exactly $m-1$ hyperedges, which form a cut isolating it, so by Theorem A.46 every pair has Berge connectivity at most $m-1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m - 1 \leq \binom{n-2}{r-2}$, Theorem A.48 below attains the floor exactly, without even needing repeated hyperedges. That theorem assumes $r \geq 3$, which leaves only $r = 2$, and there the hypothesis reads $m - 1 \leq \binom{n-2}{0} = 1$, forcing $m = 2$. The floor is then $n - 1$ and a spanning tree attains it, with no repeated edge and every local edge-connectivity equal to one. $\square$

The multiplicities in the lower bound can in fact be avoided throughout the range of the next theorem: for $r \geq 3$ and $m - 1 \leq \binom{n-2}{r-2}$, *simple* r-uniform hypergraphs already attain the same bound. This is in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

The *rank* of a uniform hypergraph is its common hyperedge size. This use is separate from the cycle rank of a graph introduced later.

Theorem A.48 (Simplicity is free for $r \geq 3$, [AI]). *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m-1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a* simple *r-uniform hypergraph on n vertices with $\lambda^{\max} \leq m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is Proposition 1.8, since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By Lemma A.51 applied with rank $r-1$ and degree cap $d = m-1$ on the $n-1$ vertices of V', there is an $(r-1)$-uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r-sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m-1$ hyperedges, which form a cut isolating w. By Theorem A.46, $\lambda^{\max}_{\mathcal{H}} \leq m-1$. □

Remark A.49 (The hypothesis is not decorative). Outside this range the displayed value can fail for simple hypergraphs, even though it stays a valid upper bound (Proposition 1.8) and, when $(r-1) \mid (n-1)$, stays exactly attained by a multihypergraph. The smallest case already shows it. At $n = r = m = 3$ there is only one 3-subset of a 3-element vertex set, so a simple hypergraph carries at most one hyperedge. The hypothesis needs $m-1 \leq \binom{n-2}{r-2} = \binom{1}{1} = 1$, and $m-1 = 2$ misses it, so the formula's value $\lfloor 2 \cdot 2/2 \rfloor = 2$ is not attained by any simple hypergraph there. It is attained by the multihypergraph star, two parallel copies of that one triple. Theorem 1.10 carries the same hypothesis at $m-1 = 2$, where it reduces to $2 < r < n$, for the same reason. Table 3.1 splits the simple and multi hypergraph rows on exactly this distinction, and the program enumerates the two models separately for the same reason, stated where each formula is used.

Construction A.50 (Twelve hyperedge copies on six vertices, [AI]). Take a hub h and five other vertices a, b, c, d, e. Include all ten triples consisting of h and two of the other vertices. Add a second copy of $\{h, d, e\}$ and one copy of $\{a, b, c\}$. The resulting 3-uniform multihypergraph has 12 hyperedge copies and $\lambda^{\max} \leq 5$. Consequently, the multihypergraph edge value at $(n, m, r) = (6, 6, 3)$ is exactly 12, even though $(r-1)$ does not divide $(n-1)$.

Proof. Initially each non-hub vertex lies in four of the ten triples. The repeated triple raises the degrees of d and e to five, while the final triple does the same for a, b and c. Every pair of distinct vertices therefore includes a non-hub vertex whose five incident hyperedge copies form a cut separating it from the other endpoint. Thus no pair has more than five hyperedge-disjoint routes. The construction has $10+1+1 = 12$ copies, matching the upper bound $\lfloor 5(6-1)/(3-1) \rfloor = 12$ from Proposition 1.8. □

The construction was derived after a longer unaided search found a twelve-copy witness in this cell. Its optimality follows from the degree cuts and the proved upper bound, not from an exhaustive search. In the vertex problem it supplies a lower bound of 12 by Whitney's inequality, but the edge upper bound does not prove vertex optimality.

The sparse hypergraph used in the proof of Theorem A.48 follows from a direct degree-smoothing argument.

Lemma A.51 (Sparse uniform hypergraphs exist)**.** *Let r, n and d be integers with $r \geq 1$, $n \geq r$ and $0 \leq d \leq \binom{n-1}{r-1}$. There is a simple r-uniform hypergraph on n vertices with exactly $\lfloor \frac{dn}{r} \rfloor$ hyperedges and maximum degree at most d.*

Proof. Put $e := \lfloor \frac{dn}{r} \rfloor \leq \frac{n}{r}\binom{n-1}{r-1} = \binom{n}{r}$, so e distinct r-sets are available. Among all families of exactly e distinct r-sets choose one minimising $\sum_x \deg(x)^2$. Suppose two vertices had $\deg(x) \geq \deg(y) + 2$. Write a for the number of family edges containing x but not y and b for the number containing y but not x, so $a \geq b + 2$. The injection $A \mapsto A - x + y$ sends the first kind into the second, so if every image were present we would have $b \geq a$. Hence some A in the family has $A - x + y$ absent, and exchanging them keeps the family simple and of size e while changing the objective by

$$(\deg(x) - 1)^2 + (\deg(y) + 1)^2 - \deg(x)^2 - \deg(y)^2 = 2\big(\deg(y) - \deg(x)\big) + 2 \leq -2,$$

contradicting minimality. So all degrees differ by at most one, and since they sum to re the largest is $\lceil re/n \rceil \leq d$, the last step because $re \leq dn$ and d is an integer. $\square$

A.9 The hypergraph vertex problem

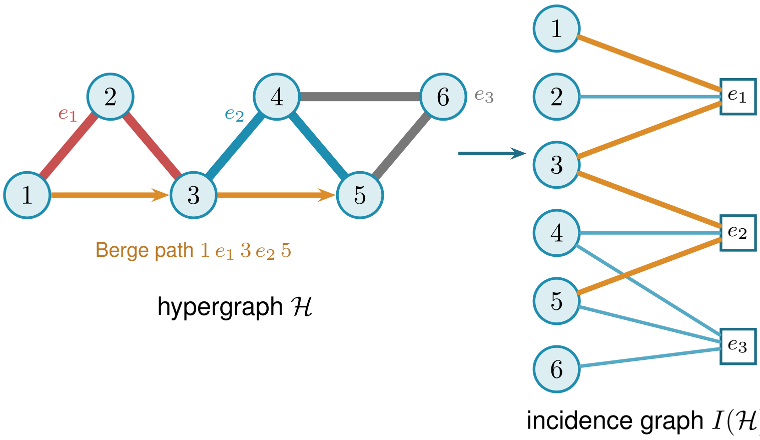
The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which Figure A.3 draws in full. Write $I(\mathcal{H})$ for the bipartite *incidence graph* of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That hybrid is what $\kappa_{\mathcal{H}}$ means throughout, and both halves are needed: dropping the hyperedge half would let two routes ride the same hyperedge, which a single Berge edge cannot serve twice. Hence

$$\kappa_{\mathcal{H}}(u, v) \;=\; \kappa_{I(\mathcal{H})}(u, v),$$

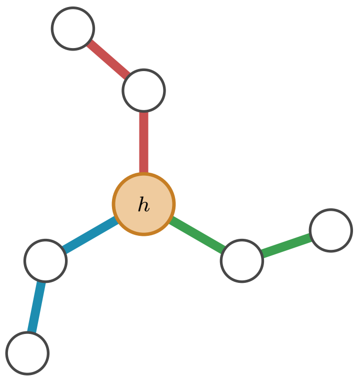
the maximum number of internally disjoint u–v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree-r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. Figure A.3a traces a single Berge path through both pictures, and Figure A.3b shows the star hypertree of Theorem 1.9, whose incidence graph is a tree.

Theorem 1.9, stated in Chapter 1, follows at once from the incidence translation.

Proof of Theorem 1.9. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \ldots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \ldots, e_2, v_2$ are two Berge v_1–v_2 paths with disjoint hyperedge sets and disjoint interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1\, e_1\, 3\, e_2\, 5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2, changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1{-}e_1{-}3{-}e_2{-}5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.3. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u,v) = \kappa_{I(\mathcal{H})}(u,v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem 1.9, whose incidence graph is a tree.

hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n+q$ nodes has at most $n+q-1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \le n+q-1$, i.e. $q \le (n-1)/(r-1)$, and q is an integer. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges and a forest as incidence graph (one tree together with any unused isolated vertices) because each non-hub vertex used in the construction has incidence degree one, while all hyperedges meet only at the hub. Thus no incidence cycle exists and the floor is attained. ☐

Proposition A.52 (General lower bound)**.** *For $r \ge 3$, $m \ge 2$ and $m-1 \le \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \;\ge\; \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of Theorem A.48 has Berge edge-connectivity at most $m-1$, and Whitney's inequality $\kappa \le \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. ☐

The simple construction uses distinct $(r-1)$-sets on the non-hub vertices. The condition $m-1 \le \binom{n-2}{r-2}$ bounds the requested degree cap by the number of such sets containing a fixed vertex. A multihypergraph does not, because it may repeat one hyperedge instead, and that closes a different range of cases.

Proposition A.53 (The multihypergraph lower bound)**.** *For $r \ge 2$, $m \ge 2$ and $(r-1) \mid (n-1)$,*

$$K_m^{(r)}(n) \;\ge\; \frac{(m-1)(n-1)}{r-1},$$

attained by the star hypertree of Figure 1.8 with every hyperedge taken at multiplicity $m-1$. At $m=3$ this meets the upper bound of Theorem 1.10, so

$$K_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \qquad \textit{whenever } (r-1) \mid (n-1).$$

For $r \ge 3$, that range neither contains nor is contained in the simple model's $2 < r < n$, so the two models close different cases.

Proof. Take the hub h, split the other $n-1$ vertices into blocks $G_1, \ldots, G_q$ of size $r-1$ with $q = (n-1)/(r-1)$, and give each hyperedge $\{h\} \cup G_i$ multiplicity $m-1$. The count is $(m-1)q$, which is the displayed value.

For the connectivity, take $u \ne w$. If both lie in one block G_i, or if one of them is h and the other lies in G_i, the $m-1$ copies of $\{h\} \cup G_i$ are $m-1$ one-step Berge routes between them. A one-step route has no interior vertex, so no two of them share one, and they ride different copies, so the family is internally vertex-disjoint and $\kappa(u,w) \ge m-1$. Those same $m-1$ copies are also every hyperedge containing the block vertex among u, w, so deleting them separates the pair and Theorem A.46 caps $\kappa(u, w)$ at $m - 1$.

If $u \in G_i$ and $w \in G_j$ with $i \neq j$, then two different blocks meet only at h, so every u–w Berge route has h in its interior. Two internally vertex-disjoint routes cannot both use h there, so $\kappa(u, w) \leq 1$. Hence $\kappa^{\max} = m - 1$ and the hypergraph is feasible.

At $m = 3$ the count is $2(n-1)/(r-1)$, which is $\lfloor 2(n-1)/(r-1) \rfloor$ under the divisibility hypothesis, and Theorem 1.10 bounds every multihypergraph by that same quantity. □

At $m = 3$ the upper-bound side closes completely, and equality follows in the construction range stated below. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\operatorname{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from Chapter 1, so we fix the words once. A graph on more than k vertices is *k-connected* if it stays connected after the removal of any $k - 1$ vertices. By Menger in its global form [30, Ch. 3] this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so k-connected means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do.

Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph: for a 2-connected graph on at least four vertices, it is the obstruction to being 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own. In a multigraph it is a single bridge edge, a two-vertex bundle of parallel edges, or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.54 (Incidence rank lemma, [AI]). *Let G be a connected loopless multigraph with a partition $V(G) = X \,\dot\cup\, Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then* $\operatorname{rank}(G) \leq |X| - 1$.

In words: this is the engine behind the $m = 3$ hypergraph result, stated for a multigraph with an independent class Z, allowing edges inside X so that suppression preserves the hypotheses.

Read X as the vertices of a hypergraph and Z as its hyperedges, so that G is the incidence graph and $\operatorname{rank}(G)$, the number of independent cycles it carries, is the quantity the hyperedge count is eventually read off from. The four hypotheses say in turn that no two hyperedge-class nodes are adjacent (i), that no hyperedge repeats a vertex (ii), that no hyperedge dangles as a leaf (iii), and that the hypergraph is feasible at $m = 3$ (iv). The conclusion caps the independent cycles by the number of vertices.

Proof. Induction on $|V| + |E|$. Step 1 handles a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z-vertex or X-vertex without changing the bound. What survives is 2-connected with minimum degree at least three. Step 4 shows that it is simple, excludes a 3-connected core, then splits at a separating pair and applies induction to two smaller torsos. *Base* $|V| \leq 2$. A single vertex has degree 0, too little for the degree-at-least-2 floor (iii) imposes on a Z-vertex, so it cannot lie in Z and must lie in X, giving $0 \leq |X| - 1$. On two vertices the same reasoning applies to either one: a vertex in Z could reach only the other vertex, and (ii) forbids that single edge from being parallel to another, so it would have degree at most 1, again short of (iii). Hence (i)–(iii) force both vertices into X. Parallel edges between them are internally disjoint paths (each has empty interior, so no two can share an interior vertex), so (iv) caps the multiplicity at 2 and $\operatorname{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \ldots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, a bridge edge, a two-vertex parallel bundle, or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G. No cycle can pass through two different blocks: the block-cut tree forces any excursion out of a block to return through the same cut vertex, which would repeat that vertex on the cycle. So the cycle rank adds up over blocks, $\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i)$, and it is enough to bound each $\operatorname{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Non-bridge blocks inherit the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X, giving $0 \leq |X \cap B| - 1$. Every other block has minimum degree at least 2, including the possible two-vertex parallel bundle, so (i) to (iv) hold inside it and the induction hypothesis gives $\operatorname{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v, so $b(v) = 1$ except at cut vertices. Counting each X-vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$. This follows from counting the edges of the block-cut tree two ways. That tree has c block-nodes and one node per cut vertex, so being a tree its edge count is one less than its total number of nodes, $c + \#\{\text{cut vertices}\} - 1$. The same edge count also equals $\sum_{v \text{ cut}} b(v)$, since each cut vertex v

is joined in the tree to exactly the $b(v)$ blocks that contain it. Equating the two expressions for the edge count and solving for c gives the identity. Subtracting this count of blocks from the previous display and separating the cut vertices into their X- and Z-parts leaves

$$\operatorname{rank}(G)=\sum_i \operatorname{rank}(B_i)\le\sum_i(|X\cap B_i|-1)=|X|-1-\sum_{z\in Z\text{ cut}}(b(z)-1)\le |X|-1,$$

where the last inequality holds because each $b(z)-1\ge 0$, so the Z-cut sum only pushes the bound further below $|X|-1$.

Step 2: a Z-vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0\in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z-vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\operatorname{rank}=|E|-|V|+1$ is unchanged. The new edge runs X to X, so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z-vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1,x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G.

Step 3: an X-vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z-vertex has degree at least 3. Suppose some $x_0\in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X|\ge 2$ to begin with, for if $|X|\le 1$ then by (i) and (ii) every z would send all its edges to one X-vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\operatorname{rank}(G)-1\le(|X|-1)-1$, that is $\operatorname{rank}(G)\le|X|-1$.

Step 4: the remaining core. Now G is 2-connected with $|V|\ge 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x,x')\ge 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G-x'$, then from w to x' inside $G-x$, both walks available because deleting a single vertex from a 2-connected graph leaves it connected, exactly as already used in Step 3. The concatenation visits x only first and x' only last, so it contains an x–x' path with at least one interior vertex, giving $\kappa(x,x')\ge 3$, against (iv). Thus G is simple, and its minimum degree at least three forces $|V|\ge 4$. If G is 3-connected, Menger puts every pair at $\kappa\ge 3$, so two vertices of X would violate the cap $\kappa(x,x')\le 2$ that (iv) imposes on every pair drawn from X, forcing $|X|\le 1$, and then every z has degree at most 1 by (i), absurd.

Otherwise G has a separating pair $\{a,b\}$. Delete it and gather the remaining components into two non-empty groups with vertex sets U and W. Since G is 2-connected, each of a and b has a neighbour in U and one in W, and each side carries an a–b path. Write $s=|X\cap\{a,b\}|$ and build two *torsos* G_U and G_W, where G_U is the subgraph induced on $U\cup\{a,b\}$ carrying one temporary a–b connection, and G_W likewise:

- if $ab \in E(G)$, retain that edge on both sides.
- if $ab \notin E(G)$ and $s \geq 1$, add one new edge ab to each side.
- if $s = 0$, add to each side a fresh X-vertex together with the two edges joining it to a and to b.

The three cases are exhaustive, since Z is independent by (i), so $ab \in E(G)$ already forces $s \geq 1$. In that case $s = 1$ exactly: two endpoints in X would give $\kappa(a, b) \geq 3$, counting the edge together with one a–b path through each side, against (iv).

Each torso satisfies (i) to (iv). Condition (i) holds because every temporary edge has at least one endpoint in X, a fresh vertex in the third case. Condition (ii) holds because a temporary edge joins a pair that carried no edge at all. For (iii), a Z-vertex inside U keeps all its neighbours, and a separator vertex in Z keeps at least one neighbour on its own side and gains the temporary connection. For (iv), a fresh X-vertex has degree two, so its local connectivity is at most two, and for two old X-vertices any internally disjoint family uses the temporary connection at most once, since internally disjoint x–x' paths can share no edge but xx' itself. Replacing that one path's temporary connection by an a–b path through the opposite side turns three internally disjoint paths in a torso into three in G, which (iv) forbids. Both torsos are smaller in $|V| + |E|$. Neither side can consist of a single vertex: in this simple graph it would have only a and b as possible neighbours, contradicting minimum degree three. Forming a torso therefore removes at least two vertices and at least three incident edges, while adding at most one vertex and two edges. This justifies the induction even in the fresh-vertex case.

The bookkeeping. Write r_U, r_W for the two ranks and X_U, X_W for the two X-sets. Direct calculation from $\mathrm{rank} = |E| - |V| + 1$ gives $\mathrm{rank}(G) = r_U + r_W$ and $|X_U| + |X_W| = |X| + 1$ in the first case, where the retained edge is counted twice, and $\mathrm{rank}(G) = r_U + r_W - 1$ with $|X_U| + |X_W| = |X| + s$ in the second and $|X| + 2$ in the third. Feeding in the two inductive bounds $r_U \leq |X_U| - 1$ and $r_W \leq |X_W| - 1$ gives $\mathrm{rank}(G) \leq |X| - 1$ in the first and third cases, and $\mathrm{rank}(G) \leq |X| + s - 3 \leq |X| - 1$ in the second, since $s \leq 2$. This exhausts the cases. □

The rank bound now yields Theorem 1.10, stated in Chapter 1. In words: at the route limit $m = 3$ a hyperedge network can hold no more hyperedges under the stricter separation than the displayed edge bound. When $2 < r < n$, that count is reached without repeating a hyperedge, so the two separations agree in this range exactly as they did at $m = 2$.

Proof of Theorem 1.10. Apply Lemma A.54 to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X-node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum $\mathrm{rank} \leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r incidences) and $n+q$ nodes, so $\sum_c \mathrm{rank} = rq - (n+q) + C$, while $\sum_c (|X_c| - 1) =$

$n-C$. The inequality reads $rq-(n+q)+C \le n-C$, hence $q(r-1) \le 2(n-1)+2(1-C) \le 2(n-1)$, the last step using $C \ge 1$. The lower bound is Proposition A.52 at $m=3$. $\square$

Remark A.55 (Edges as gates at $r=2$, and the boundary of the method). At $r=2$ the theorem speaks of multigraphs, where the incidence convention of Section 1.3.1 makes q parallel edges q distinct one-step routes. It gives $K_3(n) = 2(n-1)$, the same number as the multigraph edge value of Theorem 1.3, and Corollary A.59 below records that specialisation. The proof of Lemma A.54 is calibrated to the cap $\kappa \le 2$ at every step rather than at one. Two parallel edges beside a single detour already exceed it, which is what forces the surviving core to be simple, and three internally disjoint paths exceed it, which is what rules out a 3-connected core. The conclusion is the cap-two number too: at $\kappa \le 3$ it is already false, since two X-vertices joined by three parallel edges have cycle rank two against $|X|-1=1$. For $m=4$ the analogous question asks for a different bound relative to $\kappa \le 3$, taken apart along 3-cuts rather than along separating pairs, and none is supplied here.

A.10 The multigraph incidence problems

The theorem above is about hyperedge networks, and at $r=2$ a hyperedge network is a multigraph, so the two multigraph vertex questions are the $r=2$ case of the two hypergraph vertex questions and not restatements of their simple counterparts. This section works them out. Throughout, q parallel copies of the edge uv count as q internally disjoint u-v routes, since a single edge is a path with no interior, which is the incidence convention of Section 1.3.1. Write $K_m(n)$ for the largest total multiplicity of a multigraph on n vertices with $\kappa^{\max} \le m-1$, as in Table 1.1.

The two kinds of route never interfere, which splits the measure in two.

Lemma A.56 (Splitting the vertex measure)**.** *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0. For any two vertices $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint u-v paths of length at least two in G_0. Then*

$$\kappa_G(u,v) \;=\; \mu(u,v) \;+\; \pi(u,v).$$

Proof. A route of length one is a single copy of the edge uv, and its interior is empty, so it is internally disjoint from every other route whatever the length. We prove the two inequalities separately.

Lower bound. Take all $\mu(u,v)$ copies of uv, and take a maximum family P of pairwise internally disjoint u-v paths of length at least two in G_0, so $|P| = \pi(u,v)$. Lift each path of P into G by choosing one copy of each of its edges, which is possible because every edge of G_0 carries multiplicity at least one. Lifting leaves interior vertices unchanged, so the lifted routes are pairwise internally disjoint, and they are internally disjoint from the copies of uv because those have empty interior. That is a family of $\mu(u,v)+\pi(u,v)$ pairwise internally disjoint routes, so $\kappa_G(u,v) \ge \mu(u,v)+\pi(u,v)$.

Upper bound. Let F be any family of pairwise internally disjoint u-v routes in G, and split it into the routes F_1 of length one and the routes F_2 of length at least two. The members of F_1 are distinct copies of the edge uv and there are only $\mu(u, v)$ such copies, so $|F_1| \leq \mu(u, v)$. Send each route of F_2 to its vertex sequence, a u-v path of G_0 of length at least two. That map is injective on F_2: two routes with the same vertex sequence have the same interior, which is non-empty because the length is at least two, and F is internally disjoint, so no two such routes can both belong to it. Distinct members of F_2 have disjoint interiors, so their images are pairwise internally disjoint paths of G_0 of length at least two, giving $|F_2| \leq \pi(u, v)$. Hence $|F| \leq \mu(u, v) + \pi(u, v)$, and taking F maximum gives $\kappa_G(u, v) \leq \mu(u, v) + \pi(u, v)$.

The two bounds give the equality. □

So feasibility says exactly that $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge, and that $\pi(u, v) \leq m - 1$ on every pair. At $m = 2$ this is already decisive.

Theorem A.57 (The multigraph incidence value at $m = 2$)**.** $K_2(n) = n - 1$*, attained exactly by the spanning trees.*

Proof. By Lemma A.56, $\kappa^{\max} \leq 1$ forces $\mu(u, v) + \pi(u, v) \leq 1$ on every edge. If G_0 contained a cycle, an edge uv of that cycle would have $\mu(u, v) \geq 1$ and $\pi(u, v) \geq 1$ from the rest of the cycle, a contradiction, so G_0 is a forest and every multiplicity is one. A forest has at most $n - 1$ edges, and a spanning tree attains it. □

Remark A.58 (Why $m = 2$ is not a multigraph question at all)*.* The proof shows more than the count. At $m = 2$ the cap $m - 1$ is one, so a second copy of any edge is a second route on its own, and every feasible multigraph is simple. The multigraph problem is therefore the simple problem, $K_2(n) = k_2(n) = n - 1$, which is also what Theorem 1.9 says at $r = 2$. This is the one value of m at which the two coincide for a reason rather than by accident, and Corollary A.66 makes the same observation in the directed model.

The next value is the one the hypergraph theorem supplies, and no separate argument is needed for it.

Corollary A.59 (The multigraph incidence value at $m = 3$, [AI])**.** $K_3(n) = 2(n - 1)$ *for every* $n \geq 2$*.*

Proof. The upper bound is Theorem 1.10 at $r = 2$, where the displayed bound $q(r-1) \leq 2(n-1)$ reads $q \leq 2(n - 1)$ and q is the total multiplicity. Its proof asks only that hyperedge-nodes have degree $r \geq 2$, which holds at $r = 2$. For the lower bound take a spanning tree and give every edge multiplicity two. A tree edge uv has $\pi(u, v) = 0$, so $\kappa(u, v) = 2$ by Lemma A.56, and a non-adjacent pair is joined by the unique tree path, so $\kappa = 1$. Nothing exceeds $2 = m - 1$, and the count is $2(n - 1)$. □

At $m = 3$ this is the multigraph *edge* value $L_3(n)$ of Theorem 1.3 as well. The first threshold at which the two values are known to differ is $m = 5$: $K_5(4) = 14$ against $L_5(4) = 12$, and

$K_5(5) = 19$ against $L_5(5) = 16$, both by exhaustion. At $m = 4$ they agree for every $n \leq 8$, since $g_4(b) = 3(b-1)$ throughout the range of Table A.1, and whether they agree at every n is open. For $m \geq 3$ at least three constructions compete, and which one wins depends on how n compares with m, exactly as in every other variant of this thesis.

- *The thickened tree.* Take a spanning tree and give every edge multiplicity $m-1$. Tree edges have $\pi = 0$ and non-adjacent pairs have $\pi = 1$, so this is feasible for every $m \geq 2$, and it carries $(m-1)(n-1)$ edges, the multigraph *edge* value of Theorem 1.3.
- *The thickened complete graph.* Give every edge of K_n the same multiplicity q. Here $\pi(u,v) = n-2$, so feasibility asks $q \leq m+1-n$, and the count is $\binom{n}{2}(m+1-n)$, positive only when $n \leq m$. At $n = m+1$ the multiplicity is zero and the construction is empty.
- *The thickened theta.* Choose two *poles* and join each of the $n-2$ remaining vertices to both of them at multiplicity $m-2$, then join the poles to each other at multiplicity $m+1-n$ when that is positive. A pole and a middle vertex have $\pi = 1$, two middle vertices have $\pi = 2$, and the two poles have $\pi = n-2$, so this is feasible exactly when $n \leq m+1$, and it carries $2(n-2)(m-2) + \max(0,\, m+1-n)$ edges.

The theta family is what makes this problem different from the edge problem, and it was found by the search rather than by hand. Whenever it is feasible, which is to say for $n \leq m+1$, it matches the thickened tree exactly at $m = 4$ and, for $n \geq 3$, beats it for every $m \geq 5$. It accounts for the first cells in which the direct exhaustion exceeds the tree, although the later block sweep finds still better blocks in some larger parameter cells. Comparing these three constructions, the tree leads once $n \geq m+2$, where the other two die for lack of route budget.

An exhaustive include-or-exclude search, independently checked by a separate implementation on the cells where both completed, confirms these constructions at the small sizes used below. The statements below are structural rather than cell-by-cell.

The small-n regime $n \leq m+1$, where the theta construction is feasible, is tempting to read as the *only* place a cycle pays for itself, with the thickened tree optimal once $n \geq m+2$. It is not: a single triangle grafted anywhere onto the thickened tree already beats it, for every $m \geq 5$ and every such n, which the next theorem makes precise.

By Lemma A.56, a feasible multigraph on top of a fixed connected simple graph G_0 maximises its total multiplicity by setting $\mu(u,v) = m-1-\pi(u,v)$ on every edge, giving total $(m-1)\,|E(G_0)| - \sum_{uv \in E(G_0)} \pi(u,v)$. Writing $|E(G_0)| = n-1+\operatorname{rank}(G_0)$, the thickened tree is beaten by any G_0 with

$$\sum_{uv \in E(G_0)} \pi(u,v) \;<\; (m-1)\operatorname{rank}(G_0).$$

The obvious guess is that cycles must always pay for themselves against this bound: each independent cycle costs $m-1$ units of route capacity. The next theorem shows a single triangle already fails to pay, for $m \geq 5$, and packing many of them compounds the failure into a gap that grows with n.

Theorem A.60 (A bouquet of thickened cliques, and how it beats the thickened tree, [AI])**.** *Fix $3 \leq r \leq m$, the order of the clique block rather than the hyperedge uniformity of Table 1.1, and set $q = m+1-r \geq 1$, the multiplicity of a thickened K_r that saturates the vertex cap on r vertices (the "thickened complete graph" construction above, restricted to a block of size r rather than all of n). Let $1 \leq k \leq \lfloor (n-1)/(r-1) \rfloor$. Take k copies of K_r that all share one common vertex and are otherwise disjoint, thicken every edge of every copy to multiplicity q, and attach any leftover vertices as a pendant path at multiplicity $m-1$. This multigraph is feasible, $\kappa^{\max} \leq m-1$, and its total multiplicity is exactly*

$$(m-1)(n-1) \;+\; k \cdot \mathrm{gain}(r,m), \qquad \mathrm{gain}(r,m) := \frac{(r-1)(r-2)(m-1-r)}{2}.$$

The count therefore exceeds the thickened tree's $(m-1)(n-1)$ exactly when $\mathrm{gain}(r,m) > 0$, that is when $r \leq m-2$, which is possible for some r in range precisely when $m \geq 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. Different clique copies and pendant edges meet only at their attachment vertices. Whenever such an attachment separates two nonempty pieces, it is a cut vertex. So a path of length ≥ 2 between two vertices of the same block cannot leave that block and come back: to do so it would have to pass through the shared vertex twice. Hence for an edge uv inside a block, every route counted by $\pi(u,v)$ stays inside its K_r, where deleting uv leaves u, v joined by exactly $r-2$ pairwise internally-disjoint two-step routes, one through each of the other block vertices, and by Menger no more, since those same $r-2$ vertices are a cut of that size. So $\pi(u,v) = r-2$ and, by Lemma A.56, $\kappa_G(u,v) = q + (r-2) = m-1$ exactly, never over. For u, v in different blocks, or nonadjacent with either endpoint on the pendant path, a single cut vertex separates them, so $\kappa_G(u,v) \leq 1$. If u, v are adjacent along the pendant path, then the edge lies on no cycle, and Lemma A.56 gives $\kappa_G(u,v) = \mu(u,v) + \pi(u,v) = (m-1) + 0 = m-1$.

Count. Since $q \geq 1$, every block edge is present, so G_0 is connected and $|E(G_0)| = n - 1 + \mathrm{rank}(G_0)$. This is where the hypothesis $r \leq m$ is used: at $r = m+1$ the multiplicity q would drop to zero, the blocks would carry no edges at all, and G_0 would fall apart into isolated vertices. Only the k blocks contribute to $\mathrm{rank}(G_0)$ or to $\sum \pi$, since the pendant edges have $\pi = 0$ and lie in no cycle. Each K_r has rank $(r-1)(r-2)/2$ and contributes $\binom{r}{2}(r-2)$ to the sum, since each of its $\binom{r}{2}$ edges has $\pi = r-2$. Substituting into $(m-1)(n-1+\mathrm{rank}(G_0)) - \sum \pi$,

$$(m-1)\Big[n-1+k\tfrac{(r-1)(r-2)}{2}\Big] - k\tfrac{r(r-1)(r-2)}{2} \;=\; (m-1)(n-1) + k\,\mathrm{gain}(r,m),$$

using $(m-1)\frac{(r-1)(r-2)}{2} - \frac{r(r-1)(r-2)}{2} = \frac{(r-1)(r-2)\big[(m-1)-r\big]}{2} = \mathrm{gain}(r,m)$. □

At $r = 3$, $\mathrm{gain}(3,m) = m-4$: zero at $m = 4$, matching the absence of counterexamples at the tested orders for $m \leq 4$, and strictly positive for every $m \geq 5$. So the guess that the thickened tree is optimal once $n \geq m+2$ is false for every $m \geq 5$: it held only within the reach of the exhaustive table, $m \leq 4$. The smallest witness is small enough to check by hand. Take $m = 5$ and $n = 7$, the first size the guess covers. Put a triangle on $\{u_1, u_2, u_3\}$ at multiplicity 3 and hang a path of four further vertices off u_1 at multiplicity 4. Two triangle vertices have three parallel copies plus one route through the third vertex, so $\kappa = 4$, and every other pair is separated by a single cut vertex or joined only by its own parallel copies, so nothing exceeds $4 = m - 1$. The total multiplicity is $3 \cdot 3 + 4 \cdot 4 = 25$, one more than the $(m-1)(n-1) = 24$ the guess allows. Packing k blocks instead of one multiplies the excess by k, and k may be taken as large as $\lfloor (n-1)/(r-1) \rfloor$, so the gap over the thickened tree grows linearly in n.

Sharing a vertex lets more clique blocks fit. Disjoint copies of K_r use r vertices each, even when bridges join them, so at most $\lfloor n/r \rfloor$ copies fit. With one shared vertex, each new copy instead uses only $r-1$ new vertices. For example, at $m = 10$, $r = 6$ and $n = 11$, two shared-vertex copies carry 150 edges counted with multiplicity. Only one disjoint copy fits, carrying 75, and attaching the five remaining vertices by bridges of multiplicity 9 brings its count to 120.

Optimising r widens the gap further. Treated as continuous, the gain per added vertex, which is $\mathrm{gain}(r, m)/(r-1)$, peaks near $r \sim (m+1)/2$, where it is $\Theta(m^2)$. Packing blocks of that size therefore gives $K_m(n) \geq \big((m-1) + \Theta(m^2)\big)(n - 1 - O(m))$. In particular the per-vertex rate is quadratic in m once $n/m \to \infty$, a different order from the tree's linear rate.

The clique bouquet need not be optimal. At $m = 5$, $n = 7$ the best clique bouquet uses three triangles and gives 27 and the block computation of Table A.1 gives 29. What is not open is the order, and to say that the bouquet is a lower bound of the right order there has to be a matching upper bound. There is one, and it comes from the underlying simple graph rather than cycle rank.

Proposition A.61 (An upper bound of the same order, [AI])**.** *For all $m \geq 2$ and $n \geq 2$,*

$$K_m(n) \;\leq\; (m-1)\, k_m(n) \;<\; 2(m-1)^2\, n.$$

Proof. Let G be feasible, with underlying simple graph G_0.

Every multiplicity is at most $m-1$. By Lemma A.56, $\kappa_G(u, v) = \mu(u, v) + \pi(u, v) \geq \mu(u, v)$, so feasibility gives $\mu(u, v) \leq m - 1$ on every pair and the total is at most $(m-1)|E(G_0)|$.

G_0 is itself feasible for the simple vertex problem. Take two vertices $u \neq v$. If they are adjacent in G_0 then $\mu(u, v) \geq 1$, and the routes of G_0 between them are the single edge together with the internally disjoint routes of length at least two, so $\kappa_{G_0}(u, v) = 1 + \pi(u, v) \leq \mu(u, v) + \pi(u, v) \leq m - 1$. If they are not adjacent then $\kappa_{G_0}(u, v) = \pi(u, v) \leq m - 1$ directly. So $\kappa^{\max}_{G_0} \leq m - 1$ and $|E(G_0)| \leq k_m(n)$ by definition of k_m, which gives the first inequality.

The simple problem is linearly bounded. Suppose G_0 had average degree at least $4(m-1)$. By Mader's density theorem [25], a graph of average degree at least $4k$ contains a $(k+1)$-connected subgraph, so with $k = m - 1$ some subgraph $H \subseteq G_0$ is m-connected. An m-connected graph has more than m vertices and, by the global form of Menger's theorem [30, Ch. 3], any two of them are joined by m internally disjoint paths inside H and hence inside G_0. That contradicts $\kappa^{\max}_{G_0} \leq m - 1$. So the average degree is below $4(m-1)$, that is $2|E(G_0)|/n <$

$4(m-1)$, giving $k_m(n) < 2(m-1)n$ and the second inequality. $\square$

Combining the two directions, first letting $n/m \to \infty$ so that optimised blocks can be packed with only $O(m)$ unused vertices, and then letting $m \to \infty$, yields

$$\Big(\tfrac{1}{8} + o_m(1)\Big) m^2 n \;\le\; K_m(n) \;<\; 2m^2 n,$$

where the $o_m(1)$ vanishes as m grows, so $K_m(n) = \Theta(m^2 n)$ in that large-n regime. This qualification is necessary: for example $K_m(2) = m-1$, so no estimate of order $m^2 n$ can hold uniformly when n is fixed. The bouquet of Theorem A.60 has the right joint order once many blocks fit, and what separates the two sides is a constant factor tending to 16. Theorem A.64 below improves the lower constant and brings that factor down to $6 + 4\sqrt{2} \approx 11.66$. The upper bound is crude, since it throws away the $-\sum \pi$ correction entirely and charges every edge the full $m-1$. The first inequality, $K_m(n) \le (m-1)k_m(n)$, is the more informative half. It says that this variant can never outrun the simple vertex problem by more than the multiplicity cap, which ties the least understood of the sixteen to the one with the longest history. It is tight at $m = 2$, where both sides read $n-1$ by Remark A.58.

A.10.1 The value is a block problem

The lower bounds above are constructions and the upper bound is a density count, and neither says what an extremiser looks like. This section reduces the whole question to one about 2-connected graphs: the number of vertices in play stops being n and becomes the size of a single block.

Two observations do it. The first is already in the section, one line further than it was taken.

Lemma A.62 (The objective in closed form, [AI]). *For a simple graph G_0 on n vertices with $\kappa_{G_0}^{\max} \le m-1$, the largest total multiplicity of a feasible multigraph with underlying simple graph exactly G_0 is*

$$W_m(G_0) \;:=\; \sum_{uv \in E(G_0)} \big(m - \kappa_{G_0}(u,v)\big),$$

and every simple graph arising as the underlying graph of a feasible multigraph satisfies $\kappa_{G_0}^{\max} \le m-1$. Hence

$$K_m(n) \;=\; \max\big\{\, W_m(G_0) \;:\; G_0 \text{ simple on } n \text{ vertices},\; \kappa_{G_0}^{\max} \le m-1 \,\big\}.$$

Proof. By Lemma A.56 feasibility reads $\mu(u,v) \le m-1-\pi(u,v)$ on every edge, so the best choice is $\mu(u,v) = m-1-\pi(u,v)$, which is what the paragraph before Theorem A.60 records. On an edge $\kappa_{G_0}(u,v) = 1 + \pi(u,v)$, since the routes of G_0 between adjacent u and v are the single edge together with the routes of length at least two. Substituting turns $m-1-\pi(u,v)$ into $m - \kappa_{G_0}(u,v)$ and the total into $W_m(G_0)$.

Two things have to be checked for that maximum to be attained by a multigraph over G_0 itself. Feasibility of G_0 is the middle step of Proposition A.61. And the chosen multiplicities are all at least one, so the underlying graph really is G_0 and no edge silently disappears: $\kappa_{G_0}(u,v) \le m-1$ on an edge gives $\pi(u,v) \le m-2$, hence $\mu(u,v) = m-1-\pi(u,v) \ge 1$. Conversely any feasible multigraph is one of the multigraphs considered over its own underlying graph, so nothing is lost by maximising over G_0 first. □

That the objective is a sum over edges of a *local* quantity is what makes the second observation bite. Recall that a *block* of a graph is a maximal connected subgraph with no cut vertex of its own, so every block is either 2-connected or a single edge, every edge lies in exactly one block, and two blocks meet in at most one vertex.

Theorem A.63 (The multigraph incidence problem is a knapsack over blocks, [AI]). *For a 2-connected graph B, or a single edge, write*

$$g_m(b) \;:=\; \max\bigl\{\, W_m(B) \;:\; B \text{ 2-connected or an edge, } |V(B)| = b,\ \kappa_B^{\max} \le m-1 \,\bigr\},$$

with $g_m(b) := -\infty$ when no such B exists. Then for all $m \ge 2$ and $n \ge 2$,

$$K_m(n) \;=\; \max\Bigl\{\, \sum_i g_m(b_i) \;:\; b_i \ge 2,\ \sum_i (b_i - 1) \le n-1 \,\Bigr\},$$

the maximum over all finite multisets of block sizes. In particular the value is determined by the finitely many numbers $g_m(2), \ldots, g_m(n)$.

Proof. The objective is additive over blocks. Let G_0 be feasible and let u, v be adjacent, lying in the block B. Every u-v path of G_0 stays inside B: a path leaving B must leave through a cut vertex, and to return it would have to pass through that same cut vertex a second time, which no path does. So $\kappa_{G_0}(u,v) = \kappa_B(u,v)$, and since every edge lies in exactly one block,

$$W_m(G_0) \;=\; \sum_{\text{blocks } B} W_m(B).$$

Each block inherits feasibility, $\kappa_B^{\max} \le \kappa_{G_0}^{\max} \le m-1$ by the same identity, so $W_m(B) \le g_m(|V(B)|)$.

The sizes fit. Suppose G_0 has c components and blocks $B_1, \ldots, B_t$. Within a single component, contracting the block tree makes $\sum(|V(B_i)| - 1)$ equal to the number of vertices of that component minus one. Isolated components contribute zero to both sides, so summing gives $\sum_i (|V(B_i)| - 1) = n - c \le n-1$. With Lemma A.62 this proves $\le$.

Every multiset is realised. Given blocks $B_1, \ldots, B_t$ with $\sum(b_i - 1) \le n-1$, take one common vertex and attach the B_i to it, pairwise disjoint otherwise, leaving any spare vertices isolated. The result uses $1 + \sum(b_i - 1) \le n$ vertices. Its blocks are exactly the B_i, so by the first paragraph its κ agrees with κ_{B_i} on pairs inside a block and equals at most 1 on pairs split between two blocks, which the shared vertex separates. So the graph is feasible and scores $\sum_i W_m(B_i)$, and choosing each B_i optimal gives $\sum_i g_m(b_i)$. This proves $\ge$. □

The reduction changes what has to be searched: enumerate 2-connected simple graphs, evaluate W_m, and solve the resulting knapsack. A sweep through eight vertices gives Table A.1 and agrees with the independent exhaustive routine wherever both finish. It runs from a standalone script, `program/scripts/multi_vertex_blocks.py`, and its retained transcript is `program/logs/multi_vertex_blocks_log.txt` in the repository named in Section A.13.

	b						
m	2	3	4	5	6	7	8
2	1	—	—	—	—	—	—
3	2	3	4	5	6	7	8
4	3	6	9	12	15	18	21
5	4	9	14	19	24	29	34
6	5	12	19	26	33	40	47
7	6	15	24	33	42	52	62
8	7	18	30	42	54	66	79

Table A.1. $g_m(b)$, the largest W_m a single 2-connected block on b vertices can score, by exhaustion over every 2-connected graph of that order. A dash marks a size at which no feasible block exists, which at $m = 2$ is every $b \geq 3$. Feeding this row into the knapsack of Theorem A.63 gives $K_m(n)$ for $n \leq 8$. At $m = 5$, $n = 7$ it returns the 29 quoted above, against the 27 that Theorem A.60 reaches. The per-vertex rate $g_m(b)/(b-1)$ is still rising at $b = 8$ in every row from $m = 5$ on, while at $m = 3$ it falls and at $m = 4$ it is flat, so this table does not extrapolate: a larger n needs larger blocks enumerated.

One cell beyond the table is known. An exhaustive sweep of the 2-connected graphs on nine vertices gives $g_6(9) = 54$, attained by a block with fourteen edges, run from the standalone script `program/scripts/multi_vertex_blocks_b9.py`. A single block of that order beats every split of the same budget, the best of which reach 52, so $K_6(9) = 54$. Its rate $54/8$ exceeds the $47/7$ of $b = 8$, which is the $m = 6$ row still rising at $b = 9$ exactly as the caption warns.

The winning blocks are never complete once $b \geq 5$, and from $m = 4$ on their degree sequences are lopsided, a few vertices of high degree beside a majority at degree two or three. At $m = 3$ they are the cycles, every degree two. What the sweep does *not* show is that they are complete bipartite graphs with a few edges added. That description fits at $m = 7$ and $m = 8$, where $K_{3,4}$ or $K_{3,5}$ with two further edges inside the side of size three attains the maximum at $b = 7$ and $b = 8$. It is the unique maximiser at three of those four cells and one of six at $m = 8$, $b = 7$, where two of the six carry the shape. The description fails at $m = 4$, 5 and 6, where from $b = 7$ onwards the maximum is attained by many non-isomorphic blocks at once, up to 38 of them, and across those six cells exactly one tied block is complete bipartite, the $K_{2,5}$ at $m = 6$, $b = 7$, which scores the $st(m-s) = 40$ of the family below at $s = 2$, $t = 5$. So the family of Theorem A.64 below is a construction suggested by the largest m in reach, and it is a lower bound rather than a description of the extremisers. The script prints each maximiser's `graph6` string, its degree sequence and the count of blocks tied with it, so the reader can check which of these cells is which.

Theorem A.64 (Thickened complete bipartite blocks, [AI])**.** *Let $1 \leq s \leq t \leq m-1$. Thicken every edge of $K_{s,t}$ to multiplicity $m-s$. The result is feasible, $\kappa^{\max} \leq m-1$, on $s+t$ vertices with total* *multiplicity* $st(m-s)$*. Hanging* k *such blocks off one shared vertex gives, for* $n = k(s+t-1)+1$*,*

$$K_m(n) \;\geq\; \frac{st\,(m-s)}{s+t-1}\,(n-1).$$

Optimised over s *and* t *this rate is* $\big(3-2\sqrt{2}+o_m(1)\big)m^2$*, attained at* $t = m-1$ *and* $s \sim (\sqrt{2}-1)m$*, against the* $\big(\frac{1}{8}+o_m(1)\big)m^2$ *of Theorem A.60. The improvement is by a factor* $8(3-2\sqrt{2}) = 24-16\sqrt{2} \approx 1.373$*.*

Proof. Write S and T for the two sides, $|S| = s$ and $|T| = t$.

The connectivities of $K_{s,t}$*.* For $u \neq u'$ in S, the t two-step routes through distinct vertices of T are internally disjoint. They give $\kappa(u,u') \geq t$. Every route must also meet T, which is a cut of size t, so $\kappa(u,u') = \pi(u,u') = t$. Longer routes may exist but cannot increase this maximum. Symmetrically $\kappa(v,v') = s$ for $v \neq v'$ in T. For adjacent $u \in S$ and $v \in T$, delete the edge uv. The sets $T \setminus \{v\}$ and $S \setminus \{u\}$ both separate u from v, so $\pi(u,v) \leq \min(s-1,t-1) = s-1$. Choose distinct $u_i \in S \setminus \{u\}$ and distinct $v_i \in T \setminus \{v\}$ for $1 \leq i \leq s-1$, possible because $s \leq t$. The routes u, v_i, u_i, v attain the bound. Thus $\pi(u,v) = s-1$ and $\kappa(u,v) = s$.

Feasibility and count. By Lemma A.56 the multigraph has $\kappa = (m-s)+(s-1) = m-1$ on an edge, and $\kappa = t \leq m-1$ or $s \leq m-1$ on a non-adjacent pair, so nothing exceeds the cap. The count is st edges at multiplicity $m-s$. Equivalently, by Lemma A.62, $W_m(K_{s,t}) = st(m-s)$.

The bouquet. If $s \geq 2$, then $K_{s,t}$ is 2-connected, so Theorem A.63 applies and k copies at one shared vertex are feasible with total $k\,st(m-s)$ on $n = k(s+t-1)+1$ vertices. If $s = 1$, then $K_{1,t}$ is a tree (a single edge when $t = 1$), and after thickening every edge to multiplicity $m-1$, sharing vertices among copies still produces a thickened tree and is feasible directly. Its count is again $k\,t(m-1) = k\,st(m-s)$. Thus the displayed rate holds throughout the theorem's full range $1 \leq s \leq t$.

The optimum. For fixed $s > 1$, the factor $t/(s+t-1)$ is strictly increasing in t, and for $s = 1$ it is constant. Thus some optimum has $t = m-1$, its largest permitted value. Put $s = \alpha m$. The rate becomes

$$\frac{s\,(m-1)(m-s)}{s+m-2} \;=\; \frac{\alpha(1-\alpha)}{1+\alpha}\,m^2 \;+\; O(m),$$

and $h(\alpha) = \alpha(1-\alpha)/(1+\alpha)$ has $h'(\alpha) = (1-2\alpha-\alpha^2)/(1+\alpha)^2$, vanishing at $\alpha = \sqrt{2}-1$, where $h = 3-2\sqrt{2}$. Since $3-2\sqrt{2} > \frac{1}{8}$, the family beats the clique bouquet, by the stated factor. □

The optimum is lopsided: one side has size about $0.41m$ and the other reaches the cap $m-1$. The finite sweep of Table A.1 shows that adding a few edges inside the small side improves this clean family at $m = 7$ and $m = 8$, and that at $m = 4$, 5 and 6 the extremal blocks are mostly of other shapes, the $K_{2,5}$ tying at $m = 6$, $b = 7$ being the one that is not. The answer is therefore a block optimisation problem. In the regime $m \to \infty$ with $n/m \to \infty$, the construction gives $K_m(n)/(m^2 n) \geq 3-2\sqrt{2}-o(1)$ and the upper bound gives $K_m(n)/(m^2 n) < 2$. These inequalities do not establish that this ratio has a limit. Its possible limiting value and the blocks needed for sharper bounds remain open here.

A.10.2 The directed problem

Everything above splits the measure into copies and detours, and that split does not use the absence of orientation. The directed statement is the same one with directed routes, and it is worth writing down because it is what carries the arc results across to $K_m^{\mathrm{dir}}(n)$ without any asymptotic argument.

Lemma A.65 (Splitting the directed vertex measure, [AI])**.** *Let D be a multidigraph with multiplicity function μ and underlying simple digraph D_0. For an ordered pair $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint directed u-v paths of length at least two in D_0. Then $\kappa_D(u,v) = \mu(u,v) + \pi(u,v)$, and consequently*

$$K_m^{\mathrm{dir}}(n) \;\le\; (m-1)\, k_m^{\mathrm{dir}}(n).$$

Proof. A directed u-v route of length one is a parallel copy of the arc uv and has empty interior, and no step of Lemma A.56 uses the absence of orientation, so its proof applies with directed routes throughout. The lower bound lifts a maximum family of internally disjoint directed u-v paths of length at least two out of D_0 and adjoins the $\mu(u,v)$ copies. The upper bound splits a family into copies of the arc uv and routes of length at least two, the latter mapping injectively to directed paths of D_0 because two routes with the same vertex sequence share an interior that is non-empty. Note that a directed u-v path uses no copy of the arc vu, so the reverse multiplicity plays no part.

For the bound, let D be feasible. The identity gives $\mu(u,v) \le \kappa_D(u,v) \le m-1$ on every ordered pair, so the total multiplicity is at most $(m-1)|A(D_0)|$. And D_0 is itself feasible: on an arc, $\kappa_{D_0}(u,v) = 1 + \pi(u,v) \le \mu(u,v) + \pi(u,v) \le m-1$, and on a pair with no arc from u to v, $\kappa_{D_0}(u,v) = \pi(u,v) \le m-1$ directly. So $|A(D_0)| \le k_m^{\mathrm{dir}}(n)$ by definition. $\square$

Corollary A.66 (The directed multigraph incidence value, [AI])**.** *For every $n \ge 2$,*

$$K_2^{\mathrm{dir}}(n) \;=\; k_2^{\mathrm{dir}}(n) \;=\; \ell_2^{\mathrm{dir}}(n) \;=\; \max\!\Big(2(n-1),\ \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big),$$

and for each fixed $m \ge 2$,

$$(m-1)M(n) \;\le\; K_m^{\mathrm{dir}}(n) \;\le\; (m-1)\Big\lfloor \frac{n^2}{4} \Big\rfloor + 4(m-1)^2(n-1),$$

so that $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ and the bipartite digraph of Theorem 1.13 is asymptotically optimal for this variant too.

Proof. At $m=2$ the cap is one, so Lemma A.65 forces $\mu(u,v) \le 1$ on every ordered pair and every feasible multidigraph is simple. Hence $K_2^{\mathrm{dir}}(n) = k_2^{\mathrm{dir}}(n)$, which is $\ell_2^{\mathrm{dir}}(n)$ by Theorem 1.7 and the displayed value by Theorem 1.6.

For general m the lower bound is the arc extremiser. Theorem 1.13 supplies a multidigraph with $(m-1)M(n)$ arcs and $\lambda^{\max} \le m-1$, and $\kappa \le \lambda$, so that same multidigraph is feasible here. The upper bound is Theorem 1.14 at $r=2$, where a forward directed 2-uniform multihypergraph is a multidigraph and the displayed bound reads as stated. Both sides have leading term $(m-1)n^2/4$, since $M(n) = \lfloor n^2/4 \rfloor$ for $n \ge 7$, and the difference is linear in n for fixed m. □

The exhaustive values agree and stay strictly inside these bounds where both are known: $K_6^{\mathrm{dir}}(3) = 24$ against the lower bound $(m-1)M(3) = 20$, so the arc extremiser read through $\kappa \le \lambda$ is a lower bound for the incidence problem and not its value. What beats it at that size is a block, and the block argument of Theorem A.63 carries over to directed routes without change.

Construction A.67 (Thickened complete digraph, and its bouquet)**.** For $2 \le b \le m$, take the complete digraph on b vertices and give every arc multiplicity $m+1-b$. At $b=2$ this is one bidirected edge at multiplicity $m-1$. Hanging k such blocks off one shared vertex, pairwise disjoint otherwise, uses $k(b-1)+1$ vertices, and blocks of different sizes may be mixed.

Proposition A.68 (The directed incidence value is a block problem too, [Al])**.** *Let D be a multidigraph with underlying simple digraph D_0, and take the blocks of the underlying undirected graph of D_0. Then κ_D agrees with the value measured inside a block on every pair joined by an arc, and is at most 1 on every pair split between two blocks, so the count of Lemma A.65 is additive over blocks exactly as in Theorem A.63. Each block of Construction A.67 is feasible and carries $b(b-1)(m+1-b)$ arcs, so*

$$K_m^{\mathrm{dir}}(n) \;\ge\; \max\Big\{ \sum_i b_i(b_i-1)(m+1-b_i) \;:\; 2 \le b_i \le m,\ \sum_i (b_i - 1) \le n-1 \Big\}.$$

Proof. For the block structure, let u, v be joined by an arc of D_0 and let B be the block of the underlying undirected graph that contains it. A directed u-v path of D_0 leaves B only by crossing a cut vertex, and returning crosses that same vertex a second time, which no path does. So every route counted by $\pi(u,v)$ stays inside B. For two vertices that lie in no common block, either no path joins them or a cut vertex on the intervening block chain separates them. Every directed route must pass through that vertex, giving $\kappa_D \le 1$ there. The two blocks need not themselves share a vertex.

For one block, the internally disjoint directed u-v routes of length at least two in the complete digraph on b vertices have pairwise disjoint interiors drawn from the other $b-2$ vertices, so there are at most $b-2$ of them, and the two-step routes attain that. Hence $\pi(u,v) = b-2$, and Lemma A.65 gives $\kappa_D(u,v) = (m+1-b)+(b-2) = m-1$ on each of the $b(b-1)$ ordered pairs, which sits at the cap. The count is $b(b-1)(m+1-b)$. Attaching the blocks at one shared vertex uses $1 + \sum_i (b_i - 1)$ vertices, and any spare vertices are left isolated. □

The blocks and the arc extremiser lead on different ranges, and the grids of Section A.13.2 plot whichever is larger. At $m=3$ the best block is the bidirected edge, the bouquet is the thickened bidirected tree, and the arc extremiser takes over from $n=8$. At $m=6$ the best block has three or four vertices, the bouquet leads through $n = 8$ at 84 arcs against the extremiser's 80, and the extremiser leads from $n = 9$.

A.11 The directed hypergraph

The remaining four of the sixteen variants orient the Berge edge. They are grouped here rather than with the undirected hypergraph bounds above because the questions they raise are different ones: not which decomposition controls the rank of an incidence graph, but how much a single edge with one tail and many heads can carry, and whether the way it is oriented changes the answer. The first result gives a pair of bounds a factor of four apart, and the last closes that factor for the model the program measures.

Proposition A.69 (Directed hypergraphs: first bounds, [AI])**.** *Let $n \geq r \geq 2$ and $m \geq 2$. Model a directed r-uniform hyperedge as a tail vertex together with $r-1$ head vertices, a Berge route entering at the tail and leaving at a head. If every ordered pair of vertices has at most $m-1$ hyperedge-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \;\leq\; \frac{(m-1)\,n(n-1)}{r-1},$$

and, for every $1 \leq \alpha \leq n-r+1$, there is a feasible directed multihypergraph with

$$\alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$$

hyperedges. This construction is simple whenever $m-1 \leq \binom{n-\alpha-1}{r-2}$. Consequently, for fixed r and m, the balanced choice $\alpha = \lfloor n/2 \rfloor$ is available for all sufficiently large n and proves that the extremal value is $\Theta_{r,m}(n^2)$.

Proof. Upper bound: a hyperedge with tail u serves the $r-1$ ordered pairs (u,h), h a head. If some ordered pair (u,v) were served by m distinct hyperedges, those would be m hyperedge-disjoint one-step routes, so each ordered pair is served at most $m-1$ times. Summing that cap over all $n(n-1)$ ordered pairs counts every hyperedge once for each of its $r-1$ tail-head pairs, so the sum is $(r-1)|E|$ on one side and at most $(m-1)n(n-1)$ on the other, giving $(r-1)|E| \leq (m-1)n(n-1)$.

For the lower bound, split V into tails A, $|A| = \alpha$, and heads B, where $b := |B| = n-\alpha \geq r-1$. Put $k := r-1$, $d := m-1$ and $e := \left\lfloor \frac{db}{k} \right\rfloor$. Label the vertices of B by $0, 1, \ldots, b-1$ and read the infinite cyclic word

$$0, 1, \ldots, b-1, 0, 1, \ldots$$

in consecutive blocks of length k. Take the first e blocks as the hyperedge copies of a k-uniform multihypergraph $\mathcal{G}^*$ on B. Because $k \leq b$, each block contains k distinct vertices. Those blocks consume ek consecutive positions of the cyclic word, so every vertex occurs either $\lfloor ek/b \rfloor$ or $\lceil ek/b \rceil$ times. Since $ek \leq db$, the maximum degree is at most d.

Give every tail $a \in A$ the hyperedges (a, H) for $H \in \mathcal{G}^*$, with the same multiplicities. No vertex of B is a tail, so every route is a single step and $\lambda(a,b) = \deg_{\mathcal{G}^*}(b) \leq m-1$, while pairs inside A or starting in B have no routes. The count is $\alpha e = \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$. If $d \leq \binom{b-1}{k-1}$, use Lemma A.51 instead of the cyclic multihypergraph. Its edges are distinct, so the resulting directed hypergraph is simple. □

Construction A.70 (Bounded outgoing hyperedge count, [AI]). For $n \geq r \geq 2$ and $m \geq 2$, give every vertex $d = \min\{m-1, \binom{n-1}{r-1}\}$ distinct outgoing forward hyperedges. Their head sets may be any d distinct $(r-1)$-subsets of the other vertices. This gives a simple directed hypergraph with nd hyperedges, feasible for both route separations. If repeated hyperedges are allowed, give every vertex $m-1$ copies of any one outgoing hyperedge instead, for a total of $n(m-1)$ copies.

Proof. Every directed route from u must begin with a hyperedge whose tail is u. Deleting all outgoing copies therefore separates u from every other vertex. There are at most $m-1$ such copies, so $\lambda(u,v) \leq m-1$ for every ordered pair. The vertex-separated measure is no larger because those routes must also be hyperedge-disjoint. Different tails define different forward hyperedges, which gives the stated counts and ensures simplicity when the head sets are distinct. □

This linear construction can beat the two-layer construction at small orders. For example, at $n = m = 6$ and $r = 3$ it gives 30 hyperedges even in the simple model. The construction markers take the larger of the two counts. Neither count is asserted to be optimal in general.

The forward model is one of three ways to orient a Berge edge. In general a directed r-uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H, a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist for $r \geq 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.71 (Backward is the reverse of forward, [AI]). *Reversing every hyperarc, $(T,H) \mapsto (H,T)$, is an edge-count-preserving bijection between the forward directed r-uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u,v) = \lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r, m and n.*

In words: the backward model is the forward model seen in a mirror. Turning every hyperarc around swaps the tails with the heads and reverses every route, which leaves the number of independent routes between a pair untouched. The two models have identical extremal numbers, so nothing below needs to treat backward as a separate case.

Proof. Reversal sends a backward hyperarc ($|T| = r-1$, $|H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection between the two families. Let $u = x_0, e_1, x_1, \ldots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and

x_i a head of e_i. The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \dots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u–v routes in $\mathcal{H}$ and v–u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u,v) = \lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ. Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. $\square$

Proposition A.72 (The general model obeys the forward bound, [AI])**.** *Let a directed r-uniform hyperedge be any split into a non-empty tail set T and head set H. If every ordered pair has at most $m-1$ hyperedge-disjoint, or at most $m-1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \le \frac{(m-1)\,n(n-1)}{r-1},$$

the same bound as the forward model of Proposition A.69. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta_{r,m}(n^2)$ for fixed r and m.

Proof. Fix an ordered pair (u,v). A hyperedge (T,H) with $u \in T$ and $v \in H$ carries the one-step route $u \to v$ that enters at the tail u and leaves at the head v. Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u,v) \ge k$ and $\kappa(u,v) \ge k$, so feasibility forces $k \le m-1$. Each hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs (u,v) with $u \in T$ and $v \in H$, so summing the per-pair bound over all pairs gives $\sum_{(T,H)} |T|\,|H| \le (m-1)\,n(n-1)$. As $|T| + |H| = r$ with both parts at least one, $|T|\,|H| \ge 1 \cdot (r-1) = r-1$, whence $(r-1)|E| \le (m-1)n(n-1)$. The forward family of Proposition A.69 consists of general hyperedges, so it witnesses the same $\approx (m-1)n^2/(4(r-1))$ lower bound. $\square$

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. The rest of this section closes that gap for both models. Their exact finite values may still differ.

The two bounds are not equally tight, and the simple digraph case indicates which end of the gap the slack sits at. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n-1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n-\alpha) \le n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where Construction A.17 has a complete one-directional bipartite part and only

$O_m(n)$ additional internal arcs. There, Theorem 1.11 now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that should come down by a factor of four, not the construction that should climb, and that is what happens.

The argument of Theorem 1.11 needs a different route-counting step here. For digraphs, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r-1$ heads at once, so a pair of two-step routes with different midpoints may enter through the very same hyperedge, and the family of two-step routes is not edge-disjoint at all. The resolution is to accept the loss. A packing can always be extracted from that family at a cost of a factor $r-1$, and that factor turns out to be free, because of where it lands. The route count does not bound the hyperedges directly. It bounds the arcs of an auxiliary digraph, and there the factor $r-1$ enters only the *linear* error term of Lemma A.34, never the $n^2/4$ leading term. Meanwhile a second, independent factor of $r-1$ is gained when the auxiliary digraph is exchanged back for hyperedges, and that one does divide the leading term. The two factors are not the same factor, and only one of them is paid where it matters.

Definition A.73 (One-step shadow)**.** Let $\mathcal{H}$ be a forward directed r-uniform hypergraph on V. Its *one-step shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u, v) exactly when some hyperedge of $\mathcal{H}$ has tail u and v among its heads. Write $\operatorname{cod}(u, v)$ for the number of hyperedges realising that arc.

Theorem 1.14, stated in Chapter 1, says in words that the bipartite construction of Proposition A.69 is asymptotically the best there is. Its count already matches the leading term $\frac{m-1}{r-1}\lfloor n^2/4\rfloor$, so the factor of four that used to separate the known bounds is gone and the remaining uncertainty sits entirely in a term linear in n. The bound covers both separations at once.

Proof of Theorem 1.14 [AI]. Write $R = R(\mathcal{H})$ for the one-step shadow and note it has no loops, since a hyperedge's tail is not one of its heads.

Step 1: hyperedges against shadow arcs. Suppose $\operatorname{cod}(u, v) = k$. Those k hyperedges each give a one-step Berge route from u to v, and the routes use one hyperedge apiece and different ones, so they are pairwise hyperedge-disjoint, while having no interior vertex at all they are internally vertex-disjoint as well. Hence $k \le \kappa_{\mathcal{H}}(u, v) \le m-1$. Every hyperedge is counted by exactly its $r-1$ tail-head pairs, so summing over the arcs of R,

$$(r-1)\,|E(\mathcal{H})| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u, v) \;\le\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.69 stopped one step earlier. That proof then bounded $|A(R)|$ by the trivial $n(n-1)$, which is exactly where its factor of four was lost, and the rest of this argument replaces that step.

Step 2: the shadow is budgeted. Fix an ordered pair (u, v) of distinct vertices, and let $T \subseteq V$ collect v itself when $(u, v) \in A(R)$ together with every midpoint counted by $p_2(u, v)$, so $|T| =$

$\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Every $t \in T$ is a head of at least one hyperedge whose tail is u. Form the bipartite graph joining each $t \in T$ to each such hyperedge containing it, and form a maximal matching M in it by repeatedly adding an edge whose two endpoints are still unmatched. Here *maximal* means that no further edge can be added. Finding a matching of largest possible size is unnecessary.

Then $|M| \geq |T|/(r-1)$. To see it, take any $t \in T$ left unmatched by M. Every hyperedge containing t must already be matched, since otherwise M could be extended along that pair. So every element of T, matched or not, lies in some hyperedge used by M, and each hyperedge holds at most $r-1$ elements of T, giving $|T| \leq |M|\,(r-1)$.

Now read M as a family of routes. For a matched pair (v,e) take the one-step route $u \to v$ through e, and for a matched pair (x,e) with x a midpoint take $u \to x \to v$, entering through e and leaving through any hyperedge whose tail is x and which has v as a head, one of which exists because $(x,v) \in A(R)$. These routes are pairwise hyperedge-disjoint: the entering hyperedges are distinct because M is a matching, the leaving hyperedges have pairwise distinct tails and so are distinct from each other, and no leaving hyperedge is an entering one, since the entering ones have tail u and the leaving ones have tail a midpoint, which is not u. They are also internally vertex-disjoint, since the one-step route has empty interior and each two-step route has as its interior the single vertex it is matched to, and M matches distinct targets. Hence

$$\frac{\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)}{r-1} \;\leq\; |M| \;\leq\; \kappa_{\mathcal{H}}(u,v) \;\leq\; m-1,$$

so R is $(r-1)(m-1)$-budgeted in the sense of Definition A.28.

Step 3: assembling. Lemma A.34 with $C = (r-1)(m-1)$ gives $|A(R)| \leq \lfloor n^2/4 \rfloor + 4(r-1)(m-1)(n-1)$, and substituting that into Step 1,

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left[\left\lfloor \frac{n^2}{4} \right\rfloor + 4(r-1)(m-1)(n-1)\right] \;=\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

The lower bound is Proposition A.69, whose bipartite family has

$$\max_{1\leq\alpha\leq n-r+1} \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor \;=\; \frac{(m-1)n^2}{4(r-1)} - O_{m,r}(n)$$

hyperedges, where the balanced split $\alpha = \lfloor n/2 \rfloor$ already attains the right-hand side on its own, and which is feasible for both separations at once, since it is λ-feasible and $\kappa \leq \lambda$. For fixed $r \geq 3$, the simplicity condition of Proposition A.69 holds for large n, so the same leading term is attained by simple hypergraphs. At $r = 2$ and $m \geq 3$, repeats are needed for this leading constant, and the simple digraph value instead has leading term $n^2/4$. (The integer maximum is occasionally taken a little off balance, since the floor can favour an unbalanced split by a unit or two, but never by enough to move the n^2 term.) The upper and lower bounds agree in their n^2 term, which is the asymptotic claim. □

Three remarks on the statement. First, the two upper bounds are not comparable and neither supersedes the other: the bound of Proposition A.69 is the smaller one until n is around $16(m-1)(r-1)/3$, since the linear term here carries a constant of $4(m-1)^2$, and the theorem is an asymptotic improvement rather than a uniform one. Second, at $r=2$ the shadow is the digraph itself and cod is the multiplicity, so the theorem recovers the quadratic branch of Theorem 1.13 up to its linear error term, which is the consistency check one wants, though it does not reprove that theorem, which is exact.

Third, the general orientation model can look out of reach, but the obstacle and the way past it are both short to state. The proof above uses the forward model twice, once for the entering hyperedges and once for the leaving ones, in both cases to know that a hyperedge has a single tail. That is what makes Step 2's matching legitimate: the leaving hyperedges have pairwise distinct tails and so are automatically distinct from each other, and none of them can be an entering hyperedge, since the entering ones have tail u and the leaving ones do not. Allow several tails and both statements fail at once. Two midpoints may leave through one shared hyperedge, and from $r \geq 4$ a single hyperedge with two tails and two heads may serve as one target's entrance and another's exit. So the conflicts run both ways and no matching on one side controls them.

The repair is to take a maximal family and ask what stopped it. Here maximal means that no further route can be added while preserving the required disjointness. It need not have the largest possible size. Every target it leaves unserved must be obstructed by a hyperedge the family has already spent, and a hyperedge has only r vertices in all to obstruct with. That counts the leftovers without ever needing the two sides to be independent, which is exactly what the general model denies.

Theorem A.74 (The general orientation model shares the constant, [AI])**.** *Let $r \geq 2$ and $m \geq 2$, and let a directed r-uniform hyperedge be any split (T, H) into a non-empty tail set and a non-empty head set, as in Proposition A.72. Every such multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \frac{4(2r+1)(m-1)^2}{r-1}\,(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$. Hence for fixed r and m the general model has the same leading term $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$ as the forward model, under both separations, and the bipartite construction of Proposition A.69 is asymptotically optimal for it too.

Allowing arbitrary nonempty tail and head sets in addition to forward hyperedges does not change the leading term. The general model still contains the forward construction that supplies the lower bound. This does not assert the same constant for a model restricted to one prescribed tail-set size. With Lemma A.71, the result compares the forward, backward and general models to first order.

Proof. Let R be the one-step shadow read in the general model, so $(u, v) \in A(R)$ exactly when some hyperedge has u among its tails and v among its heads, and $\operatorname{cod}(u,v)$ counts the hyperedges realising it. A hyperedge's tail set and head set are disjoint, so R has no loops.

Step 1: hyperedges against shadow arcs. The $\operatorname{cod}(u,v)$ hyperedges realising an arc give that many one-step routes, one hyperedge apiece and different ones, with no interior vertex, so they are pairwise hyperedge-disjoint and internally vertex-disjoint and $\operatorname{cod}(u,v) \le \kappa_{\mathcal{H}}(u,v) \le m-1$. A hyperedge (T,H) is counted by exactly the $|T|\,|H|$ ordered pairs it fills, and $|T|+|H| = r$ with both parts non-empty gives $|T|\,|H| \ge r-1$. Hence

$$(r-1)\,|E(\mathcal{H})| \;\le\; \sum_{(T,H)\in E(\mathcal{H})} |T|\,|H| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

This is the count of Proposition A.72 stopped one step early, exactly as Step 1 above stopped Proposition A.69 one step early.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v) of distinct vertices and let $\mathcal{T}$ collect v itself when $(u,v) \in A(R)$ together with every midpoint counted by $p_2(u,v)$, so $|\mathcal{T}| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Call a hyperedge e an *entrance* for $t \in \mathcal{T}$ if u is a tail of e and t a head, and an *exit* for a midpoint t if t is a tail of e and v a head. Every $t \in \mathcal{T}$ has at least one entrance, and every midpoint at least one exit, by the definition of R. No hyperedge is both an entrance and an exit for the *same* t, since that would put t in its tail set and its head set at once.

Consider families of routes indexed by distinct targets: for the target v the one-step route through an entrance, and for a midpoint t the two-step route $u \to t \to v$ through an entrance and an exit. Such a family is admissible when its hyperedges are pairwise distinct. Build such a family greedily until no route with an unused target and unused hyperedges can be added. Let $\mathcal{F}$ be the resulting maximal family, of size M, serving the target set S, and let U be the hyperedges it uses, so $|U| \le 2M$. The routes of $\mathcal{F}$ are pairwise hyperedge-disjoint by construction and internally vertex-disjoint because their interiors are the empty set and distinct singletons, so

$$M \;\le\; \kappa_{\mathcal{H}}(u,v) \;\le\; m-1.$$

Now take $t \in \mathcal{T} \setminus S$. If some entrance $e \notin U$ existed and, for a midpoint, some exit $f \notin U$ as well, then $e \ne f$ by the paragraph above and the route through them could be added to $\mathcal{F}$, serving a target not yet served and using no hyperedge already spent. That contradicts maximality. So every entrance of t lies in U, or t is a midpoint all of whose exits lie in U. Either way t lies in the tail set or the head set of some member of U, and therefore

$$\mathcal{T} \setminus S \;\subseteq\; \bigcup_{e\in U} (T_e \cup H_e), \qquad \text{so} \qquad |\mathcal{T}| \;\le\; M + \sum_{e\in U} \bigl(|T_e| + |H_e|\bigr) \;=\; M + r\,|U|.$$

The last equality is where the two sides are counted together rather than separately, and it is what keeps the constant at r rather than $2(r-1)$. With $|U| \le 2M$ this gives $|\mathcal{T}| \le (2r+1)M \le (2r+1)(m-1)$, so R is $(2r+1)(m-1)$-budgeted in the sense of Definition A.28.

Step 3: assembling. Lemma A.34 with $C = (2r+1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor +$

$4(2r+1)(m-1)(n-1)$, and substituting into Step 1 gives the stated bound. The hypothesis $\lambda^{\max} \le m-1$ implies $\kappa^{\max} \le m-1$ by Whitney, so the bound holds there too. For the lower bound, the bipartite family of Proposition A.69 consists of general hyperedges, so it witnesses $\approx (m-1)n^2/(4(r-1))$ here as well, and the upper and lower bounds agree in their n^2 term. $\square$

Two remarks on that proof. The constant $2r+1$ is not optimised and is not claimed to be sharp: it is what the crude estimate $|U| \le 2M$ gives, and a search over deliberately adversarial instances, in which every target shares both its entrance and its exit with another, never drives $|\mathcal{T}|/\kappa(u,v)$ above $r-1$, the forward model's own constant. That search runs from the standalone script `program/scripts/general_orientation_check.py`. Nothing rests on the difference, since Lemma A.34 places this constant in the linear term, where it never touches the $n^2/4$ that carries the leading constant. The second point is what the theorem does to the orientation axis of Section 3.3.2. Forward and backward were already known equal, exactly, by Lemma A.71. The general model was known only to sit between the same two bounds a factor of four apart, which left room for it to be denser to first order. It is not. All three orientation models have the same leading term, and the computed differences between them at $n \approx r$ are a small-size effect that the asymptotics cannot see.

Part IV. The audit

The bookkeeping that keeps the search's connectivity checks cheap, and the record of what was computed, with what, and to what standard.

A.12 The search's cheap connectivity bookkeeping

Chapter 2 states that the search avoids most calls to the checker by tracking a bound on $\lambda^{\max}$ rather than recomputing it. This section records how the shortcut works and how it is checked against the value the naive implementation would have returned.

Here the *binding connectivity* is $\lambda^{\max}$, the largest local connectivity over all pairs. A *capped predicate* is the yes-or-no test that asks whether this value exceeds a prescribed cap and may stop as soon as one violating pair is found. The *energy* is the numerical objective used by the search to compare two proposed states. Its connectivity penalty is defined in the next paragraph.

Proposition A.4's two parts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0,\ \lambda^{\max}-(m-1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \le m-2$, also stays feasible. Part (2) caps the rise from a single added unit at one, so the graph can climb at most to $\lambda^{\max} = m-1$, which is still inside the feasible region. The excess is zero again, and again no measurement is needed. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip a feasible graph over. What the implementation tests is the two shortcuts rather than that boundary, so it runs a capped pairwise sweep, stopping at the first pair for which it finds one route too many, on every proposal clearing neither: an addition whose maintained bound has not been resynchronised down to $\lambda_{\mathrm{ub}} \leq m-2$, and a removal applied to a graph that is already infeasible. Both are supersets of the steps that can actually be infeasible, so the sweep is sometimes run where monotonicity would have settled the step, never the reverse. The exact connectivity value is recomputed only on the rare step whose proposal is infeasible, where the penalty term needs it.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at a fixed step cadence. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

For the same initial state, random seed and fixed number of steps, the energy returned on every step is, by Proposition A.4, exactly the value the naive implementation would have computed. The accept-or-reject decisions and final graph are therefore unchanged under that fixed-step comparison. Under a time limit, however, a faster checker can complete more steps and return a different witness. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. The witness the search reports is always certified exactly, either by the monotonicity deductions above or by the capped predicate, and the full binding-connectivity value is recomputed only when the energy or a recorded trace needs it.

Where the method is needed differs by variant. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu shortcut of Theorem A.2, which encodes every pairwise edge-connectivity in $n - 1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let the same matrix-search machinery serve every graph variant at speed.

A.13 Computational audit

The authoritative code and audit instructions are maintained in the versioned repository at `https://github.com/louisvandenbruwaene/thesis`, which serves this document and the `program/` directory it refers to. The source revision recorded for this PDF build is `aebed4ee1c4c` (tag `submitted-12`, 2026-09-07). Before an unrecorded working build, the revision marker is reset to "not recorded". For a recorded build, `record_revision.sh` is run after committing the sources, so the printed tag identifies the source snapshot. Everything this thesis refers to sits in one directory, `program/`: the unified program and its test suite, the standalone scripts behind the block sweeps and the conjecture checks (`program/scripts/`), the recorded machine values (`program/data/`) and the run transcripts (`program/logs/`). This document and that directory are the whole of it.

Three commands cover the main audit. Run them from `program/`. The first performs the principal checks, the second runs the regression suite and the third regenerates the computational figures.

```
../.venv/bin/python3 erdos915_unified.py
../.venv/bin/python3 -m unittest discover -s tests -v
../.venv/bin/python3 make_figures.py
```

The four variant grids require several hundred `solve` calls. Their values are stored in `program/data/machine_values.json`. This readable file records what each run returned. It lets the grids redraw in seconds. It also ensures that the document uses the recorded points rather than the outcome of a new timed search.

Rendering and recomputing are separate operations. The third command draws only the figures and tables used by this thesis, using the frozen observations and stated formulas. It does not run searches or refresh archived experiments. A missing observation is an error, not permission to compute a replacement. Passing `--rebuild` performs a new experiment and writes a candidate file beside the published record. Passing `--compare` lists the differences. `--promote` replaces the record only after review. Promotion is refused if two completed exact values disagree. Such a disagreement is a defect to investigate, not a timing fluctuation.

For new rebuilds, each run records its method, requested and elapsed time, completion status and witness, together with source and environment information. The source fingerprint now covers the full program, the C source and the experiment driver. The earlier chapter-four cutoff omitted flow helpers defined later in the program. The historical observations below predate these additional per-run fields. Their missing information cannot be reconstructed by editing metadata, so it remains unknown.

The block sweep is independent of that file and of the main program. Its self-contained implementation is `program/scripts/simple_vertex_blocks.py`. It requires nauty's `geng` on the path. From `program/`, the quoted range is reproduced by `BMAX=9 ../.venv/bin/python3 scripts/simple_vertex_blocks.py`. The recorded environment used nauty 2.9.3.

The three commands name the virtual environment that holds the dependencies. It is machine-specific and is not included in the repository. Starting from Python 3.14.6, create one at the repository root with `python3 -m venv .venv`, then run `.venv/bin/python3 -m pip install -r program/requirements-lock.txt`. The required dependency ranges are listed in `program/requirements.txt`, and `program/requirements-lock.txt` pins the exact verified versions. They are Python 3.14.6, NumPy 2.4.6, SciPy 1.18.0, and PuLP 3.3.2. The two optional roles use Matplotlib 3.11.0 and NetworkX 3.6.1. Installing from the lock file reproduces that environment.

A.13.1 Equal-budget search records

The protocol is stated in Section 2.6. Its raw records are in `program/data/equal_budget_2026-09-06/`. The original checkpoints are in `trials/` and timing replacements are in replacements/trials/. Both sets include full witnesses. The replacement manifest records which original record each new trial replaces and the clock change used to run it.

The report separates original outcomes from replacement outcomes and names the selected record for every seed. An independent validator rebuilds the flow networks from the saved witnesses using a separate gate construction and NetworkX's implementation of the Edmonds and Karp algorithm. It calls neither the thesis connectivity checker nor its compiled helper. Agreement is a cross-check of feasibility and counts, not a proof of search optimality.

The construction comparison used in the tables is frozen separately in manuscript_baseline.json. It includes constructions added during the audit and is distinct from the earlier comparison snapshot in comparison_baseline.json. Neither file is read by the search runner. The historical observations in Section A.13.2 remain unchanged.

From program/, the reporting commands are:

```
../.venv/bin/python3 scripts/benchmark_report.py data/equal_budget_2026-09-06
../.venv/bin/python3 scripts/validate_benchmark.py data/equal_budget_2026-09-06
../.venv/bin/python3 scripts/benchmark_tables.py data/equal_budget_2026-09-06
```

The table renderer refuses incomplete timing coverage, unvalidated selected witnesses or a search count above a recorded upper bound. Its provenance file records the hashes of the report, comparison snapshot and witness audit.

Each row below contains exactly three timing-qualified runs. Search counts are printed without adjustment, even when a supplied construction is better.

n	Search min	Median	Search max	Construction	Seeds
simple graph, undirected, edge					
6	7	7	7	7	3
8	10	10	10	10	3
10	13	13	13	13	3
12	16	16	16	16	3
simple graph, undirected, vertex					
6	7	7	7	7	3
8	10	10	10	10	3
10	13	13	13	13	3
12	16	16	16	16	3
simple graph, directed, arc					
6	15	15	15	15	3
8	21	21	21	21	3
10	27	27	27	30	3
12	33	33	33	42	3
simple graph, directed, vertex					
6	15	15	15	15	3
8	21	21	21	21	3
10	27	27	27	30	3
12	33	33	33	42	3
multigraph, undirected, edge					
6	10	10	10	10	3
8	14	14	14	14	3
10	18	18	18	18	3
12	22	22	22	22	3
multigraph, undirected, vertex					
6	10	10	10	10	3
8	14	14	14	14	3
10	18	18	18	18	3
12	22	22	22	22	3
multigraph, directed, arc					
6	20	20	20	20	3
8	28	28	28	32	3
10	36	36	36	50	3
12	44	44	44	72	3
multigraph, directed, vertex					
6	20	20	20	20	3
8	28	28	28	32	3
10	36	36	36	50	3
12	44	44	44	72	3

Table A.2. Equal-budget graph results at $m = 3$. Each row receives three 150-second searches. Construction counts are supplied lower bounds, not search inputs or assertions of optimality.

n	Search min	Median	Search max	Construction	Seeds
simple graph, undirected, edge					
6	15	15	15	15	3
8	21	21	21	21	3
10	27	27	27	27	3
12	32	33	33	33	3
simple graph, undirected, vertex					
6	15	15	15	15	3
8	21	21	21	21	3
10	27	27	27	27	3
12	33	33	33	33	3
simple graph, directed, arc					
6	30	30	30	30	3
8	40	40	40	42	3
10	50	50	50	54	3
12	60	60	60	66	3
simple graph, directed, vertex					
6	30	30	30	30	3
8	40	40	40	42	3
10	50	50	50	54	3
12	60	60	60	66	3
multigraph, undirected, edge					
6	25	25	25	25	3
8	35	35	35	35	3
10	45	45	45	45	3
12	55	55	55	55	3
multigraph, undirected, vertex					
6	32	32	32	33	3
8	46	46	46	47	3
10	54	56	56	59	3
12	64	64	64	73	3
multigraph, directed, arc					
6	50	50	50	50	3
8	70	70	70	80	3
10	90	90	90	125	3
12	110	110	110	180	3
multigraph, directed, vertex					
6	58	58	60	60	3
8	70	70	70	84	3
10	90	90	90	125	3
12	91	91	110	180	3

Table A.3. Equal-budget graph results at $m = 6$, under the same protocol as Table A.2.

n	Search min	Median	Search max	Construction	Seeds
simple hypergraph, undirected, edge					
6	5	5	5	5	3
8	7	7	7	7	3
10	9	9	9	9	3
12	10	11	11	11	3
simple hypergraph, undirected, vertex					
6	5	5	5	5	3
8	7	7	7	7	3
10	9	9	9	9	3
12	10	11	11	11	3
simple hypergraph, directed, arc					
6	12	12	12	12	3
8	17	17	17	16	3
10	20	20	20	25	3
12	24	24	24	36	3
simple hypergraph, directed, vertex					
6	12	12	12	12	3
8	16	17	17	16	3
10	20	20	20	25	3
12	24	24	24	36	3
multihypergraph, undirected, edge					
6	5	5	5	5	3
8	7	7	7	7	3
10	9	9	9	9	3
12	10	11	11	11	3
multihypergraph, undirected, vertex					
6	5	5	5	5	3
8	7	7	7	7	3
10	9	9	9	9	3
12	10	11	11	11	3
multihypergraph, directed, arc					
6	13	13	13	12	3
8	16	17	17	16	3
10	20	20	21	25	3
12	24	24	24	36	3
multihypergraph, directed, vertex					
6	13	13	13	12	3
8	16	17	17	16	3
10	20	20	21	25	3
12	24	24	24	36	3

Table A.4. Equal-budget rank-three hypergraph results at $m = 3$. Directed hypergraphs use the forward convention. Each row receives three 150-second searches and construction counts are reported separately.

n	Search min	Median	Search max	Construction	Seeds
simple hypergraph, undirected, edge					
6	11	11	11	11	3
8	16	16	16	17	3
10	19	20	20	22	3
12	23	23	23	27	3
simple hypergraph, undirected, vertex					
6	11	11	11	11	3
8	16	16	16	17	3
10	19	20	20	22	3
12	23	23	23	27	3
simple hypergraph, directed, arc					
6	30	30	30	30	3
8	40	40	40	40	3
10	50	50	50	60	3
12	60	60	60	90	3
simple hypergraph, directed, vertex					
6	30	30	30	30	3
8	40	40	40	40	3
10	50	50	50	60	3
12	60	60	60	90	3
multihypergraph, undirected, edge					
6	12	12	12	12	3
8	16	16	16	17	3
10	19	20	20	22	3
12	22	22	23	27	3
multihypergraph, undirected, vertex					
6	12	12	12	12	3
8	16	16	16	17	3
10	19	20	20	22	3
12	22	22	24	27	3
multihypergraph, directed, arc					
6	30	30	30	30	3
8	40	40	40	40	3
10	50	50	50	60	3
12	60	60	60	90	3
multihypergraph, directed, vertex					
6	31	32	34	30	3
8	40	41	41	40	3
10	50	50	50	60	3
12	60	60	60	90	3

Table A.5. Equal-budget rank-three hypergraph results at $m = 6$, under the same protocol as Table A.4.

A.13.2 Detailed values across the sixteen variants

The four grids below record the finite data behind Figure 3.3 and Table 3.1. Every hypergraph panel and comparison table in this section uses $r = 3$. Directed hypergraphs use the forward orientation, with one tail and two heads. A blue curve is proved. A filled square is a value established by completed enumeration. In particular, the directed multigraph arc formula is drawn as a theorem curve, not as a supposedly independent enumeration result. An open circle is the value returned by the unaided search. A red plus is the value of a separately supplied construction. Both give lower bounds, but they answer different questions: the circle measures what the program found, while the plus records what is known to be feasible independently of that run. The search is not raised to the construction, to a completed enumeration or to a value found at another order or in another variant. A downward step in the circles can therefore reflect a weaker timed run. It does not mean that the true extremal function decreases. Open cases have no interpolated curve. This keeps parity effects in the discrete data from looking like a proposed formula.

Budgets and the limits of the record. These are historical experiments, not a controlled equal-time benchmark. Every matrix search starts from the empty graph and uses tabu search. Every hypergraph search uses randomised greedy growth. The figure driver uses initial seed zero. Most search cases were allocated 0.4 seconds each. The open simple undirected vertex panel and open undirected hypergraph vertex panels use 4 seconds, as do all directed hypergraph panels. Here a *case* means one fixed variant, n and m, not a whole row of sizes. The comparison tables below print the applicable limit on every search row.

Enumeration has a separate budget: 60 seconds per case by default, 1800 seconds for simple directed graphs and 120 seconds for multigraph vertex separation. Enumeration stops sampling a panel after its first timeout. This is a sampling rule, not a proof that every larger case would also time out. The standalone computation at $(m, n, r) = (6, 6, 3)$ is a separate experiment.

The original record saves values and requested budgets, but not each run's elapsed time or witness. Its limits are cooperative stopping targets: a connectivity check already in progress can finish after the deadline. Consequently, neither exact one-hour runs nor equal computational effort can be inferred from these data. Fixed seeds do not make timed runs repeatable across different machines or loads. The frozen file makes the *reported observations* reproducible. Re-running the searches is a new experiment.

`program/data/search_evidence.json` exports the raw search and supplied construction values in distinct fields for every plotted order. Missing historical elapsed times and witnesses are explicitly marked as unknown. The companion tables distinguish the same sources numerically. A construction entry of zero means that the selected formula supplies no positive lower bound in that parameter range, not that the optimum is zero.

Proposition 1.8 bounds every simple hypergraph from above. It does not assert attainment outside its stated range. At $m = 6$, $n = 6$, and $r = 3$, all $125\,970$ simple twelve-hyperedge candidates are infeasible, and an eleven-hyperedge family is not. The true maximum is there-

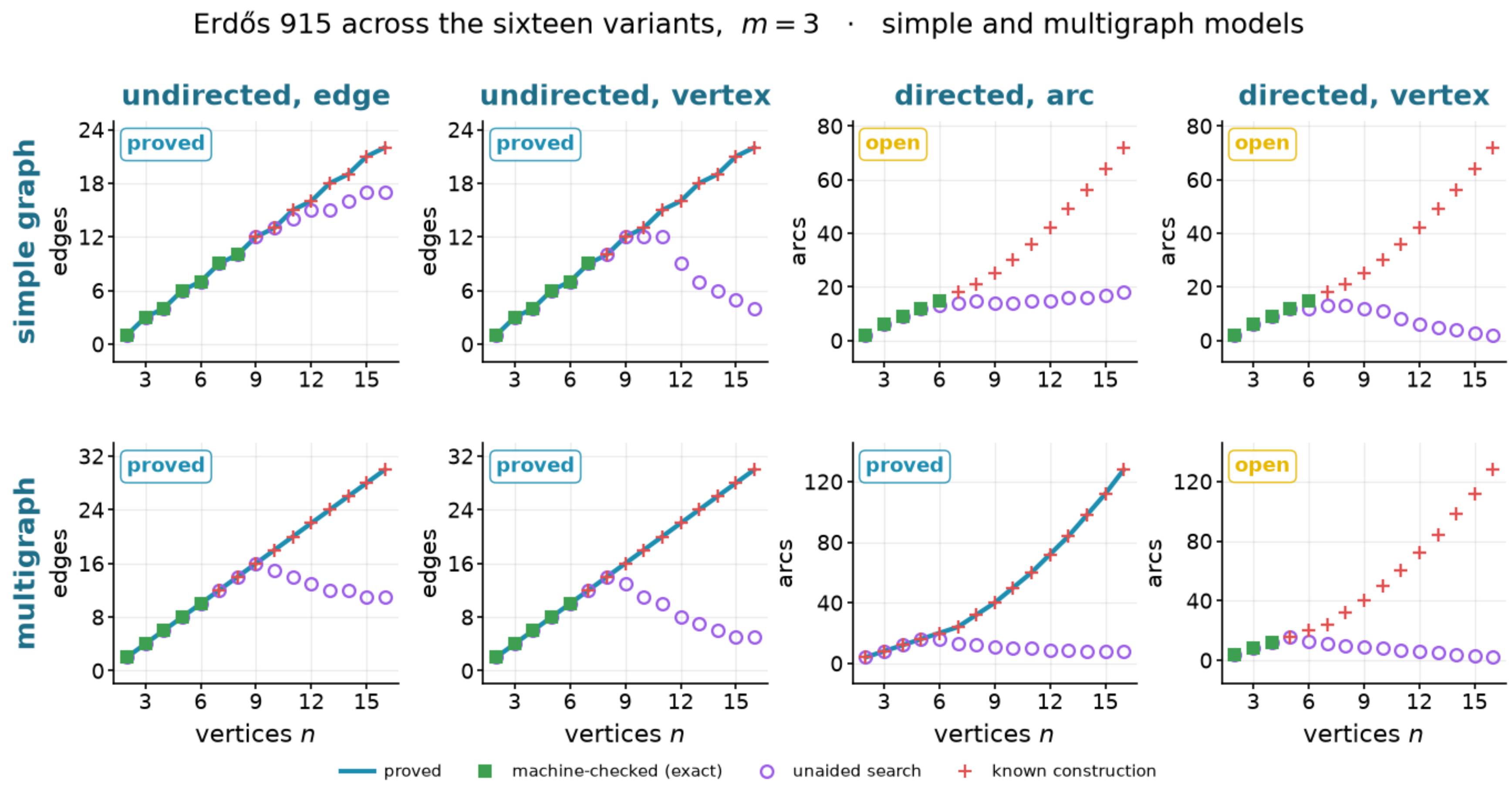


Figure A.4. The eight graph-model variants at $m = 3$. The simple directed arc and vertex panels show verified lower bounds and completed exact values only. The construction marks assert feasibility, not optimality. No conjectured finite-order formula is drawn.

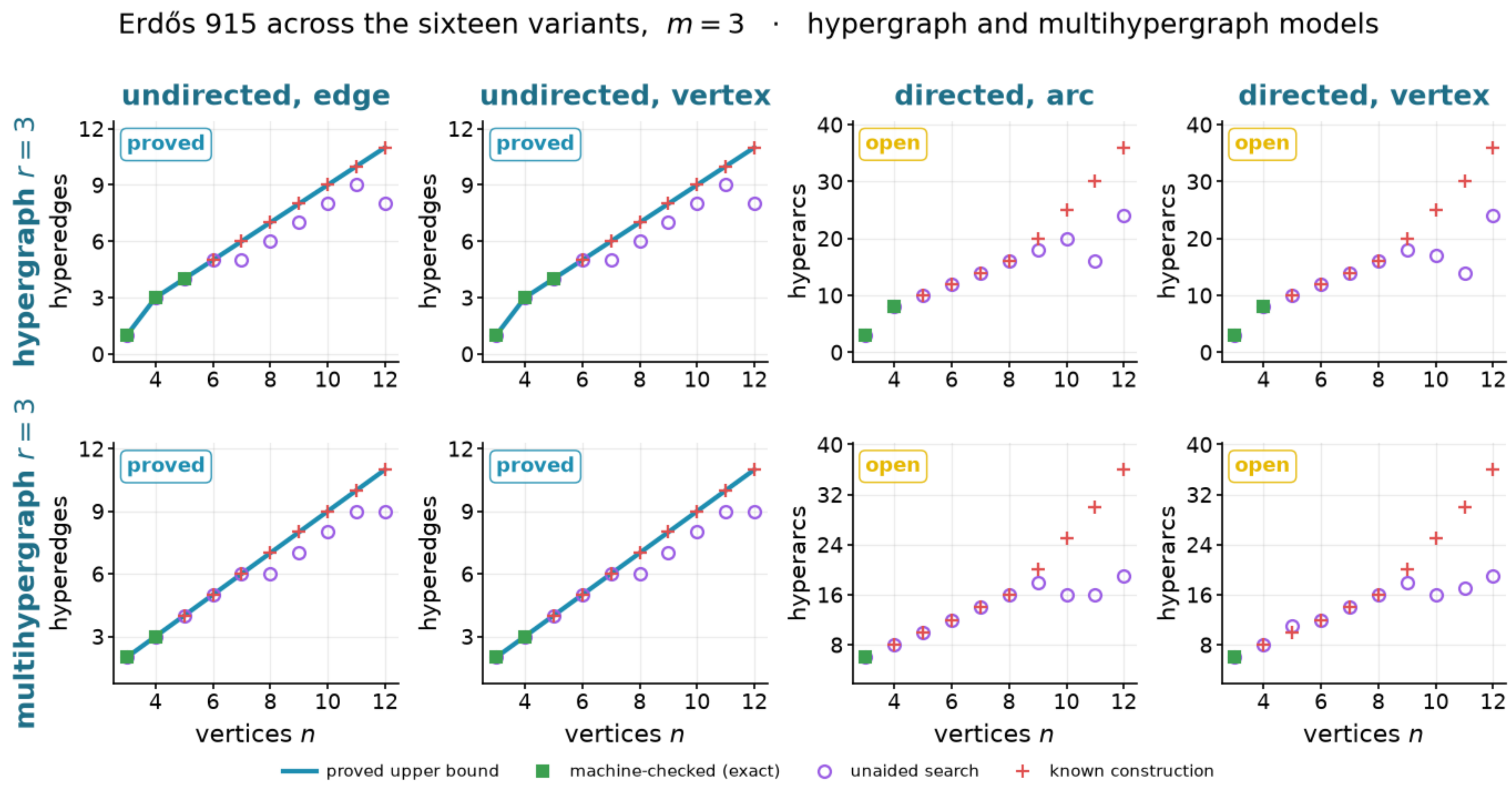


Figure A.5. The eight hypergraph-model variants at $m = 3$. The blue curve is the proved upper bound of Proposition 1.8, not always a proved value. A lower-bound point below it does not alone prove that the bound is unattainable. Only an exact square below it establishes such a gap.

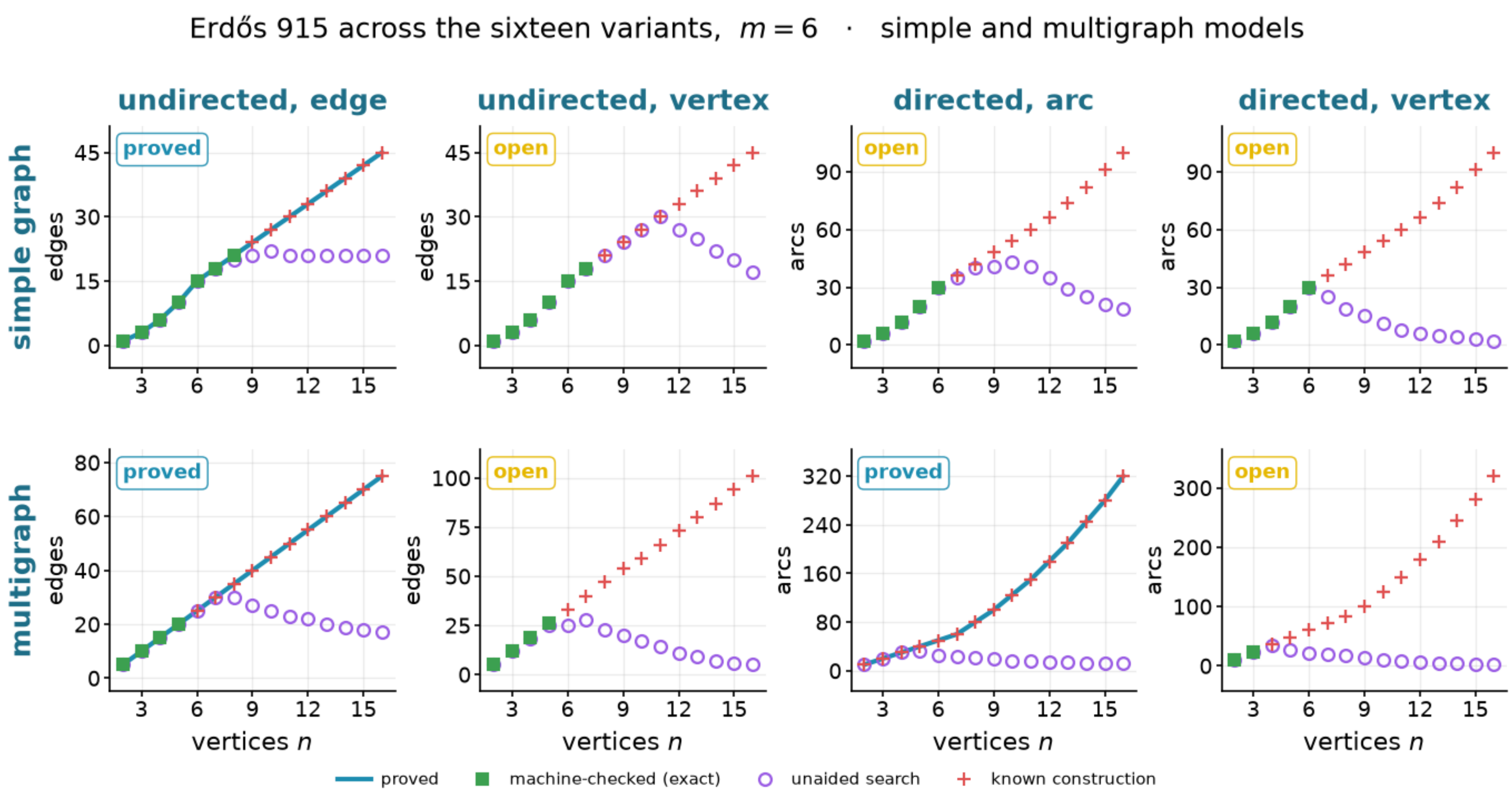


Figure A.6. The eight graph-model variants at $m = 6$. Fewer exact squares appear because exhaustive search reaches fewer vertices as the feasible range grows.

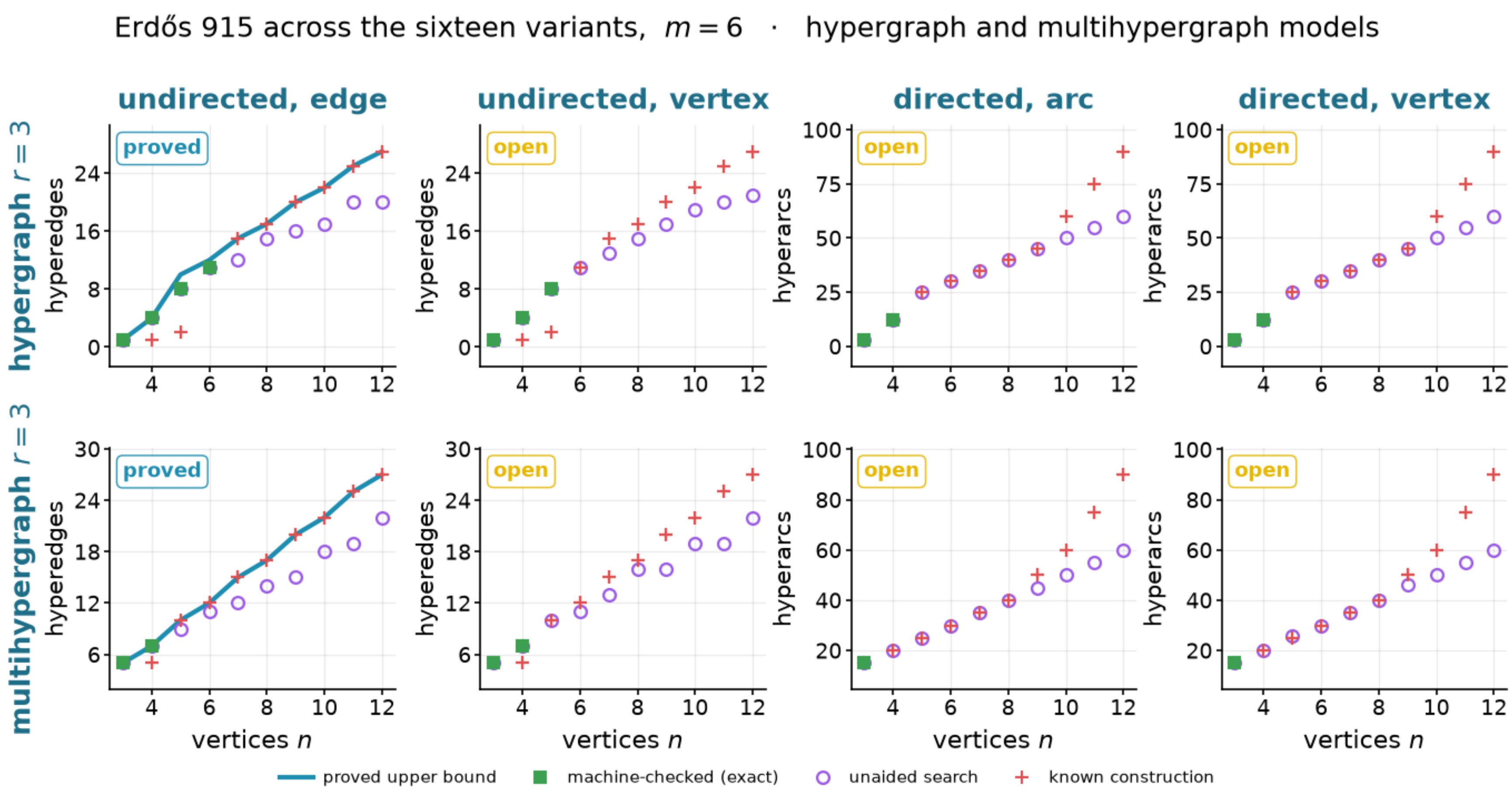


Figure A.7. The eight hypergraph-model variants at $m = 6$. In the simple undirected edge panel the proved upper bound is 10 at $n = 5$ and 12 at $n = 6$, while the true maxima are 8 and 11. Both are drawn as squares. The 8 comes from the exhaustive solve, which finished at $n = 5$ and not at $n = 6$, and the 11 from the twelve-hyperedge enumeration below.

fore 11, one below the bound of 12. That enumeration runs from its own standalone script, `program/scripts/hyper_edge_m6_n6_gap.py`, because the main program's branch and bound does not finish this cell inside the figure budget. It supplies the exhaustive computation the branch and bound could not, so the value is recorded as exact. The smallest analogous gap occurs at $n = r = m = 3$. A simple hypergraph then has one hyperedge, while a multihypergraph attains the bound with two copies.

Table A.6 gives the same data numerically. Blue entries are proved, and gold entries mark open cases. A bold green entry is machine-checked exact. The prefix $\leq$ marks a proved upper bound that no recorded exact value matches at that point. The prefix $\geq$ marks a verified construction without a matching upper bound. A blank hypergraph cell at $n = 2$ omits the degenerate zero value: no 3-element hyperedge can be formed. The simple hypergraph cell at $m = 6, n = 6$ reads a bold 11 against the bound of 12 drawn as the curve above it, which is the gap the enumeration just described settles. The corresponding multihypergraph *edge* cell is 12 by Construction A.50. It is blue rather than a green square because its exactness comes from an explicit construction matching the proved upper bound, not from completed enumeration. The historical search value is unchanged. The new construction also raises the multihypergraph vertex lower bound in this cell to 12, without settling its optimum.

Search performance versus supplied constructions. The next four tables report the two lower-bound sources separately. Unlike Table A.6, they do not replace either source with a proved optimum. For example, a search value of s below a construction value of c says that the unaided run found s edges although a feasible example with c edges is known. It is a limitation of that run, not a contradiction. These tables show orders through eight. The data file includes the full plotted ranges.

In the audit below, *branch and bound* means exhaustive generation in which a partial candidate is extended along several branches and a proved bound discards, or *prunes*, a branch that cannot reach the target. A search *node* is one such partial candidate.

The directed $m = 2$ values at $n \leq 7$ were settled independently by a pruned exhaustive search that is exact and complete at these sizes, so each completed row is an upper-bound proof at that n. The driver was run twice, once over all simple digraphs with $\kappa^{\max} \leq 1$ and once with $\lambda^{\max} \leq 1$, only the inner feasibility test swapped, and Table A.11 records both. These rows were the base cases of the vertex-deletion induction that Section A.5 used to carry. That induction has been replaced by the reachability-skeleton argument, which closes every n in one step, so the table no longer supports either theorem and both columns now stand as an independent check on them. The general theorems do not depend on these computations. The finite values explicitly labelled as exhaustive results do. What the two runs share by coincidence is their search-tree size, the node counts coinciding at every n as a computed fact.

The historical cut-counting optimisation closes the directed multigraph checks through $n = 6$ but emits no independently replayable matching bound, so no theorem rests on it. Its raw log is `program/logs/certificate_log.txt`. Exact flow measures a supplied object, completed enumeration proves a finite optimum, and heuristic search supplies only a checked witness and

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
$m=3$								
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), open	2	6	9	12	15	≥18	≥21
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), open	2	6	9	12	15	≥18	≥21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($K_m(n)$), proved	2	4	6	8	10	12	14
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($K_m^{\mathrm{dir}}(n)$), open	4	8	12	≥16	≥20	≥24	≥32
hypergraph $r=3$	undirected, edge, proved		1	3	4	5	6	7
	undirected, vertex, proved		1	3	4	5	6	7
	directed, arc, open		3	8	≥10	≥12	≥14	≥16
	directed, vertex, open		3	8	≥10	≥12	≥14	≥16
multihypergraph $r=3$	undirected, edge, proved		2	3	4	5	6	7
	undirected, vertex, proved		2	3	4	5	6	7
	directed, arc, open		6	≥8	≥10	≥12	≥14	≥16
	directed, vertex, open		6	≥8	≥11	≥12	≥14	≥16
$m=6$								
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	≥21
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), open	2	6	12	20	30	≥36	≥42
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), open	2	6	12	20	30	≥36	≥42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($K_m(n)$), open	5	12	19	26	≥33	≥40	≥47
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($K_m^{\mathrm{dir}}(n)$), open	10	24	≥36	≥48	≥60	≥72	≥84
hypergraph $r=3$	undirected, edge, proved		1	4	8	11	15	17
	undirected, vertex, open		1	4	8	≥11	≥15	≥17
	directed, arc, open		3	12	≥25	≥30	≥35	≥40
	directed, vertex, open		3	12	≥25	≥30	≥35	≥40
multihypergraph $r=3$	undirected, edge, proved		5	7	10	12	15	17
	undirected, vertex, open		5	7	≥10	≥12	≥15	≥17
	directed, arc, open		15	≥20	≥25	≥30	≥35	≥40
	directed, vertex, open		15	≥20	≥26	≥30	≥35	≥40

Table A.6. Finite values and bounds for all sixteen variants at $m=3$ and $m=6$. The colours and markers match the four grids above.

Variant	Source	Limit (s)	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
simple graph, undirected, edge	search	0.4	1	3	4	6	7	9	10
	construction	–	1	3	4	6	7	9	10
simple graph, undirected, vertex	search	0.4	1	3	4	6	7	9	10
	construction	–	1	3	4	6	7	9	10
simple graph, directed, arc	search	0.4	2	6	9	12	13	14	15
	construction	–	2	6	9	12	15	18	21
simple graph, directed, vertex	search	0.4	2	6	9	12	12	13	13
	construction	–	2	6	9	12	15	18	21
multigraph, undirected, edge	search	0.4	2	4	6	8	10	12	14
	construction	–	2	4	6	8	10	12	14
multigraph, undirected, vertex	search	0.4	2	4	6	8	10	12	14
	construction	–	2	4	6	8	10	12	14
multigraph, directed, arc	search	0.4	4	8	12	16	16	13	12
	construction	–	4	8	12	16	20	24	32
multigraph, directed, vertex	search	0.4	4	8	12	16	13	11	10
	construction	–	4	8	12	16	20	24	32

Table A.7. Raw search and separately supplied construction counts for the graph-model variants at $m = 3$. The search limit is per $(n, m, \text{variant})$ case. A dash means no search budget applies to the construction.

Variant	Source	Limit (s)	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
hypergraph, undirected, edge	search	0.4	1	3	4	5	5	6
	construction	–	1	3	4	5	6	7
hypergraph, undirected, vertex	search	0.4	1	3	4	5	5	6
	construction	–	1	3	4	5	6	7
hypergraph, directed, arc	search	4	3	8	10	12	14	16
	construction	–	3	8	10	12	14	16
hypergraph, directed, vertex	search	4	3	8	10	12	14	16
	construction	–	3	8	10	12	14	16
multihypergraph, undirected, edge	search	0.4	2	3	4	5	6	6
	construction	–	2	3	4	5	6	7
multihypergraph, undirected, vertex	search	0.4	2	3	4	5	6	6
	construction	–	2	3	4	5	6	7
multihypergraph, directed, arc	search	4	6	8	10	12	14	16
	construction	–	6	8	10	12	14	16
multihypergraph, directed, vertex	search	4	6	8	11	12	14	16
	construction	–	6	8	10	12	14	16

Table A.8. Raw search and separately supplied construction counts for the hypergraph-model variants at $m = 3$. The search limit is per $(n, m, \text{variant})$ case. A dash means no search budget applies to the construction.

Variant	Source	Limit (s)	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
simple graph, undirected, edge	search	0.4	1	3	6	10	15	18	20
	construction	–	1	3	6	10	15	18	21
simple graph, undirected, vertex	search	4	1	3	6	10	15	18	21
	construction	–	1	3	6	10	15	18	21
simple graph, directed, arc	search	0.4	2	6	12	20	30	35	40
	construction	–	2	6	12	20	30	36	42
simple graph, directed, vertex	search	0.4	2	6	12	20	30	25	19
	construction	–	2	6	12	20	30	36	42
multigraph, undirected, edge	search	0.4	5	10	15	20	25	30	30
	construction	–	5	10	15	20	25	30	35
multigraph, undirected, vertex	search	0.4	5	12	18	25	25	28	23
	construction	–	5	12	19	26	33	40	47
multigraph, directed, arc	search	0.4	10	20	30	33	25	23	21
	construction	–	10	20	30	40	50	60	80
multigraph, directed, vertex	search	0.4	10	24	34	28	22	19	17
	construction	–	10	24	36	48	60	72	84

Table A.9. Raw search and separately supplied construction counts for the graph-model variants at $m = 6$. The search limit is per $(n, m, \text{variant})$ case. A dash means no search budget applies to the construction.

Variant	Source	Limit (s)	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
hypergraph, undirected, edge	search	0.4	1	4	8	11	12	15
	construction	–	1	1	2	11	15	17
hypergraph, undirected, vertex	search	4	1	4	8	11	13	15
	construction	–	1	1	2	11	15	17
hypergraph, directed, arc	search	4	3	12	25	30	35	40
	construction	–	3	12	25	30	35	40
hypergraph, directed, vertex	search	4	3	12	25	30	35	40
	construction	–	3	12	25	30	35	40
multihypergraph, undirected, edge	search	0.4	5	7	9	11	12	14
	construction	–	5	5	10	12	15	17
multihypergraph, undirected, vertex	search	4	5	7	10	11	13	16
	construction	–	5	5	10	12	15	17
multihypergraph, directed, arc	search	4	15	20	25	30	35	40
	construction	–	15	20	25	30	35	40
multihypergraph, directed, vertex	search	4	15	20	26	30	35	40
	construction	–	15	20	25	30	35	40

Table A.10. Raw search and separately supplied construction counts for the hypergraph-model variants at $m = 6$. The search limit is per $(n, m, \text{variant})$ case. A dash means no search budget applies to the construction.

n	arc-disjoint	vertex-disjoint	$M(n)$	nodes	status
2	2	2	2	5	complete
3	4	4	4	22	complete
4	6	6	6	537	complete
5	8	8	8	17 823	complete
6	10	10	10	788 101	complete
7	12	12	12	45 122 129	complete

Table A.11. The directed $m = 2$ values at small n under both separations: exhaustive branch and bound over all simple digraphs on n vertices with $\lambda^{\max} \leq 1$, and the same run with $\kappa^{\max} \leq 1$. Same driver, same prunes, only the inner feasibility test changes. Both columns agree with $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ at every size, and every run completed. Neither column is load-bearing. These were the base cases of an earlier induction, which Section A.5 has replaced by an argument from Lemma A.19 that runs at every n at once, so both columns now corroborate rather than prove. The node counts coincide as a computed fact, not by construction: the feasibility prune switches between the arc and the vertex flow test with the separation, and at these sizes the two happen to cut the tree in the same places. The wall times do not coincide, with the vertex run at $n = 7$ taking 3125 s against the arc run's 718 s. Full logs: `program/logs/basecase_search_log.txt` and `program/logs/basecase_search_vertex_log.txt`.

lower bound. The file `program/README.md` records the dependencies, commands, entry points, and archival checklist.

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